

ORIGINAL ARTICLE

A survey on retrieval-augmentation generation (RAG) models for healthcare applications

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Abstract

Retrieval-augmented generation (RAG) models have become crucial in healthcare applications, significantly enhancing the relevance and reliability of AI-driven insights by combining the generative capabilities of large language models (LLMs) with retrieval-based methods. As healthcare data demand precision and accountability, RAG models address critical limitations of LLMs, such as the tendency to “hallucinate” or produce inaccurate information—by incorporating real-time retrieval from trusted medical knowledge bases and clinical literature. This dual process of retrieving and generating ensures that responses are both contextually accurate and aligned with the latest clinical evidence, making RAG models especially valuable for medical question answering, diagnostics, and treatment planning. This paper presents an in-depth exploration of RAG models and LLMs, specifically within healthcare contexts, to meet the unique demands of medical data processing and decision support. It reviews various RAG architectures, including Naive, Advanced, and Modular RAG approaches, discussing how each framework optimizes retrieval depth, response quality, and computational efficiency. By reducing errors in patient-care recommendations, RAG models play an essential role in scenarios that require high precision and accountability. Additionally, this survey addresses the ethical considerations and transparency requirements for deploying RAG models in healthcare, identifying current challenges and future directions, such as enhancing source interpretability and adapting RAG frameworks for specialized medical fields. By systematically analyzing RAG techniques, this paper provides a comprehensive guide to the state-of-the-art in RAG applications within healthcare, positioning RAG models as a transformative tool for advancing AI-supported healthcare outcomes.

Keywords Generative AI · Large language model · Contrastive language-image pre-training · Multimodal · Retrieval-augmented generation · Natural language processing · Knowledge retrieval

1 Introduction

The rapid advancement of artificial intelligence (AI), particularly in natural language processing, has ushered in a new era of possibilities for healthcare. Large language models (LLMs) have emerged as transformative tools, demonstrating remarkable capabilities in understanding, generating, and reasoning with medical language [1, 2]. From general-purpose models like GPT-3 and GPT-4 to specialized medical models such as Med-PaLM [3], these AI systems are reshaping clinical practice by enhancing diagnostic accuracy, improving patient-provider communication, and supporting personalized treatment planning.

However, the application of LLMs in healthcare presents unique challenges that demand specialized solutions. The high-stakes nature of medical decision-making, where errors can have life-altering consequences, requires an



unprecedented level of accuracy and reliability. Traditional LLMs, while powerful, face several critical limitations in healthcare settings:

- **Static Knowledge Problem:** Medical knowledge evolves rapidly, with new research, clinical guidelines, and treatment protocols emerging constantly [4]. LLMs trained on fixed datasets quickly become outdated, potentially providing recommendations based on superseded information.
- **Hallucination Risk:** The tendency of LLMs to generate plausible-sounding but factually incorrect information is particularly dangerous in healthcare contexts, where accuracy is paramount [5].
- **Domain-Specific Nuances:** Medical language contains specialized terminology, abbreviations, and contextual dependencies that general-purpose LLMs may misinterpret [6].
- **Evidence-Based Practice:** Modern medicine requires decisions to be grounded in verifiable evidence, yet standard LLMs often lack transparent sourcing for their outputs [7].

To address these challenges, Retrieval-Augmented Generation (RAG) models have emerged as a promising solution [8]. RAG architectures combine the generative capabilities of LLMs with dynamic retrieval from authoritative medical knowledge bases, creating systems that can:

- Access up-to-date clinical guidelines and research findings
- Ground responses in verifiable sources
- Adapt to specialized medical subdomains
- Provide transparent rationales for clinical recommendations

The potential applications of RAG in healthcare are vast and transformative:

Clinical Decision Support: RAG systems can assist clinicians by retrieving relevant case studies, treatment guidelines, and drug interaction information in real-time, reducing diagnostic errors and improving treatment outcomes [9].

Medical Education: These models can serve as intelligent tutors, providing medical students with accurate, context-aware explanations drawn from the latest literature [10].

Patient Engagement: By generating personalized health information grounded in authoritative sources, RAG systems can improve health literacy and shared decision-making [11].

Administrative Efficiency: Automating documentation tasks like clinical note generation and coding with retrieval-verified accuracy can reduce clinician burnout [12].

This paper provides a comprehensive examination of RAG models in healthcare, analyzing their architectural variations, performance characteristics, and practical implementations. We systematically review:

1. The evolution of RAG architectures from naive to advanced modular implementations
2. Performance benchmarks across medical specialties and use cases
3. Ethical considerations and implementation challenges
4. Emerging applications and future directions

Our analysis reveals that while RAG models show tremendous promise in healthcare applications [13], their successful implementation requires careful consideration of:

- Retrieval accuracy and relevance in specialized medical contexts
- Integration with clinical workflows and electronic health record systems
- Privacy and security protections for sensitive patient data
- Regulatory compliance and liability frameworks
- Bias mitigation and equity considerations

As healthcare systems worldwide face increasing pressure to deliver high-quality, cost-effective care, RAG models offer a pathway to augment clinical expertise with AI-powered knowledge retrieval and synthesis [14]. This review aims to provide researchers, clinicians, and healthcare administrators with the insights needed to harness this potential responsibly and effectively.

The subsequent sections will delve deeper into the technical architectures of medical RAG systems, their performance across various healthcare applications, the datasets and evaluation metrics used to assess them, and the ethical and practical considerations for their deployment in real-world clinical settings. Through this comprehensive analysis, aim to chart the current state of RAG in healthcare and illuminate promising directions for future research and development in this critical intersection of AI and medicine.

1.1 Contributions

This survey paper makes the following key contributions:

- **Systematic Taxonomy of RAG Architectures:** A hierarchical classification of Naive, Advanced, and Modular RAG frameworks with comparative analysis of their components and performance characteristics in medical applications.
- **Data-to-Model Pipeline Analysis:** Comprehensive examination of the medical AI pipeline from foundational datasets (textual, visual, and multimodal) to implemented systems, highlighting distinct challenges and solutions for different data modalities.
- **Medical Application Spectrum:** Coverage of RAG implementations across clinical domains including diagnostic support, treatment planning, medical education, and multimodal interpretation of imaging data.
- **Rigorous Model Evaluation:** Standardized assessment of RAG systems using both quantitative metrics (accuracy, BLEU, ROUGE) and qualitative frameworks (S.C.O.R.E.) across medical benchmarks.
- **Implementation Challenges:** Critical analysis of deployment barriers including data privacy, algorithmic bias, regulatory requirements, and clinical workflow integration.
- **Emerging Research Frontiers:** Identification of promising directions including real-time knowledge integration, specialty-specific architectures, and validation protocols for clinical adoption.

This work establishes a foundational reference for AI researchers, medical practitioners, and healthcare technologists developing next-generation RAG systems for medicine.

2 Comprehensive applications of retrieval-augmented generation in healthcare [15–19]

Retrieval-Augmented Generation (RAG) models represent a transformative advancement in healthcare AI by synergizing the generative capabilities of Large Language Models (LLMs) with real-time access to authoritative medical knowledge sources. This integration addresses critical challenges such as outdated training data and the propensity for AI hallucination, ensuring that outputs are both accurate and grounded in the latest clinical evidence. The following subsections explore the diverse applications of RAG in the medical domain, highlighting its impact on clinical workflows, diagnostic precision, and patient care.

2.1 Enhanced clinical documentation and decision support

RAG models excel in processing and summarizing vast volumes of clinical data, including electronic health records (EHRs), research literature, and patient notes. By distilling this information into concise summaries, they empower healthcare professionals to quickly grasp complex cases. For instance, RAG-powered question-answering (QA) systems retrieve evidence-based responses from trusted repositories like PubMed or clinical guidelines, enabling clinicians to make informed decisions without manually sifting through extensive databases.

Key applications include:

- **Automated Clinical Summarization:** Generation of patient summaries by synthesizing information from multiple EHR sections, laboratory results, and imaging reports
- **Real-time Clinical Guidelines Retrieval:** Dynamic access to updated treatment protocols and best practices during patient encounters
- **Medication Reconciliation:** Cross-referencing patient medications with drug interaction databases and formulary guidelines
- **Discharge Planning:** Automated generation of discharge summaries with personalized patient instructions based on diagnosis and treatment history

This capability is particularly valuable in time-sensitive scenarios, such as emergency medicine or ICU care, where rapid access to relevant clinical information can significantly impact patient outcomes.

2.2 Precision in named entity recognition (NER) and clinical data extraction

In healthcare, extracting structured information from unstructured text—such as diagnoses, medications, or laboratory results—is essential for automating workflows and maintaining data quality. RAG models leverage medical ontologies and knowledge graphs to enhance NER accuracy, identifying and categorizing key clinical entities with high precision.

Advanced applications include:

- **Automated ICD-10 and CPT Coding:** Extraction and assignment of appropriate diagnostic and procedural codes from clinical notes
- **Clinical Trial Matching:** Identification of eligible patients by extracting relevant criteria from EHRs and matching against trial protocols
- **Adverse Event Detection:** Automated identification and reporting of adverse drug reactions and medical errors from clinical documentation
- **Quality Measure Calculation:** Extraction of data elements required for healthcare quality reporting and performance metrics

These systems can automatically populate EHRs with diagnostic codes or flag potential drug interactions by cross-referencing patient data with up-to-date pharmacological databases, thereby reducing administrative burdens and minimizing errors in data entry.

2.3 Advanced diagnostic and treatment planning

RAG models support differential diagnosis and personalized treatment planning by integrating multimodal data, including patient histories, imaging reports, genomic profiles, and population health data. The systems provide clinicians with evidence-based recommendations while maintaining transparency through source attribution.

Clinical applications encompass:

- **Differential Diagnosis Support:** Generation of ranked differential diagnoses with supporting evidence from medical literature
- **Personalized Treatment Recommendations:** Integration of patient-specific factors with current treatment guidelines and emerging therapies
- **Drug Dosing and Selection:** Optimization of medication choices based on patient characteristics, drug interactions, and efficacy data
- **Prognostic Assessment:** Risk stratification and outcome prediction using integrated clinical and research data

In oncology, for instance, a RAG-augmented LLM could retrieve the latest research on targeted therapies while correlating it with a patient's genetic markers, tumor characteristics, and treatment history. Multi-turn dialog systems powered by RAG enable dynamic interactions, allowing clinicians to refine queries iteratively—such as

exploring alternative diagnoses or comparing treatment efficacy—based on real-time retrieval of relevant studies or case reports.

2.4 Multimodal applications in medical imaging and pathology

RAG extends its utility to visual data interpretation, bridging the gap between imaging analysis and textual medical knowledge. These systems integrate computer vision capabilities with medical literature retrieval to provide comprehensive diagnostic support.

Key implementations include:

- **Radiological Report Generation:** Automated creation of structured radiology reports by combining image analysis with relevant diagnostic criteria and case studies
- **Pathology Image Analysis:** Correlation of histopathological findings with literature on rare diseases and diagnostic patterns
- **Clinical Photography Assessment:** Analysis of dermatological conditions, wound healing progress, and surgical outcomes
- **Medical Image Education:** Generation of educational content by linking imaging findings to relevant medical literature and case studies

In radiology, these models generate detailed descriptions of X-rays, CT scans, or MRIs by retrieving similar cases and current diagnostic criteria, aiding radiologists in detecting anomalies and reducing interpretation time. For pathology, RAG can correlate histopathology images with literature on rare diseases, improving diagnostic confidence and supporting subspecialty consultation. Multimodal QA systems further allow clinicians to submit image-based queries (e.g., “Identify this skin lesion”) and receive synthesized insights combining visual and textual evidence.

2.5 Clinical research and evidence synthesis

RAG models significantly enhance clinical research capabilities by automating literature reviews, protocol development, and data analysis workflows. These applications support evidence-based medicine and accelerate the translation of research findings into clinical practice.

Research applications include:

- **Systematic Literature Reviews:** Automated identification, screening, and synthesis of relevant research papers for meta-analyses
- **Clinical Protocol Development:** Generation of research protocols by integrating regulatory requirements with best practices
- **Grant Writing Support:** Assistance in literature review sections and methodology development for research proposals
- **Biomarker Discovery:** Integration of genomic, proteomic, and clinical data to identify potential therapeutic targets

2.6 Patient education and communication

RAG systems enhance patient-provider communication by generating personalized educational materials and facilitating patient engagement. These tools ensure that patient education is both accurate and tailored to individual health literacy levels.

Communication applications encompass:

- **Personalized Patient Education:** Generation of condition-specific educational materials adapted to patient reading level and cultural background
- **Informed Consent Support:** Creation of comprehensive yet understandable consent documents for procedures and treatments
- **Medication Counseling:** Automated generation of medication guides with drug-specific instructions and precautions
- **Symptom Checker Systems:** Interactive tools that provide evidence-based health information while encouraging appropriate care-seeking behavior

2.7 Population health and epidemiological surveillance

RAG models contribute to population health management by analyzing trends, identifying outbreaks, and supporting public health decision-making through integration of diverse data sources including clinical records, laboratory data, and epidemiological reports.

Public health applications include:

- **Disease Surveillance:** Real-time monitoring of disease patterns and outbreak detection through EHR analysis
- **Health Disparities Analysis:** Identification of care gaps and health inequities across different population groups
- **Preventive Care Optimization:** Analysis of screening rates and development of targeted intervention strategies
- **Health Policy Support:** Evidence synthesis for healthcare policy development and resource allocation decisions

2.8 Ethical and practical considerations

While RAG mitigates hallucinations and improves transparency through source citation, several challenges remain in healthcare implementation:

- **Data Privacy and Security:** Ensuring HIPAA compliance and protecting patient confidentiality in retrieval systems
- **Bias Mitigation:** Addressing potential biases in retrieved content and ensuring equitable care recommendations
- **Clinical Workflow Integration:** Seamless incorporation of RAG systems into existing EHR platforms and clinical workflows
- **Regulatory Compliance:** Meeting FDA requirements for AI/ML-based medical devices and maintaining clinical validation
- **Liability and Accountability:** Establishing clear frameworks for responsibility when AI systems contribute to clinical decisions

Future advancements may focus on real-time updates from federated knowledge bases, robust validation frameworks to align AI outputs with evolving medical standards, and the development of specialized RAG architectures optimized for specific medical specialties.

3 Retrieval-augmented generation (RAG) architectures [15–18, 20, 21]

RAG has emerged as a critical approach for grounding large language models (LLMs) in external knowledge sources, particularly in healthcare applications where accuracy and factual grounding are paramount. This section provides a comprehensive analysis of three distinct RAG paradigms: Naive, Advanced, and Modular RAG, examining their architectural components, capabilities, and performance characteristics (Fig. 1).

3.1 Naive retrieval-augmented generation (RAG)

Naive RAG provides a foundational framework for integrating retrieval systems with large language models (LLMs), representing the simplest form of RAG without the sophistication needed to handle complex queries. This framework follows a “Retrieve-Read” methodology: first indexing relevant data, then retrieving information based on user queries, and finally generating responses using the retrieved content.

3.1.1 Core architectural components

Indexing Phase: The process begins with cleaning and converting raw data from various sources, such as PDFs, HTML, and Word documents, into a uniform text format. This data preprocessing stage involves removing formatting artifacts, normalizing text encoding, and ensuring consistent document structure. Texts are then segmented into smaller chunks using basic splitting strategies, typically based on fixed character counts or simple sentence boundaries. These chunks are encoded as vector representations using an embedding model, creating dense numerical representations that capture semantic content. These vectors are stored in a vector database, which enables efficient retrieval in subsequent stages through similarity search operations.

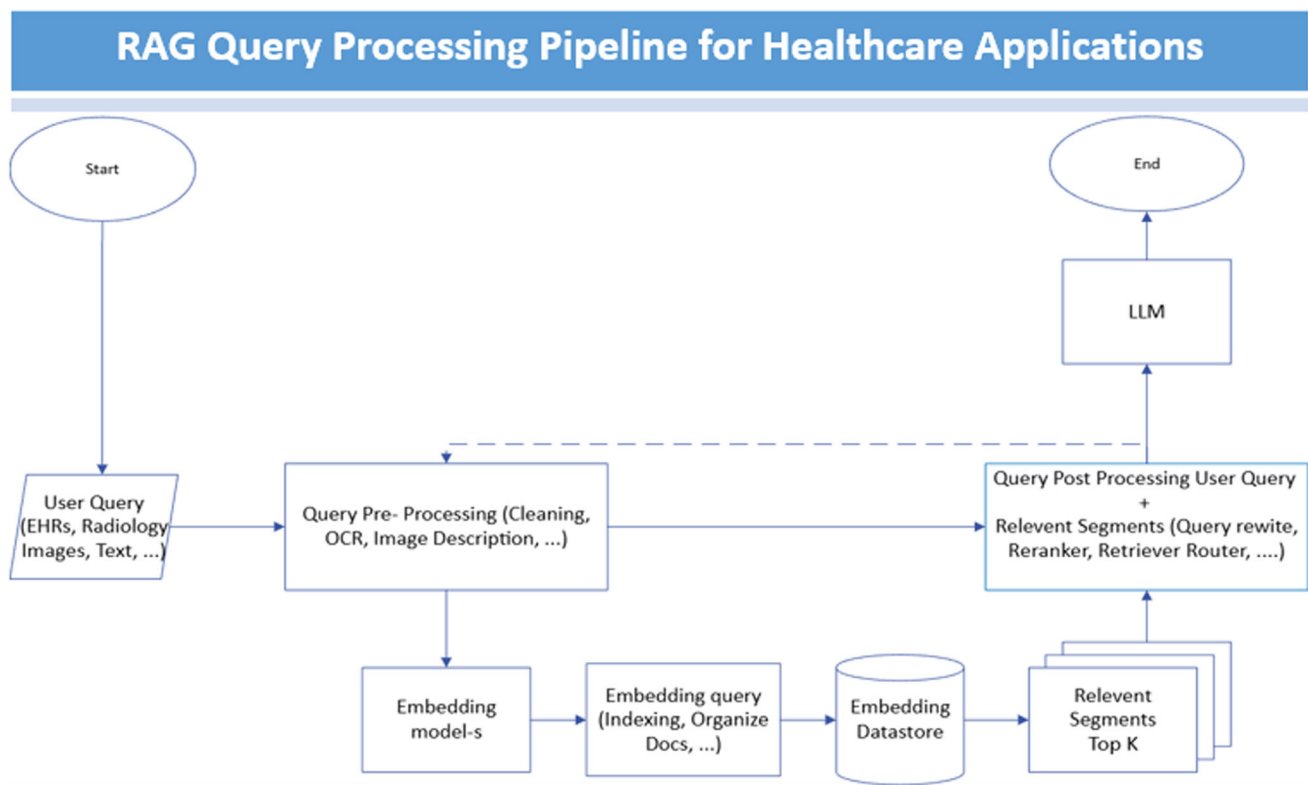


Fig. 1 Illustration diagram for RAG Query processing pipeline

Retrieval Phase: The retrieval process starts when a user submits a query. The system encodes this query into a vector using the same embedding model employed during the indexing phase, ensuring consistency in the vector space representation. The system then compares the query vector to the pre-encoded chunks in the database using similarity metrics, typically cosine similarity or Euclidean distance. Using these similarity computations, the system identifies and retrieves the top K chunks that are most relevant to the query, where K is a predefined parameter. These retrieved chunks serve as an extended context for the model, allowing it to generate a more contextually grounded response.

Generation Phase: This phase involves constructing a prompt that includes both the original query and the retrieved context. The retrieved chunks are typically concatenated with the user query in a straightforward manner, often using simple templates or delimiters. An LLM then generates a response, which can rely on either the model's internal parametric knowledge or strictly adhere to the retrieved information. This approach also supports multi-turn interactions by incorporating conversational history, allowing the model to engage in ongoing dialog while maintaining context from previous exchanges.

3.1.2 Key characteristics and limitations

The Naive RAG framework operates on basic principles without sophisticated optimization mechanisms. The chunking strategy is typically uniform and static, without consideration for semantic boundaries or content coherence. The retrieval mechanism relies purely on embedding similarity without additional ranking or filtering mechanisms. The generation process uses simple concatenation strategies without context optimization or relevance filtering.

Significant limitations exist within the Naive RAG framework. Retrieval inaccuracies can result in the selection of irrelevant chunks, which may impact the quality of generated responses due to the lack of sophisticated relevance assessment. Additionally, the generation phase can suffer from hallucinations, where the model produces information not substantiated by the retrieved content, as there are no mechanisms to ensure strict adherence to retrieved information. The fixed pipeline architecture cannot adapt to different query types or complexity levels, limiting its effectiveness across diverse use cases.

3.1.3 Performance analysis

Healthcare implementations of Naive RAG have demonstrated significant improvements over foundational models alone. Geet et al. showed enhanced specificity in liver disease guidance generation using GPT-3.5-turbo with embedded clinical guidelines. Similarly, ChatENT demonstrated reduced hallucinations in otolaryngology applications compared to vanilla ChatGPT-4.0. Zakka et al.'s Almanac system, augmented with clinical knowledge vectors, provided more accurate and factual responses across 130 clinical cases evaluated by domain experts.

Quantitative improvements are substantial: GPT-4 accuracy increased from 73.44% to 79.97% on medical multiple-choice questions, while GPT-3.5 and Mixtral showed improvements from 60.69% to 71.57% and 61.42% to 69.48% respectively when augmented with external medical corpora.

3.1.4 Limitations and challenges

Despite its effectiveness, Naive RAG exhibits several critical limitations. Retrieval inaccuracies can result in irrelevant chunk selection, directly impacting response quality. The generation phase remains susceptible to hallucinations where models produce information not substantiated by retrieved content. Additionally, the static retrieval mechanism lacks sophistication for handling complex queries requiring multi-faceted information integration.

3.2 Advanced retrieval-augmented generation (RAG)

Advanced RAG introduces sophisticated techniques to enhance the retrieval and generation process beyond the basic Naive RAG framework. This paradigm addresses the fundamental limitations of Naive RAG by incorporating multiple optimization stages throughout the retrieval and generation pipeline.

3.2.1 Enhanced indexing strategies

Advanced RAG begins with improved indexing strategies that create more structured and semantically meaningful data representations. Fine-grained segmentation techniques replace the simple chunking approach of Naive RAG, utilizing semantic boundaries and content-aware splitting strategies. Sliding windows methodology enables overlapping chunks that maintain contextual continuity across segment boundaries, reducing the risk of losing critical information that spans multiple chunks.

Metadata inclusion represents a significant advancement, where chunks are enriched with contextual information such as document titles, sections, timestamps, authors, and topical categories. This metadata tagging—such as with dates, chapters, or document types—enables targeted and efficient retrieval based on both semantic similarity and contextual relevance. Advanced indexing strategies include adjusting chunk sizes dynamically based on content density and using sentence-based window retrieval, which allows for flexible data representation and facilitates more precise alignment with user needs.

3.2.2 Pre-retrieval phase optimizations

In the pre-retrieval phase, Advanced RAG applies comprehensive data and query optimizations to clarify user intent and increase retrieval precision. Query rewriting techniques transform user queries into more effective search formulations, while query expansion strategies add relevant terms and synonyms to broaden the search scope without losing specificity.

Enhanced data granularity through rigorous data cleaning processes ensures higher quality indexed content. This includes removing noise, standardizing formats, and enriching content with domain-specific terminology. The pre-retrieval phase also implements query analysis mechanisms that categorize queries by type, complexity, and domain to select appropriate retrieval strategies.

3.2.3 Post-retrieval refinement processes

Post-retrieval processes represent a critical innovation in Advanced RAG, refining retrieved content to ensure relevance and coherence before generation. Re-ranking mechanisms recalculate relevance scores using sophisticated algorithms that consider multiple factors including semantic similarity, contextual alignment, and domain-specific importance weights. This re-ranking process can utilize different similarity metrics, cross-encoder models, or even small specialized ranking models.

Context compression techniques remove unnecessary data while preserving critical information, addressing the challenge of fitting relevant content within LLM context windows. This involves intelligent summarization, redundancy removal, and priority-based content selection. Context fusion strategies combine information from multiple retrieved chunks in coherent ways, resolving conflicts and synthesizing complementary information.

By organizing data optimally before it reaches the language model, Advanced RAG enhances response quality, particularly for high-stakes domains like healthcare and law, where accuracy is paramount. Integration within established frameworks like LangChain and HayStack provides robust infrastructure for implementing these advanced techniques while maintaining system reliability.

3.2.4 Framework integration and hallucination mitigation

The combined optimizations in Advanced RAG effectively minimize hallucination risks, where a model may generate unsupported content. This is achieved through strict content grounding mechanisms that ensure generated responses maintain fidelity to retrieved information. Citation tracking and source attribution capabilities enable verification of generated content against source materials.

Advanced RAG implementations integrate seamlessly with modern framework ecosystems, providing APIs and interfaces that support complex workflow orchestration. These frameworks enable rapid prototyping and deployment of Advanced RAG systems while ensuring scalability and maintainability. The structured approach to retrieval precision and advanced refinement makes Advanced RAG particularly suitable for applications requiring high accuracy and reliability in critical-use scenarios.

3.2.5 Healthcare applications and performance

Advanced RAG implementations in healthcare demonstrate superior performance in high-stakes scenarios. Rau et al.'s context-aware radiology chatbot using LlamaIndex and GPT-3.5 provided more accurate imaging decision recommendations compared to generic chatbots and expert radiologists across 50 case files. Similarly, FraCChat achieved 57% and 83% accuracy in fracture classification using GPT-3.5-Turbo and GPT-4, respectively.

Knowledge graph integration has shown particular promise, with Soman et al. reporting a 71% improvement in LLaMA-2 performance when augmented with biomedical context from the SPOKE knowledge engine. MedRAG's diagnostic knowledge graph integration enabled identification of subtle diagnostic differences in conditions with similar manifestations.

3.2.6 Complexity-performance trade-offs

While Advanced RAG provides substantial performance improvements, it introduces increased implementation complexity and computational overhead. The multiple optimization stages require careful tuning and maintenance, with performance gains dependent on the sophistication of pre- and post-processing techniques employed (Fig. 2).

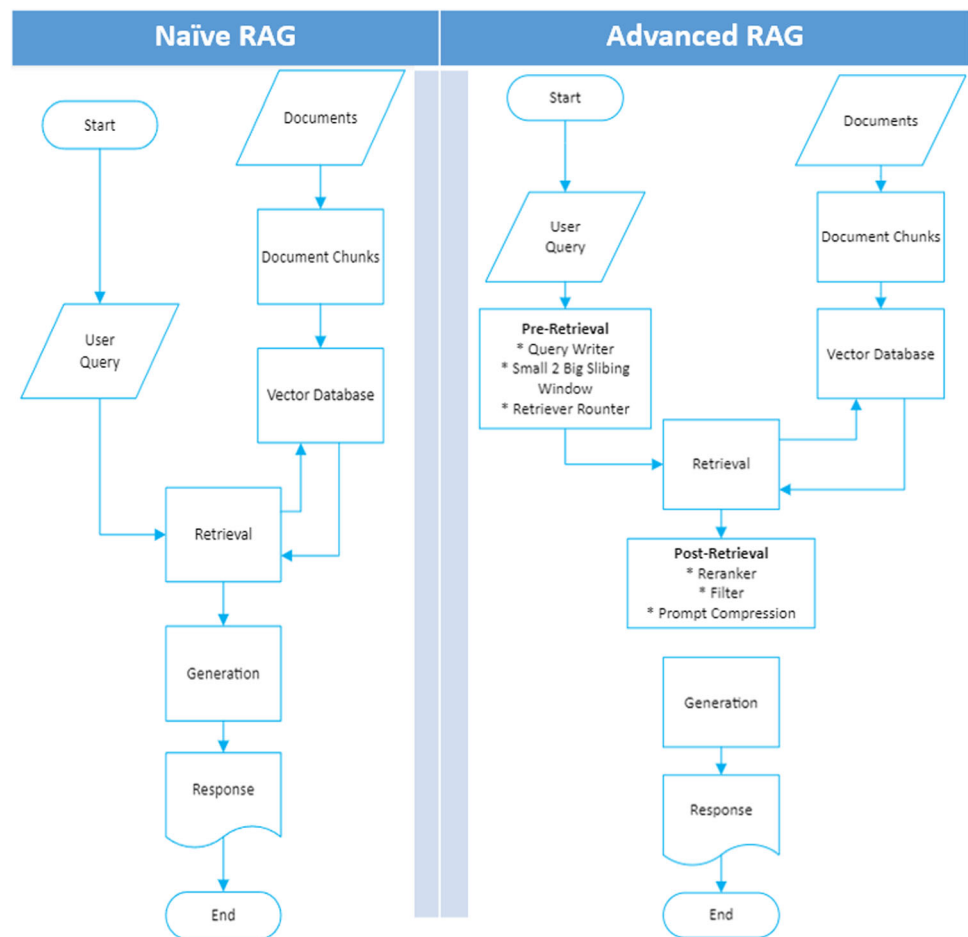
3.3 Modular retrieval-augmented generation (RAG)

Modular RAG represents a significant advancement over previous RAG paradigms, characterized by its adaptability and versatility. It introduces a modular architecture that allows for dynamic customization of its components, providing more flexibility than the fixed "Retrieve-Read" processes of Naive and Advanced RAG. Innovations in this model include restructured modules and rearranged pipelines, which enhance sequential processing and allow integrated end-to-end training. This modularity enables the system to be tailored to specific use cases, making it particularly well-suited for complex applications that require fine-tuned retrieval and generation processes.

3.3.1 Specialized module architecture

One of the key features of Modular RAG is the introduction of specialized modules that can be dynamically combined and configured based on task requirements. The Search module adapts to different scenarios by conducting direct searches across various data sources like databases, knowledge graphs, or external APIs. This module can implement multiple search strategies simultaneously, including semantic search, keyword matching,

Fig. 2 A Simple Comparison between naïve and advanced RAG workflow



and graph traversal algorithms, allowing the system to leverage the most appropriate search method for each query component.

The Memory module leverages the LLM’s memory capabilities, creating a vast memory pool that aligns more closely with data distributions over time. This module maintains both short-term conversational context and long-term knowledge persistence, enabling the system to build upon previous interactions and maintain consistency across extended dialogs. The memory component can store user preferences, domain-specific contexts, and learned patterns from previous retrieval sessions.

The Task Adapter module optimizes RAG for specific downstream tasks by automating prompt retrieval and creating task-specific retrievers. This module analyzes the nature of incoming queries and automatically configures the retrieval and generation pipeline to match task requirements. It can switch between different retrieval strategies, adjust ranking algorithms, and modify generation templates based on the identified task type, which improves the system’s performance across a wide range of contexts.

3.3.2 RAG flow and orchestration

Modular RAG offers remarkable advantages, particularly in terms of flexibility and scalability. Users can adaptively combine various modules and operators according to specific requirements of data sources and task scenarios. The architecture’s modularity allows for streamlined processes, enabling systems to effectively meet diverse user demands. The orchestration of modules and operators forms the RAG Flow, a flexible construct that can express current RAG methodologies and accommodate new research directions and applications.

Orchestration in Modular RAG systems plays a pivotal role in managing the entire retrieval and generation workflow. The incorporation of decision-making mechanisms allows for dynamic routing based on query characteristics, enabling the system to adaptively select appropriate processing pathways. This orchestration layer analyzes query complexity, domain requirements, and available resources to determine the optimal combination of modules and processing strategies. The flexibility provided by this orchestration is critical for improving the overall efficacy and user satisfaction of RAG systems, allowing them to handle a wide array of tasks while maintaining high-quality outputs.

3.3.3 Advanced indexing and query processing

The architecture introduces an innovative indexing process, which is essential for optimizing information retrieval. Challenges such as incomplete content representation, inaccurate chunk similarity search, and unclear reference trajectories necessitate advanced strategies for chunk optimization and structural organization. Techniques like hierarchical indexing create multi-level representations that capture both local semantic relationships and global document structure, enabling more sophisticated retrieval strategies that can navigate complex information hierarchies.

The use of knowledge graphs enhances the retrieval process by ensuring semantic accuracy and reducing errors associated with mismatched content. Knowledge graph integration enables relationship-aware retrieval, where the system can traverse entity relationships and infer relevant connections that pure vector similarity might miss. Multi-query and sub-query strategies enable the system to efficiently manage and decompose complex queries, breaking them into manageable components that can be processed by specialized modules before synthesis into comprehensive responses.

3.3.4 Novel operational patterns

In addition to new components, Modular RAG introduces novel patterns of operation that provide unprecedented flexibility in how retrieval and generation tasks are approached. The Rewrite-Retrieve-Read model refines queries using feedback loops, where initial retrieval results inform query modifications that lead to improved subsequent retrievals. This iterative refinement process continues until satisfactory results are achieved or predefined iteration limits are reached.

The Generate-Read approach relies on LLM-generated content to replace traditional retrieval in certain scenarios. This pattern generates hypothetical documents or answers that guide the retrieval process, particularly useful in zero-shot scenarios where traditional retrieval might fail due to query-document mismatch. Hybrid retrieval strategies combine keyword, semantic, and vector searches, allowing the system to handle diverse queries with greater precision by leveraging the strengths of different search methodologies.

The flexibility of these patterns makes Modular RAG more robust and adaptable, especially for handling knowledge-intensive tasks or dealing with complex, multi-turn interactions. As Modular RAG continues to develop, it stands poised to address emerging challenges in retrieval and generation, paving the way for future innovations in the field through its adaptable and extensible architecture.

3.3.5 Healthcare applications and performance

Modular RAG implementations in healthcare demonstrate exceptional versatility and performance improvements. Wang, Ma, and Chen's medical knowledge-augmented LLM achieved 11.4% to 13.2% improvement in open-medical QA tasks compared to GPT-4-Turbo without RAG. Jin et al.'s feature extraction framework combining RAG with XGBoost algorithms surpassed GPT-3.5, GPT-4, and fine-tuned LLaMA-2 in disease prediction tasks.

Multi-agent implementations show particular promise, with Lozano et al.'s Clinfo.ai system employing four specialized LLMs in a coordinated chain for literature search, relevance classification, summarization, and task-specific response generation.

3.3.6 Computational considerations

While offering unparalleled flexibility and performance, Modular RAG introduces significant computational overhead due to dynamic orchestration requirements and multiple specialized modules. This presents a critical trade-off between system capability and resource efficiency that must be carefully managed in production deployments (Fig. 3).

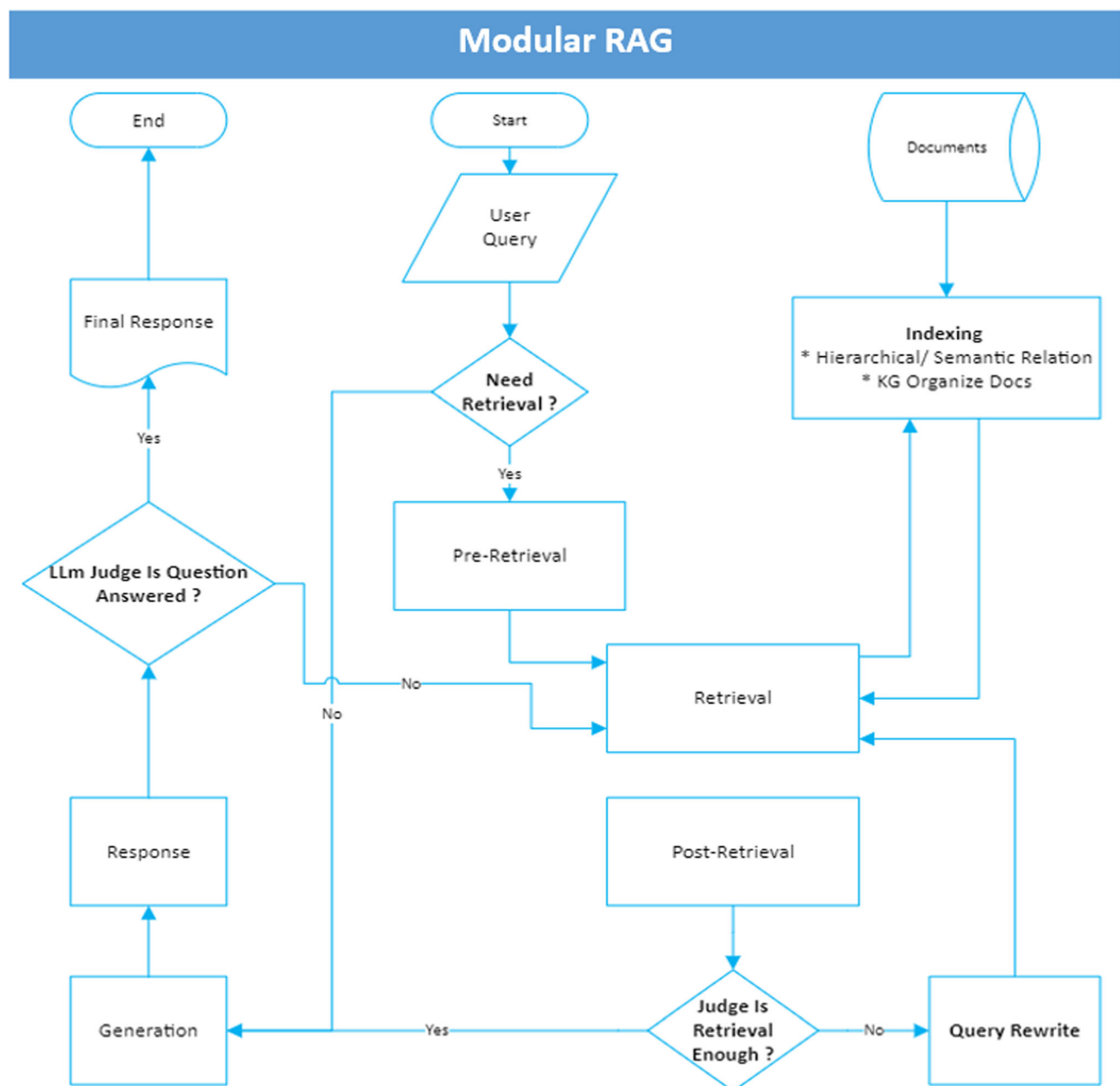


Fig. 3 A simple modular RAG workflow

4 Comprehensive analysis of RAG architectures

This section provides a detailed analysis of the three primary RAG architectures identified in the literature: Naive RAG, Advanced RAG, and Modular RAG. Each architecture represents an evolutionary step in addressing the limitations of RAG systems in medical applications.

4.1 Naive RAG

Naive RAG represents the foundational approach to retrieval-augmented generation, implementing a straightforward retrieve-then-generate paradigm.

4.1.1 Core characteristics

Naive RAG systems employ a simple pipeline consisting of three sequential steps: (1) query processing, where user queries are directly used for retrieval without modification; (2) document retrieval, utilizing basic similarity matching techniques such as cosine similarity or TF-IDF; and (3) response generation, where retrieved documents are concatenated with the original query and fed to the language model. This approach maintains simplicity in both implementation and computational requirements.

4.1.2 Advantages

- **Simplicity of Implementation:** The straightforward architecture requires minimal preprocessing and can be rapidly deployed with existing embedding models and vector databases.
- **Low Computational Overhead:** The single-pass retrieval mechanism minimizes computational costs, making it suitable for resource-constrained environments.
- **Interpretability:** The direct mapping between queries and retrieved documents provides clear traceability of information sources.
- **Baseline Performance:** Establishes a foundational performance benchmark that can be systematically improved upon.

4.1.3 Limitations

- **Limited Retrieval Precision:** Basic similarity matching often fails to capture complex semantic relationships, particularly in specialized medical contexts where terminology precision is critical.
- **Context Window Constraints:** Fixed retrieval limits may exclude relevant information or include irrelevant content, leading to suboptimal context utilization.
- **Inability to Handle Complex Queries:** Multi-hop reasoning and complex medical scenarios requiring synthesized information from multiple sources remain challenging.
- **Hallucination Susceptibility:** Without sophisticated filtering mechanisms, irrelevant retrieved content can increase the likelihood of generating inaccurate information.

4.2 Advanced RAG

Advanced RAG systems introduce sophisticated mechanisms to address the limitations of naive approaches through enhanced retrieval strategies and dynamic adaptation.

4.2.1 Core characteristics

Advanced RAG architectures implement several key innovations: (1) dynamic retrieval mechanisms that adaptively determine when and what to retrieve based on generation context; (2) multi-stage retrieval processes that employ coarse-to-fine filtering strategies; (3) confidence-based retrieval triggering, as demonstrated in FLARE [22], where low-confidence tokens initiate retrieval actions; and (4) self-reflective mechanisms, exemplified by SELF-RAG [23], that employ reflection tokens to assess retrieval necessity and output quality.

4.2.2 Representative systems

Several advanced RAG systems demonstrate significant improvements over naive approaches. FLARE achieves superior performance through forward-looking retrieval, attaining F1-scores of 59.7% on WikiMultihopQA and 46.2% EM score on ASQA-hint. SELF-RAG demonstrates robust improvements with accuracies of 54.9% on PopQA and 72.4% on PubQA for the 7B model. Med-R2 shows 14.87% average improvement over vanilla RAG across five medical datasets, with lightweight models achieving performance gains up to 79.72%.

4.2.3 Advantages

- **Adaptive Retrieval:** Dynamic retrieval mechanisms optimize information gathering based on contextual needs, improving relevance and reducing noise.
- **Enhanced Accuracy:** Self-reflective mechanisms and confidence-based retrieval significantly improve factual accuracy and reduce hallucinations.
- **Efficient Resource Utilization:** Selective retrieval based on necessity reduces computational overhead compared to exhaustive retrieval approaches.
- **Multi-hop Reasoning Capability:** Advanced systems can handle complex queries requiring information synthesis from multiple sources.
- **Quality Assessment Integration:** Built-in mechanisms for evaluating retrieval quality and output relevance enhance overall system reliability.

4.2.4 Limitations

- **Increased Complexity:** Multiple retrieval stages and reflection mechanisms significantly increase system complexity and potential failure points.
- **Computational Overhead:** Dynamic retrieval and self-reflection processes require additional computational resources, particularly for reflection token generation.
- **Dependency on Threshold Tuning:** Confidence-based retrieval mechanisms require careful threshold calibration, which may not generalize across different domains or datasets.
- **Limited Domain Adaptability:** Advanced mechanisms may require domain-specific fine-tuning to achieve optimal performance in specialized medical contexts.

4.3 Modular RAG

Modular RAG represents the most sophisticated architectural approach, implementing specialized components optimized for specific aspects of the RAG pipeline.

4.3.1 Core characteristics

Modular RAG systems decompose the RAG pipeline into specialized, interchangeable modules: (1) query processing modules that employ techniques such as query reformulation, classification, and expansion based on domain-specific taxonomies; (2) specialized retrieval modules incorporating hybrid approaches combining dense and sparse retrieval, knowledge graph integration, and multi-source retrieval strategies; (3) reranking and filtering modules that employ evidence-based scoring, credibility assessment, and relevance optimization; and (4) generation modules with specialized reasoning capabilities, including chain-of-thought prompting and evidence integration mechanisms.

4.3.2 Representative systems

Modular RAG systems demonstrate state-of-the-art performance across various medical applications. Med-GraphRAG achieves 8% improvement over standard RAG in medical QA and 10% in fact-checking tasks through graph-based retrieval. Omni-RAG demonstrates 8% average accuracy improvement, reaching 76.98% on MedQA through multi-source knowledge integration. MedRAG achieves superior diagnostic accuracy with 79.25% on CPDD and 88.65% on DDXPlus through knowledge graph-elicited reasoning. CLEAR achieves 90% F1-score on Stanford MOUD dataset while reducing inference time by 72% through clinical entity-augmented retrieval.

4.3.3 Advantages

- **Domain Specialization:** Modular components can be optimized for specific medical domains, terminology, and reasoning patterns, achieving superior domain-specific performance.
- **Flexible Architecture:** Interchangeable modules enable system customization for different medical applications and allow for iterative improvement of individual components.
- **Multi-source Integration:** Capability to seamlessly integrate diverse knowledge sources, including clinical guidelines, research papers, knowledge graphs, and electronic health records.
- **Enhanced Reasoning:** Specialized reasoning modules, such as knowledge graph traversal and evidence-based medical reasoning, enable sophisticated clinical decision support.
- **Scalability:** Modular design facilitates horizontal scaling and enables efficient resource allocation based on specific component requirements.
- **Quality Assurance:** Dedicated modules for fact-checking, safety assessment, and credibility evaluation enhance output reliability and clinical safety.

4.3.4 Limitations

- **Architectural Complexity:** The modular design significantly increases system complexity, requiring sophisticated orchestration and error handling mechanisms.
- **Integration Challenges:** Ensuring seamless communication and data flow between specialized modules requires careful interface design and potential format conversions.
- **Maintenance Overhead:** Multiple specialized components require individual maintenance, updates, and performance monitoring, increasing operational complexity.
- **Resource Requirements:** Specialized modules may require dedicated computational resources, potentially increasing overall system resource demands.
- **Development Complexity:** Building and optimizing specialized modules requires domain expertise and extensive testing across different medical scenarios.

4.4 Comparative analysis

See Table 1.

The evolution from Naive to Advanced to Modular RAG represents a trade-off between simplicity and performance. While Naive RAG provides a solid foundation with minimal complexity, Advanced RAG introduces intelligent mechanisms that significantly improve performance at moderate complexity cost. Modular RAG achieves state-of-the-art performance through specialized components but requires substantial development and maintenance investment. The choice of architecture should align with specific application requirements, available resources, and performance objectives in medical contexts.

4.5 Implications for medical applications

The architectural choice for medical RAG systems should consider several critical factors: (1) **Clinical Safety Requirements**: Modular RAG's comprehensive quality assurance mechanisms make it most suitable for high-stakes clinical applications; (2) **Resource Constraints**: Naive RAG may be appropriate for resource-limited healthcare settings where basic information retrieval suffices; (3) **Domain Complexity**: Advanced or Modular RAG architectures are essential for complex medical specialties requiring sophisticated reasoning; and (4) **Regulatory Compliance**: Modular RAG's transparency and traceability features align well with medical regulatory requirements for explainable AI systems.

4.5.1 Performance comparison

Table 2 provides a comprehensive comparison of the three RAG architectures across multiple dimensions:

4.5.2 Evolution trajectory and design principles

The progression from Naive to Advanced to Modular RAG represents a systematic approach to addressing fundamental challenges in retrieval-augmented generation:

1. **Retrieval Precision**: Evolution from basic similarity matching to sophisticated re-ranking and multi-strategy retrieval approaches.
2. **Context Utilization**: Advancement from simple concatenation to intelligent compression and task-adaptive synthesis.
3. **System Flexibility**: Progression from rigid pipelines to dynamic, customizable architectures.
4. **Domain Adaptation**: Enhancement from general-purpose retrieval to specialized, healthcare-optimized implementations.

Table 1 Comparative analysis of RAG architectures

Aspect	Naive RAG	Advanced RAG	Modular RAG
Implementation Complexity	Low	Medium	High
Computational Overhead	Low	Medium	High
Domain Adaptability	Limited	Moderate	Excellent
Performance Accuracy	Baseline	Enhanced	Superior
Maintenance Requirements	Minimal	Moderate	Extensive
Resource Efficiency	High	Medium	Variable
Scalability	Limited	Good	Excellent
Clinical Safety	Basic	Enhanced	Comprehensive

Table 2 Comprehensive comparison of RAG architectures

Aspect	Naive RAG	Advanced RAG	Modular RAG
Core Methodology	Basic “Retrieve-Read” pipeline	Enhanced pipeline with pre/post-processing	Dynamic modular orchestration
Indexing Strategy	Simple chunking and vectorization	Fine-grained segmentation, metadata integration	Hierarchical indexing, knowledge graphs
Retrieval Mechanism	Similarity-based top-K selection	Re-ranked, compressed context	Multi-strategy hybrid retrieval
Query Processing	Direct query encoding	Query rewriting and expansion	Dynamic routing and decomposition
Context Integration	Basic concatenation	Intelligent reordering and compression	Task-adaptive context synthesis
Performance Accuracy	Moderate improvement over baseline	Significant improvement with reduced hallucinations	Highest accuracy with task-specific optimization
Implementation Complexity	Low - straightforward implementation	Medium - requires optimization tuning	High - complex orchestration required
Computational Overhead	Low	Medium	High
Scalability	Limited by static architecture	Improved through optimizations	Excellent through modular design
Healthcare Applications	Basic QA, simple summarization	Diagnostic support, clinical decision aids	Complex multi-turn dialog, evidence-based medicine

5 Related work

This section is divided into six main categories: Text Datasets, Multi-Modal Datasets, Evaluation Matrix, LLM Models, Multi-Models, RAG Models and Comprehensive RAG analysis. Each of these subsections will discuss the previous research and advancements made in their respective areas. The Datasets subsection will cover the various datasets that have been employed in training and evaluating models, including their characteristics, data sources, and the preprocessing techniques used. The LLM Models subsection will focus on the development and application of Large Language Models, highlighting key works that have contributed to the progress in natural language understanding and generation. The CLIP Models subsection will explore previous work on Contrastive Language-Image Pretraining, which combines textual and visual data, addressing advancements in multimodal learning. Lastly, the RAG Models subsection will review the literature on RAG models, emphasizing how integrating external knowledge retrieval has enhanced generation tasks. These sections will collectively summarize the state of the art in each area, providing insights into how prior research has shaped current methodologies and approaches.

5.1 Large language dataset

This section provides an overview of the datasets utilized in this research, each selected for its unique contribution to the development and evaluation of models in medical natural language processing and understanding. These datasets encompass a wide range of medical topics, evaluation methodologies, and question-answering paradigms, reflecting the diverse challenges inherent in medical information retrieval and interpretation.

CAIS/MMLU (Massive Multitask Language Understanding) [24]: A challenging and comprehensive benchmark designed to evaluate large language models based on the knowledge they acquire during pretraining, focusing exclusively on zero-shot and few-shot settings. This evaluation methodology mirrors human-like assessment by requiring models to generalize their understanding to unseen tasks without further fine-tuning.

The benchmark covers 57 subjects, spanning a wide range of domains, including STEM, the humanities, the social sciences, and professional fields. MMLU’s content is divided into questions of varying difficulty, from elementary-level to advanced professional-level, enabling comprehensive evaluation of both general world

knowledge and specialized problem-solving skills. Traditional subjects, such as mathematics and history, are featured alongside specialized fields like law, ethics, and, notably, areas in medicine and the biological sciences.

The inclusion of medical domains such as Clinical Knowledge, College Medicine, Medical Genetics, Professional Medicine, and Anatomy makes MMLU particularly useful for assessing a model's capabilities in understanding and applying scientific and clinical concepts. This level of granularity and subject breadth ensures that the benchmark is well-suited for identifying potential knowledge gaps and blind spots in large language models, providing insights into how well these models can generalize across diverse, real-world tasks.

- **Clinical Knowledge:** focuses on real-world clinical decision-making with case-based questions on diagnosis, treatment, and patient management.
- **College Biology:** covers foundational undergraduate biology topics, including cell biology, genetics, and physiology.
- **College Medicine:** aligns with preclinical medical education, testing knowledge in anatomy, pathology, and pharmacology.
- **Medical Genetics:** explores the genetic basis of diseases, hereditary patterns, and modern genomic technologies like CRISPR.
- **Professional Medicine:** addresses advanced knowledge required by practicing physicians, including clinical guidelines and medical ethics.
- **Anatomy:** tests detailed knowledge of the human body's structure and clinical relevance.

The CAIS/MMLU dataset is particularly valuable for applications in medical education, clinical decision support, AI in healthcare research, and exam preparation for standardized medical licensing exams, providing a rigorous evaluation framework for multitask learning in medical domains.

MedMCQA (Medical Multiple-Choice Question Answering) [25]: A large-scale benchmark specifically designed to evaluate the performance of models in answering real-world medical entrance exam questions. The dataset comprises over 194,000 high-quality multiple-choice questions (MCQs) sourced from the AIIMS and NEET PG entrance exams, which are critical assessments for medical school admissions in India. These questions cover 2400 healthcare topics and span 21 medical subjects, making MedMCQA one of the most diverse and comprehensive medical QA datasets available.

Each question in the dataset includes not only the correct answer but also several distractor options, demanding models to exhibit deeper language comprehension and contextual reasoning. The questions test over 10 different reasoning abilities, such as logical reasoning, clinical judgment, and knowledge recall, challenging models to handle the complexities of medical language and decision-making. The dataset has an average token length of 12.77, indicating the nuanced and detailed nature of the questions.

MedMCQA's extensive coverage of medical subjects, ranging from anatomy and physiology to pharmacology and surgery, along with the diversity of topics, makes it an invaluable resource for training and evaluating models in medical question-answering tasks.

PubMedQA dataset [26]: A specialized benchmark designed for answering research questions in the biomedical domain, using abstracts from scientific publications. The task involves providing responses to research queries in the form of yes, no, or maybe, based on the information contained within corresponding abstracts from the PubMed database. An example question might be, Do preoperative statins reduce atrial fibrillation after coronary artery bypass grafting?

The dataset is composed of 1000 expert-labeled question-answer pairs, 61,200 unlabeled instances, and 211,300 artificially generated QA pairs. This variety in data allows for robust training and evaluation of models, with expert-labeled examples ensuring high-quality supervision and the large number of unlabeled and artificially generated instances offering extensive coverage for semi-supervised and unsupervised learning methods. PubMedQA focuses on evaluating a model's ability to comprehend biomedical research and make informed decisions based on the given abstracts, making it particularly valuable for the development of question-answering systems in the field of biomedical literature analysis and clinical decision support.

COVID-QA-DeepSet dataset [27]: A specialized Question Answering (QA) dataset developed to facilitate the extraction of accurate information from scientific literature related to COVID-19. It consists of 2019 question-answer pairs, which were carefully annotated by volunteer biomedical experts based on articles and studies about COVID-19. This dataset addresses the growing need for reliable, domain-specific QA systems capable of processing and extracting meaningful insights from the rapidly expanding body of COVID-19 research.

To evaluate the utility of the dataset, a comparison was performed between a RoBERTa base model fine-tuned on the SQuAD dataset and the same model fine-tuned on both SQuAD and COVID-QA. The results demonstrated significant performance improvements when the model was exposed to the additional COVID-19-specific data, highlighting the importance of domain-specific fine-tuning for biomedical QA tasks.

The COVID-QA dataset and the trained models have been open-sourced to support further research and development in natural language processing (NLP) for biomedical literature. The project aimed to enhance QA and search methods related to COVID-19, providing trustworthy and expert-annotated answers to complex questions about the pandemic.

Medical Meadow Collection [28]: The Medical Meadow dataset is a comprehensive collection of medical tasks compiled for fine-tuning and evaluating the performance of large language models (LLMs) in the medical domain. It is structured into two main categories: a collection of established medical NLP tasks reformatted into instruction-tuning formats, and a curated crawl of various internet resources. Each dataset within Medical Meadow addresses different facets of medical knowledge and practice, providing a diverse and robust framework for training and evaluation.

Key Datasets within Medical Meadow:

- **Flash Cards Used by Medical Students:** This dataset leverages the Anki Medical Curriculum flashcards, which are widely used by medical students to master essential topics such as anatomy, physiology, pathology, and pharmacology. The flashcards provide concise summaries and mnemonics for key concepts. Using GPT-3.5-Turbo, the flashcards were converted into structured question-answer pairs for model training. These pairs are brief and focused, capturing the essence of medical concepts essential for clinical practice and knowledge retention.
- **Stack Exchange Medical Sciences:** This dataset comprises 52,475 question-answer pairs from five Stack Exchange forums relevant to biomedical sciences: Academia, Bioinformatics, Biology, Fitness, and Health. The data focus on high-quality, upvoted responses that provide valuable insights into medical research methodologies, core biological concepts, health maintenance, and patient care. This dataset introduces a wide variety of health-related information, enriching the model's ability to navigate both theoretical and practical medical inquiries.
- **WikiDoc:** Medical Meadow includes question-answer pairs extracted from WikiDoc, a collaborative platform for medical professionals. The dataset covers medical specialties from the "Living Textbook" section, where headings were rephrased into questions, and the content served as answers. Additionally, the "Patient Information" section naturally provides questions in its structure, making it a rich resource for clinical and patient-facing scenarios.
- **Medical NLP Benchmarks:** To further enhance model evaluation, Medical Meadow integrates data from several established NLP benchmarks: CORD-19 (COVID-19 Open Research Dataset Challenge), containing over a million scholarly articles related to COVID-19.

The Medical Meadow dataset offers a rich, multi-dimensional training environment that spans theoretical knowledge, practical applications, patient care, and research methodologies.

MedQuAD (Medical Question Answering Dataset) [29]: The MedQuAD (Medical Question Answering Data) dataset is a comprehensive collection of 47,457 question-answer pairs derived from 12 authoritative websites associated with the National Institutes of Health (NIH), including sources, the Genetic and Rare Diseases Information Center (GARD), and MedlinePlus Health Topics. This dataset addresses a wide spectrum of medical information, categorized into 37 distinct question types, such as Treatment, Diagnosis, Side Effects, and more.

The question-answer pairs span various medical entities, including diseases, drugs, diagnostic tests, and treatments, making it a valuable resource for the development of healthcare-related question-answering systems and natural language processing applications.

MedQuAD's focus on trustworthy, expert-reviewed sources ensures high-quality information, making it particularly suitable for tasks in clinical decision support, patient education, and biomedical research. The diversity of question types and the range of medical conditions it covers enable the dataset to serve as a rich foundation for training models in areas such as medical information retrieval, automated medical dialog systems.

ChatDoctor dataset [30]: This dataset Developed to enhance the medical knowledge and accuracy of large language models (LLMs) in providing medical advice. This dataset comprises 100,000 patient-doctor dialogs sourced from a widely-used online medical consultation platform. The dialogs were carefully cleaned and anonymized to ensure privacy and confidentiality, offering a rich resource of real-world patient interactions.

The primary aim of this dataset is to address the limitations of existing LLMs, such as ChatGPT, by refining a specialized LLM, LLaMA [31], to improve its performance in the medical domain. This was achieved through fine-tuning the model using the ChatDoctor dataset, enabling the model to better understand patient needs, medical conditions, and provide accurate medical guidance.

One notable feature of the ChatDoctor model is the incorporation of a self-directed information retrieval mechanism. This allows the model to access real-time information from reliable online sources such as Wikipedia and curated offline medical databases, significantly enhancing the accuracy and relevance of its responses. This capability ensures that the model can retrieve up-to-date medical information, a critical component in providing accurate and context-aware advice.

The integration of real-world patient-doctor dialogs in the training process, combined with real-time information retrieval, allows ChatDoctor to deliver more precise, reliable, and contextually appropriate medical advice. This makes the model particularly valuable for applications where accuracy and reliability are paramount, such as in clinical decision-making and patient consultations.

MIMIC-III (Medical Information Mart for Intensive Care III) [32]: A database of de-identified electronic health records (EHR) from patients admitted to the intensive care units (ICUs) at the Beth Israel Deaconess Medical Center in Boston, Massachusetts, between 2001 and 2012. This dataset, developed through a collaboration between the Massachusetts Institute of Technology (MIT) and Beth Israel, includes a wealth of clinical data from 53,423 distinct adult hospital admissions across five specialized ICU departments: Coronary Care, Cardiac Surgery Recovery, Medical Intensive Care, Surgical Intensive Care, and Trauma Surgical Intensive Care.

MIMIC-III is unique due to its granularity and temporal depth, containing millions of time-stamped physiological measurements, laboratory results, and clinical notes. It integrates data from hospital EHR systems, automated critical care information systems (Philips CareVue and iMDsoft MetaVision), and the Social Security Administration Death Master File. The dataset includes a wide variety of data classes, such as: Patient demographics (age, gender, ethnicity), Vital signs (heart rate, blood pressure, respiratory rate), Laboratory test results, Prescription medications, Billing information, Unstructured clinical notes (doctor and nurse progress notes), Physiological measurements (ventilator settings, oxygen levels, etc.)

Due to its large size and diversity, MIMIC-III supports a wide range of clinical and research applications. The dataset has become a foundational resource for artificial intelligence (AI) and machine learning (ML) research, enabling the development of predictive models to improve ICU care. Researchers utilize MIMIC-III for projects such as predicting patient outcomes, analyzing disease progression, and assessing the effectiveness of different treatment strategies. The rich text data within MIMIC-III is particularly valuable for natural language processing (NLP) tasks, such as extracting insights from clinical notes.

MIMIC-III has facilitated advances in critical care, healthcare disparities, and automated clinical decision support systems, making it a crucial tool in the field of clinical informatics.

MeDAL (Medical Dataset for Abbreviation Disambiguation for Natural Language Understanding) [33]: An extensive medical text dataset specifically crafted for tackling abbreviation ambiguity in clinical contexts. Aimed at improving natural language understanding (NLU) within the healthcare domain, Addresses a crucial gap in

publicly available resources needed for training models in clinical NLP. Constructed from over 14 million PubMed abstracts, each containing an average of three abbreviations, Comprises more than 24 million abbreviation-expansion instances, with 24,005 abbreviation-expansion mappings and approximately 5886 unique abbreviations. Each abbreviation in the dataset averages four possible expansions, providing a rich and varied set of examples that reflect real-world clinical usage.

To create these training samples at scale, MeDAL employs a reverse substitution method, which replaces full medical terms with their corresponding abbreviations based on predefined mappings, ensuring that each abbreviation has multiple valid expansions to introduce ambiguity. Additionally, expansions are probabilistically substituted with abbreviations at a rate of 0.3, allowing models to encounter both full terms and abbreviations within various contexts, thus simulating real clinical documentation. This approach allows the dataset to be generated without manual annotation, making it both scalable and well-suited to training large models for disambiguation tasks. Mappings where abbreviations map to a single expansion are excluded to enhance the dataset's complexity and utility for disambiguation.

To manage computational demands while maintaining class balance, MeDAL employs an iterative sampling strategy to extract a balanced subset of 5 million samples, divided into 3 million training samples, and 1 million each for validation and testing. This strategy randomly samples classes with a frequency cap to control the distribution of high-frequency labels, ensuring sufficient representation of all classes, including those with low occurrence. The selected subset maintains critical diversity, supporting effective model training by preserving variability across abbreviations and expansions.

CLEVA [34]: Is a comprehensive evaluation platform designed specifically for assessing Chinese Large Language Models (LLMs). It addresses critical gaps in existing benchmarks by offering a standardized, multi-dimensional framework that evaluates models across 31 tasks grouped into ability evaluation (e.g., language understanding, reasoning, bias detection) and application assessment (e.g., translation, dialog generation). With 370 K test instances (9 M queries after augmentation) from 84 datasets, CLEVA is the largest Chinese benchmark to date. Notably, 33.98% of its data is newly collected to mitigate train-test contamination, a common issue in LLM evaluation. The platform also introduces Chinese-specific tasks like Pinyin transliteration and classical Chinese understanding, ensuring culturally and linguistically relevant assessments.

A key innovation of CLEVA is its standardized prompting methodology, which provides multiple prompt templates per task to reduce variability and enable robustness analysis. The platform supports diverse evaluation settings, including few-shot learning, chain-of-thought reasoning, and chatbot-style interactions. Additionally, CLEVA extends beyond conventional accuracy metrics by incorporating diversity (e.g., n-gram variability) and privacy (e.g., PII leakage detection), offering a holistic view of model performance. Its leaderboard is regularly updated with unique test subsets to maintain fairness and minimize data leakage risks.

DR.BENCH [35]: Is a comprehensive benchmark designed to evaluate clinical natural language processing (cNLP) models on diagnostic reasoning tasks. Introduced by Gao et al., it addresses the gap in existing cNLP benchmarks, which primarily focus on information extraction rather than complex clinical decision-making. The benchmark comprises six tasks—including medical natural language inference (MedNLI), electronic medical record question answering (EmrQA), and problem summarization—that assess a model's ability to understand clinical text, reason with medical knowledge, and generate diagnostic conclusions. These tasks are derived from ten publicly available datasets, such as MIMIC-III and MedQA, ensuring diverse and realistic clinical scenarios.

The benchmark leverages generative models, particularly variants of Google's T5 (Text-to-Text Transfer Transformer), fine-tuned with domain adaptation on medical corpora like PubMed, UMLS, and EHR progress notes. Experiments demonstrate that while models perform well on structured tasks like MedNLI (84.88% accuracy), they struggle with abstractive reasoning, such as summarizing patient problems (ROUGE-L scores as low as 2.14%). The study highlights the challenges in clinical NLP, including the need for better knowledge integration, multi-hop reasoning, and efficient model architectures to reduce computational costs.

Comparative Analysis of Medical NLP Datasets for LLM Training Table 3 presents a systematic comparison of datasets used for training medical large language models, highlighting their unique characteristics and

Table 3 Comparative analysis of medical NLP datasets for LLM training

Dataset	Modality	Clinical focus	Scale	Access
MIMIC-III	EHR	Intensive care	58 K admissions	Need Submission
UMLS	Knowledge Graph	Terminology	2 M entities	Open
MedDialog	Conversations	General medicine	3.66M dialogs	Open
CovidDialog	Consultations	Infectious disease	603 cases	Open
Medical Flashcards	Q&A pairs	Medical education	33 K pairs	Open
Wikidoc	Articles	Clinical reference	67 K articles	Open
CORD-19	Research papers	COVID-19	1.06M papers	Open
ChatDoctor	Doctor-patient	Clinical practice	215 K dialogs	Open
MeDAL	Abstracts	Multilingual	14.4M abstracts	Open
CLEVA	Clinical notes	Multi-domain	1.2M notes	Open
DR.BENCH	Benchmark tasks	Diagnostic reasoning	6 tasks (10 datasets)	Open

Note: Modality indicates primary data type; Clinical Focus shows medical specialty; Scale represents dataset size; Access denotes availability (Open/Restricted)

applications in RAG systems. The analysis focuses on four key dimensions: data modality, clinical relevance, scale, and accessibility.

5.2 Multi modal dataset

In this section, a comprehensive overview of the datasets utilized in our study, each contributing unique attributes and capabilities to enhance the development of machine learning models in the medical domain. The datasets range from multimodal resources that integrate visual and textual data to those focused specifically on visual question answering (VQA) tasks.

Roco V2 Dataset [36]: An enhanced version of the original Roco dataset, and adds 35,705 new images added to PMC since 2018. It further provides manually curated concepts for imaging modalities with additional anatomical and directional concepts for X-rays. The dataset consists of 79,789 images, Roco V2 supports tasks such as image-captioning, VQA, and multimodal understanding in healthcare, incorporating a wider array of medical fields and conditions. The expanded dataset is tailored for training advanced machine learning and deep learning models, pushing the boundaries of visual-textual integration in the medical domain.

Roco V1 Dataset [37]: Is a large-scale medical image dataset designed for multimodal tasks such as visual question answering (VQA) and image captioning in the medical domain. It contains over 81,000 images paired with captions from various medical modalities including X-rays, MRIs, and CT scans. These captions describe anatomical structures, medical conditions, and diagnostic procedures, offering a wide range of contexts for visual and textual understanding in the healthcare domain. The Roco dataset enables the development of deep learning models capable of analyzing medical images in combination with textual data to generate answers or captions, improving the interpretability of medical visual information, including Computer Tomography, Ultrasound, X-Ray, Fluoroscopy, Positron Emission Tomography, Mammography, Magnetic Resonance Imaging, Angiography. All images in ROCO have corresponding caption, keywords, Unified Medical Language Systems Concept Unique Identifiers and Semantic Type. An out-of-class set with 6k images ranging from synthetic radiology figures to digital arts is provided, to improve prediction and classification performance.

MedVidQA Dataset [38]: A visual question answering (VQA) resource containing open-ended natural language questions related to images. It includes corresponding answers to these questions and can be used to train machine learning and deep learning models for VQA tasks. The latest VQA dataset, from April 2017, features 204,721 images, 1,105,904 questions, and 11,059,040 ground truth answers that are not specific to any domain. MedVidQA inspired the creation of various VQA datasets in the medical domain and is useful for developing multimodal medical question answering models. Medical VQA datasets are typically smaller compared to general

VQA datasets like DocVQA, MUST-VQA, and VQA. The Medical Video Question Answering dataset consists of 3010 manually curated health-related questions and timestamps linked to visual answers from trusted video sources, such as medical schools, health institutes, and medical practitioners. MedVidQA was developed as part of the medical video question answering challenge for the BioNLP workshop at ACL 2022.

VQA-Med-2021 Dataset [39]: This dataset is an extension of VQA-Med-2020 and consists of 4500 radiology images with 4500 question-answer pairs for VQA data. Additionally, it includes 85 radiology images with 200 questions based on visual content for VQG (Visual Question Generation) data. The 2021 edition focuses on answering questions related to abnormalities in radiology images.

SLAKE Dataset [40]: SLAKE (Semantically-Labeled Knowledge-Enhanced Dataset for Medical Visual Question Answering) is a bilingual dataset containing comprehensive semantic labels annotated by expert physicians. It features 642 images and 14,028 question-answer pairs in both English and Chinese. Experienced doctors created the questions using a template-based annotation system.

VQA-Med-2020 Dataset [41]: The third edition of VQA-Med contains 4000 radiology images and 4000 question-answer pairs as training data. This version also introduced visual question generation (VQG), aimed at generating natural language questions related to the content of radiology images.

PathVQA Dataset [42]: PathVQA includes 4998 pathology images and 32,799 open-ended questions manually verified for accuracy. The dataset is intended to achieve a level of competency equivalent to that of a board-certified pathologist, aiming to pass the American Board of Pathology (ABP) certification examination. The images and captions were extracted from digital pathology textbooks and online libraries using tools like PyPDF2, PDFMiner, and the Stanford CoreNLP toolkit.

RadVisDial Dataset [43]: This dataset focuses on visual dialog in radiology, particularly for chest X-ray images. It is derived from the MIMIC-CXR database, containing 91,060 images after discarding those without condition information. It includes two subsets: a silver-standard dataset with synthetically created dialogs and a gold-standard dataset with natural dialogs between two expert radiologists about a specific chest X-ray.

VQA-Med-2019 Dataset [44]: This dataset features 4200 radiology images and 15,292 question-answer pairs. It is a larger and more enhanced version of VQA-Med 2018, drawing inspiration from the VQA-RAD dataset. It includes CT/MRI images from the MedPix database and targets the most frequent question categories, with filters based on modality, locality, diagnosis methods, and more.

VQA-RAD Dataset [45]: VQA-RAD consists of 315 radiological images designed for VQA tasks in the medical domain, with a focus on radiology. The dataset contains radiology images from the MedPix database and was manually created and validated by clinicians. Its goal is to provide a more specific and focused dataset for VQA tasks in radiology, addressing unique features and challenges in medical images.

VQA-Med-2018 Dataset [44]: This dataset is designed for visual question answering in the medical domain, pairing medical images such as X-rays and CT scans with questions about the images. It aims to enable machine learning models to generate answers based on the image content. The dataset contains 6,413 question-answer pairs and 2866 medical images extracted from PubMed Central articles, covering a wide range of medical conditions and modalities.

Visual Genome Dataset [46]: The Visual Genome dataset bridges language and vision by providing rich annotations for images created through crowdsourcing. These annotations include objects, attributes, relationships, and scene graphs, offering a deeper understanding of visual content. Visual Genome supports various computer vision tasks such as object recognition, image captioning, and visual question answering.

MITOS WSI CMC Dataset [47]: Addresses the gaps in existing breast cancer datasets by providing extensive mitotic figure annotations. It involves a multi-expert review with a majority vote for resolving annotation disagreements. To enhance coverage, a two-stage machine learning pipeline was used, combining a customized RetinaNet for object detection with a secondary classifier for cell verification. This approach increased the dataset's coverage by 6.06% and improved annotation consistency. The use of clustering and representation learning (using ResNet-18 and UMAP) further refined label quality, achieving a high mean F1 score of 0.791.

Additionally, the dataset showed good cross-species transferability, achieving an F1 score of 0.696 on the human TUPAC16 dataset, making it valuable for both veterinary and human cancer research.

Chest ImaGenome Dataset [48] Represents a significant advancement in the automatic interpretation of chest X-ray (CXR) images through the integration of deep learning and natural language processing (NLP) techniques. Constructed from the MIMIC-CXR dataset, this dataset uniquely employs a scene graph data structure to annotate and describe CXR images, addressing a critical gap in localized labeling for various findings. It includes 242,072 frontal CXRs, with detailed annotations that establish anatomical relationships and comparison relations between sequential exams, enhancing explainability and supporting clinical reasoning. By offering 1256 combinations of relation annotations among 29 anatomical locations and their attributes, along with over 670,000 localized comparison relations, the Chest ImaGenome Dataset enables a more nuanced understanding of imaging findings and fosters the development of algorithms that better emulate radiologists' reasoning processes. This innovative approach not only enhances the interpretability of AI models in radiology but also paves the way for future research in multimodal clinical applications, ultimately aiming to improve patient outcomes through more informed decision-making.

MedTrinity-25 M [49]: Is the largest multimodal medical dataset to date, comprising 25 million image-ROI-description triplets spanning 10 imaging modalities (including X-ray, CT, and MRI) and covering over 65 diseases. Unlike previous datasets that rely on manually paired image-text data, it introduces an automated pipeline combining domain-specific expert models and retrieval-augmented generation to produce multigranular annotations from unpaired images. Each triplet includes regions of interest (ROIs) with bounding boxes/segmentation masks and structured descriptions detailing disease types, regional abnormalities, and anatomical relationships.

The dataset addresses key limitations of existing resources through its scale and annotation quality. While MIMIC-CXR offers 227 K radiology reports and PMC-OA provides 1.65M image-caption pairs, MedTrinity-25 M delivers 10–100× more samples with finer-grained ROI-level descriptions. Its construction pipeline first identifies abnormalities using expert models (e.g., SAT for CT, CheXmask for X-rays), then enriches descriptions with medical knowledge from PubMed/StatPearls, achieving 89% alignment with human reports as evaluated by GPT-4V.

MedTrinity-25 M has demonstrated significant utility in pretraining medical vision-language models. When used for pretraining, it enables models to achieve state-of-the-art performance on VQA-RAD (86.0%) and PathVQA (99.0%), outperforming general-domain MLLMs. The dataset's combination of scale, automated annotation, and clinical detail makes it particularly valuable for developing multimodal medical AI systems, supporting tasks ranging from lesion segmentation to detailed report generation.

Comparative Analysis of Medical Datasets for Visual Question Answering and Multimodal Tasks Table 4 summarizes publicly available medical datasets for visual question answering (VQA) and multimodal tasks, including their scope, size, and accessibility. These datasets are critical for training and evaluating (LLM) and (RAG) systems in the medical domain.

5.3 Evaluation metrics

To assess the performance of RAG system and LLM, it is necessary to understand several standard evaluation metrics from information retrieval and natural language processing.

5.3.1 Quantitative metrics

Classification Metrics For tasks involving classification or binary judgments (e.g., correct/incorrect answers), use:

- **Accuracy:** The proportion of correct predictions among all predictions made.

Table 4 Overview of medical datasets for visual question answering and multimodal tasks

Data Name	Type	Size	Link
MedTrinity-25 M	Medical Text & Images	25 M text-image pairs	github
Roco V2 Dataset	Medical Images & Text Captions	79,789 images pairs	zenodo
Roco V1 Dataset	Medical Images & Text Captions	81,000+ images	github
MedVidQA Dataset	Visual Question Answering (VQA)	204,721 images, 1,105,904 questions, 11,059,040 answers	visualqa
VQA-Med-2021	Radiology Images & VQA	4500 images, 4500 Q-A pairs	github
SLAKE Dataset	Bilingual VQA (Medical Images)	642 images, 14,028 Q-A pairs	med-vqa
VQA-Med-2020	Radiology Images & VQA	4000 images, 4000 Q-A pairs	github
PathVQA Dataset	Pathology Images & VQA	4998 images, 32,799 questions	huggingface
RadVisDial Dataset	Chest X-ray & Visual Dialog	91,060 images	github
VQA-Med-2019	Radiology Images & VQA	4200 images, 15,292 Q-A pairs	github
VQA-RAD Dataset	Radiology Images & VQA	3515 Q-A pairs on 315 images	huggingface
VQA-Med-2018	Medical Images & VQA	2866 images, 6413 Q-A pairs	imageclef
Visual Genome	Image Annotations (General)	108 K images, 5.4M descriptions	huggingface
MITOS WSI CMC	Medical Image & Text Caption	13,907 annotated mitotic figures	github
Chest ImaGenome	Chest Medical Image	242,072 frontal CXRs	physionet

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

- **Precision:** The proportion of relevant instances among the retrieved instances.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

- **Recall (Sensitivity):** The proportion of relevant instances that were successfully retrieved.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

- **Specificity:** The proportion of true negatives correctly identified.

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (4)$$

- **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure.

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

Text Generation Metrics For evaluating the quality of generated text, employ:

- **BLEU (Bilingual Evaluation Understudy):** Measures n-gram precision between generated text and reference texts, with a brevity penalty for short outputs [50].

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right) \quad (6)$$

where BP is the brevity penalty, w_n are weights (typically $1/N$), and p_n is the modified n-gram precision.

- **ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation - Longest Common Subsequence):** Measures the longest common subsequence (LCS) between generated and reference texts [51].

$$R_{\text{LCS}} = \frac{\text{LCS}(X, Y)}{m}, \quad P_{\text{LCS}} = \frac{\text{LCS}(X, Y)}{n}, \quad F_{\text{LCS}} = \frac{(1 + \beta^2)R_{\text{LCS}}P_{\text{LCS}}}{R_{\text{LCS}} + \beta^2 P_{\text{LCS}}} \quad (7)$$

where X is the reference (length m), Y is the generated text (length n), and β controls recall/precision balance.

- **METEOR (Metric for Evaluation of Translation with Explicit ORdering):** Aligns generated text to reference text using exact, stem, synonym, and paraphrase matches [52].

$$\text{METEOR} = (1 - \text{Penalty}) \times \frac{F_{\text{mean}}}{\alpha \times P + (1 - \alpha) \times R} \quad (8)$$

where F_{mean} is the harmonic mean of precision and recall, Penalty accounts for fragmentation, and α controls precision/recall balance.

RAG-Specific Metrics (Ragas) For evaluating RAG systems *without relying on ground-truth annotations*, adopt the **Ragas** framework [53], which measures three critical dimensions:

- **Faithfulness:** Assesses whether the generated answer is factually grounded in the retrieved context. Given a question q , context $c(q)$, and answer $a_s(q)$, Ragas decomposes $a_s(q)$ into atomic statements S and verifies their support in $c(q)$:

$$F = \frac{|V|}{|S|}$$

where $|V|$ is the number of statements verified by the LLM as context-supported, and $|S|$ is the total statements extracted.

- **Answer Relevance:** Measures how directly the answer addresses the question, penalizing incompleteness or redundancy. For answer $a_s(q)$, Ragas generates n hypothetical questions $\{q_i\}_{i=1}^n$ from $a_s(q)$ and computes:

$$\text{AR} = \frac{1}{n} \sum_{i=1}^n \text{sim}(q, q_i)$$

where $\text{sim}(\cdot)$ is the cosine similarity between embeddings of q and q_i .

- **Context Relevance:** Evaluates the focus of the retrieved context $c(q)$ by quantifying the fraction of sentences critical to answering q :

$$\text{CR} = \frac{\# \text{ relevant sentences in } c(q)}{\# \text{ total sentences in } c(q)}$$

Sentences are extracted via LLM prompting, with low CR indicating noisy retrieval.

5.3.2 Qualitative and manual evaluations

Beyond quantitative metrics, employ human evaluations to assess aspects that are difficult to quantify automatically:

- **Manual Accuracy:** Where human evaluators judge the correctness of system outputs against ground truth. This is particularly valuable for complex outputs where automated metrics may be insufficient.
- **S.C.O.R.E. Framework:** [54] Evaluate:
 - **Safety:** Does the output contain harmful, biased, or dangerous content?
 - **Consensus:** Does the output align with established knowledge from multiple sources?
 - **Objectivity:** Is the output free from subjective bias or unsupported claims?
 - **Reproducibility:** Could another researcher verify the output's validity?
 - **Explainability:** Is the output's reasoning clear and traceable?
 Each dimension is rated on a Likert scale (1–5) by domain experts, with inter-rater reliability measured using Cohen's κ .

5.4 Large language model

Recent advancements in medical large language models (LLMs), like Med-PaLM 2 [3], Med-Gemini [55], and MedAlpaca [28], have significantly contributed to healthcare by integrating specialized architectures, design goals, and functionalities. These models are driving innovation in Med-LLMs, enhancing their practical application in healthcare, and laying a foundation for future research and medical practice.

HuatuoGPT [56]: A Chinese large language model (LLM) specifically designed for medical consultations, harnessing advanced techniques in reinforcement learning to enhance its performance. By utilizing a reinforcement learning framework integrated with AI feedback, HuatuoGPT continuously refines its capabilities, aligning its responses with both distilled training data and real-world medical knowledge through a sophisticated reward model.

To evaluate the model effectiveness, the model was rigorously benchmarked against three established question-answer datasets: cMedQA2, webMedQA, and the extensive Huatuo-26 M dataset, which comprises 26 million question-answer pairs—the largest medical QA dataset to date. This benchmark allows for a comprehensive assessment of both retrieval and generation tasks, providing insights into the model's performance in practical medical settings.

The model's outputs are evaluated using various metrics, including BLEU (Bilingual Evaluation Understudy), GLEU (Google Language Understanding Evaluation), ROUGE (Recall-Oriented Understudy for Gisting Evaluation), and DISTINCT. These metrics enable a multifaceted analysis of the model's response quality, fluency, and diversity. Additionally, the quality of generated responses is further validated using GPT-4, serving as an external standard for assessing contextual and factual accuracy.

Experimental results indicate that many existing models underperform relative to expectations when benchmarked against the Huatuo-26 M dataset, which remains challenging even in the era of pre-trained language models. The research highlights the dataset's potential across several dimensions, including:

- **Zero-Shot Learning:** Models trained on Huatuo-26 M demonstrate significant capabilities in answering questions from other QA datasets without prior exposure, showcasing the dataset's versatility and applicability.
- **Retrieval-Augmented Generation (RAG):** Huatuo-26 M serves as a critical external knowledge resource, enhancing retrieval-based systems by improving their ability to provide relevant and accurate responses to medical inquiries.

- **Pre-Training Enhancement:** Incorporating the QA pairs from Huatuo-26 M into the training corpus of existing pre-trained language models improves their performance and contextual understanding, highlighting the dataset's utility in the continued development of LLMs.

ClinicalT5 [57]: Developed by Lu et al. (2022), which is a T5-based generative language model specifically designed for clinical text data. This model addresses the unique characteristics of clinical language, which often includes complex medical terminology, abbreviations, and specialized writing styles.

The foundation of the model is the T5 (Text-to-Text Transfer Transformer) architecture, introduced by Raffel et al. T5 is built on a large-scale pre-trained model designed to approach a wide array of NLP tasks as text-to-text transformations. It employs the original Transformer encoder-decoder architecture as proposed by Vaswani et al., consisting of a stack of blocks where each block contains a self-attention layer and a feed-forward network. In the T5-Base model, both the encoder and decoder comprise 12 blocks. Each feed-forward network has a dense layer outputting 3072 dimensions, followed by a ReLU activation and another dense layer. The attention mechanism employs multiple heads (12 per layer), with the key and value matrices maintaining an inner dimensionality of 64. Overall, T5-Base encompasses approximately 220 million parameters, distributed similarly across its encoder and decoder components, comparable to the BERT-Base model.

The model capitalizes on this robust architecture by being pre-trained on a comprehensive corpus of clinical text data sourced from the MIMIC-III database. This extensive training enables the model to effectively understand and generate clinical language, achieving a high degree of proficiency in various clinical NLP tasks.

ClinicalT5 exemplifies the potential of domain-specific language models in clinical NLP, offering significant advancements in understanding and generating clinical text.

ClinicalGPT [58]: Designed specifically for clinical tasks, addressing the unique challenges faced in the medical domain by harnessing extensive medical datasets and incorporating advanced training techniques. The model is based on the BLOOM-7B architecture, which supports multilingual processing and is pre-trained on diverse medical texts, electronic health records, and curated datasets such as cMedQA2, cMedQA-KG, MD-EHR, MEDQA-MCMLE, and MedDialog.

The training phase follows a multi-stage approach, beginning with supervised fine-tuning (SFT) on medical question-answer pairs and dialog datasets to optimize the model for generating contextually appropriate responses to medical queries. It further employs instruction tuning, utilizing manually designed templates to transform structured medical knowledge into usable question-answer pairs, thereby enhancing its ability to follow specific instructions. The model also incorporates reinforcement learning from human feedback (RLHF), employing a reward model trained with human feedback to optimize response quality. This phase includes Proximal Policy Optimization (PPO) to ensure that responses are accurate, safe, and aligned with human preferences.

To evaluate the model performance, multiple dimensions are assessed, including medical knowledge question-answering, multi-turn medical conversations, and clinical record analysis. It demonstrates its capability to synthesize information and generate actionable insights based on historical data by converting patient records into structured prompts. Preliminary results indicate that the model outperforms existing LLMs in various medical applications, achieving enhanced accuracy and relevance in its outputs through a combination of supervised fine-tuning, instruction tuning, and reinforcement learning. Its parameter-efficient training approach, utilizing techniques like LoRA and ZeRO-2, ensures computational feasibility while delivering state-of-the-art performance.

ChiMed-GPT [59]: A benchmark large language model (LLM) developed specifically for the Chinese medical domain, addressing the growing demand for advanced medical services and highlighting discrepancies in healthcare infrastructure. This model adopts a comprehensive training approach that includes pre-training, supervised fine-tuning (SFT), and reinforcement learning from human feedback (RLHF). While traditional SFT enhances an LLM's ability to comprehend and respond to medical instructions, it often falls short in imparting essential domain knowledge and aligning the model's responses with human values and preferences.

To overcome these limitations, ChiMed-GPT undergoes extensive pre-training on a substantial corpus of medical text, followed by SFT tailored to specific medical tasks. The integration of RLHF further aligns the model's behavior with human expectations, ensuring its outputs are both clinically relevant and ethically sound.

The model's performance is evaluated on various tasks, including information extraction, question answering, and dialog generation, demonstrating significant improvements over general-purpose LLMs.

ChiMed-GPT's architecture is publicly available. This work not only advances the capabilities of LLMs in the Chinese medical context but also highlights the importance of creating domain-specific models to meet the unique needs of healthcare systems.

BioGPT [60]: A generative Transformer language model tailored specifically for biomedical text generation and mining. In contrast to BERT-based models such as BioBERT and PubMedBERT, which excel primarily in understanding tasks but lack generative capabilities, the model is optimized for generating coherent and contextually relevant biomedical text. Built on the GPT-2 architecture and trained with the causal language modeling (CLM) objective, BioGPT distinguishes itself by providing advanced generation capabilities that are particularly beneficial in the biomedical domain.

BioGPT is based on the GPT-2 medium configuration, which includes 24 layers, 1024 hidden units, and 16 attention heads, resulting in a total of 347 million parameters. This architecture was specifically chosen for its demonstrated efficacy in generating coherent text in various domains. To adapt the model for biomedical applications, modifications were made to the embedding and output projection layers to accommodate the domain-specific vocabulary, leading to a slightly reduced parameter count compared to the original GPT-2 medium model.

The study also investigated the effectiveness of soft prompts—continuous embeddings learned during the fine-tuning process—compared to hard prompts, which are discrete language phrases manually designed to guide the model toward specific tasks. Results indicated that soft prompts generally outperformed hard prompts, with the best-performing soft prompt achieving an F1 score of 38.60% on the KD-DTI dataset, compared to 38.16% for the best hard prompt. The success of soft prompts is attributed to their flexibility and ability to capture complex patterns in data without requiring expert knowledge for design.

BioGPT's performance across various biomedical NLP tasks is notable, achieving an F1 score of 89.2% for NER on the BC5CDR dataset, an accuracy of 86.5% for QA on the BioASQ dataset, and an F1 score of 92.1% for relation extraction on the ChemProt dataset.

PubMedBERT [61]: A specialized large language model designed for natural language processing (NLP) tasks in the biomedical domain. Unlike general-purpose models, which are first pretrained on broad-domain corpora, PubMedBERT is pretrained from scratch using a massive collection of biomedical literature. This unique pretraining strategy enables model to achieve superior performance on a variety of biomedical NLP tasks compared to models that undergo continual pretraining on general-domain data.

PubMedBERT is based on the BERT (Bidirectional Encoder Representations from Transformers) [62] architecture, which has revolutionized NLP by enabling models to learn deep bidirectional contextual embeddings. However, unlike BERT, which is initially pretrained on general-domain corpora such as books and Wikipedia, the model is pretrained exclusively on biomedical text.

The pretraining corpus consists of 14 million PubMed abstracts, representing approximately 3.2 billion words. By focusing solely on biomedical text, the model develops a vocabulary and set of embeddings that are finely tuned to the terminology, syntax, and semantic structures typical of biomedical literature. This domain-specific focus allows the model to achieve better representation and understanding of biomedical language, as well as improved generalization across various downstream biomedical tasks.

After pretraining, the model is fine-tuned on multiple biomedical NLP tasks, including Named Entity Recognition (NER), relation extraction, biomedical question answering, and text classification. PubMedBERT consistently achieves state-of-the-art results on these tasks and holds top scores on benchmarks such as the Biomedical Language Understanding and Reasoning Benchmark (BLURB).

GatorTron [63]: A large language model designed specifically for clinical natural language processing (NLP) tasks. Developed from scratch, the model is trained on a massive corpus of over 90 billion words, including more than 82 billion words of de-identified clinical text. This extensive dataset allows model to learn clinical language representations that are highly relevant and precise for healthcare applications.

Builds on the architecture of transformer-based models like BERT and GPT, but it is designed to scale up both in terms of model parameters and the size of the training data. The GatorTron family includes models that range in size from 110 million to 8.9 billion parameters. This scaling is critical, as it allows the model to capture increasingly complex patterns and relationships in clinical text, leading to significant improvements in task performance across multiple clinical NLP applications.

The model is pretrained on a combination of general-domain and clinical-domain text, with the latter comprising de-identified clinical notes and reports. By training on over 82 billion words of clinical text, GatorTron develops a deep understanding of medical terminology, clinical jargon, and the nuanced semantics of healthcare documentation. This extensive pretraining on a domain-specific corpus enables the model to outperform general-domain language models, particularly on tasks requiring specialized medical knowledge.

GatorTron is systematically evaluated on five core clinical NLP tasks, demonstrating substantial performance improvements across the board:

- **Clinical Concept Extraction:** Accurately identifies key clinical entities, such as diseases, symptoms, and medications, from unstructured text, providing crucial support for medical record analysis and clinical decision-making.
- **Medical Relation Extraction:** By understanding relationships between clinical entities, such as drug-disease interactions or symptom-disease associations, Enhances the automatic extraction of medical insights from clinical narratives.
- **Semantic Textual Similarity (STS):** Excels at identifying the semantic similarity between pairs of clinical texts, which is vital for tasks like summarization, deduplication of medical records, and clinical trial matching.
- **Natural Language Inference (NLI):** Significantly improves the accuracy of inferring logical relationships between clinical statements, achieving a 9.6% improvement in accuracy over previous models, which is essential for diagnostic reasoning and clinical question answering.
- **Medical Question Answering (MQA):** Demonstrates a 9.5% improvement in accuracy for answering medical questions, making it a powerful tool for assisting clinicians and patients in retrieving relevant medical information.

Med-PaLM 2 [3]: A large language model specifically designed for the medical domain, representing a key advancement in AI-driven medical question answering. Originally built upon Google's PaLM architecture, Med-PaLM was the first model to surpass the "passing" threshold for the US Medical Licensing Examination (USMLE)-style questions, achieving a score of 67.2% on the MedQA dataset. Despite this achievement, early versions of Med-PaLM demonstrated significant room for improvement, particularly when comparing the model's answers to those provided by clinicians. This motivated the development of Med-PaLM 2, which incorporates substantial advancements in both the base language model and domain-specific fine-tuning.

Builds on the underlying improvements of PaLM 2, Google's next-generation LLM, which includes enhanced language understanding and reasoning capabilities. In addition to base LLM enhancements, Med-PaLM 2 leverages medical domain-specific fine-tuning and innovative prompting strategies, including a novel ensemble refinement approach, which helps to further refine the model's output. As a result, Med-PaLM 2 set a new state-of-the-art in medical question answering, achieving an impressive score of 86.5% on the MedQA dataset, marking a substantial improvement of over 19% from the original Med-PaLM. Furthermore, Med-PaLM 2 demonstrated strong performance across several other benchmarks, including MedMCQA, PubMedQA, and MMLU clinical topics datasets, approaching or exceeding existing state-of-the-art results.

In addition to benchmark performance, Med-PaLM 2 underwent rigorous human evaluations to assess its applicability in real-world clinical settings. A detailed comparative study involving 1066 consumer medical questions showed that physicians preferred Med-PaLM 2's answers over those generated by physicians on eight out of nine axes related to clinical utility (with statistical significance, $p < 0.001$). Furthermore, the model demonstrated significant improvements on all evaluation axes compared to Med-PaLM, especially when tested

against a set of newly introduced long-form “adversarial” questions designed to challenge the model’s limitations.

MedAlpaca [28]: A large language model specifically fine-tuned for tasks in the medical domain, building upon the LLaMA (Large Language Model Meta AI) [31] architecture. MedAlpaca is available in two versions: medalpaca-7b, which contains 7 billion parameters, and medalpaca-13b, fine-tuned for medical applications with 13 billion parameters. The primary objective is to enhance the performance of question-answering (QA) and medical dialog tasks, positioning it as a specialized model for healthcare-related interactions.

The architecture of MedAlpaca benefits from the underlying structure of LLaMA [31], a model designed for efficient scaling with large amounts of data and parameters. By fine-tuning this architecture for the medical domain, MedAlpaca is able to leverage the advantages of LLaMA’s efficient computation while gaining specialized knowledge in medical contexts.

MedAlpaca’s training data is curated from a variety of sources. First, the model utilizes Anki flashcards to automatically generate medical question-answer pairs, extracting questions from the front of the cards and answers from the back. Second, MedAlpaca uses content from Wikidoc, a medical resource platform. Medical question-answer pairs were generated by extracting relevant paragraphs with headings and using ChatGPT 3.5 to create questions from the headings, with corresponding paragraphs used as answers. The accuracy of these question-answer pairs is estimated to be around 70%, with ongoing work to improve the dataset. Third, data was collected from StackExchange, specifically from five categories: Academia, Bioinformatics, Biology, Fitness, and Health, where top-rated questions were paired with their corresponding highly-rated answers. Lastly, MedAlpaca also incorporates a dataset from ChatDoctor, consisting of 200,000 question-answer pairs.

BioMedLM [64]: A 2.7 billion parameter autoregressive language model based on GPT architecture, specifically developed for biomedical applications. This model is trained exclusively on PubMed abstracts and full articles, which allows it to effectively capture the unique terminology, style, and knowledge of the biomedical field. With a custom Byte-Pair Encoding (BPE) tokenizer designed to preserve biomedical terminology, The model maintains important domain-specific terms as single tokens, enhancing the accuracy of downstream tasks. This design choice results in more consistent and meaningful representations for biomedical terms, which traditional tokenizers might split, thus improving its performance on complex tasks like biomedical question-answering.

BioMedLM has demonstrated strong results on multiple-choice question-answering benchmarks in the biomedical domain. It achieved scores of 57.3% on the MedMCQA development set and 69.0% on the MMLU Medical Genetics exam, showcasing its ability to match or even exceed the performance of larger models in specific tasks. Additionally, the model exhibits robust generative capabilities when fine-tuned, providing comprehensive answers to open-ended medical questions relevant to patient care. The model also outperforms general-purpose LLMs like GPT-Neo on various biomedical tasks, such as BioASQ, PubMedQA, and MedQA, underscoring its specialized training advantages in terms of accuracy and reliability.

Clinical Camel [65]: An open medical language model fine-tuned from LLaMA-2 and optimized to deliver high performance across clinical research tasks. Using the innovative Dialog-Based Knowledge Encoding (DBKE) method, Clinical Camel transforms dense medical literature into synthetic dialogs, enhancing the model’s ability to understand and generate contextually accurate responses in conversational settings. This DBKE approach improves Clinical Camel’s capacity for multi-turn dialog, allowing it to perform complex tasks such as generating plausible clinical notes and reasoning through patient cases effectively.

Clinical Camel leverages QLoRA, a parameter-efficient fine-tuning approach, enabling its development on a single GPU while maintaining competitive performance. This model demonstrates superior performance on standardized medical benchmarks, achieving 64.3% accuracy on the USMLE Sample Exam, 77.9% on PubMedQA, 60.7% on MedQA, and 54.2% on MedMCQA in five-shot settings, surpassing GPT-3.5 in all evaluated tasks. The 70B parameter model performs particularly well, showing significant gains over proprietary counterparts in tasks requiring domain-specific reasoning and language understanding. These outcomes showcase

Clinical Camel's capabilities in handling specialized medical knowledge without the need for extensive computational resources.

Med42 [66]: A specialized large language model (LLM) designed for the medical domain, fine-tuned from the Llama-2 architecture and tailored to enhance knowledge retrieval, clinical reasoning, and question-answering capabilities. The model was developed using two fine-tuning approaches: full-parameter fine-tuning and parameter-efficient tuning, with the latter leveraging Low-Rank Adaptation (LoRA) to optimize computational efficiency without compromising performance. Med42's design addresses the specific needs of medical applications, offering improved processing of complex medical queries and the ability to deliver contextually accurate responses across various clinical tasks.

The model was extensively evaluated on multiple well-known medical benchmarks, including MedQA, MedMCQA, HeadQA, PubMedQA, and the USMLE datasets. Med42 achieved an accuracy of 72% on the USMLE dataset, setting a new standard for open-access medical LLMs. In tests comparing the parameter-efficient (PE-FT) and full-parameter fine-tuned (FP-FT) models, the full-parameter version of Med42 consistently outperformed other Llama-2-based models and reached notable accuracy levels: 76.7% on the MMLU clinical topics and 76.8% on PubMedQA. The results underscore Med42's high efficacy in handling diverse medical knowledge tasks and highlight the impact of fine-tuning methodologies in maximizing model accuracy.

Med42 demonstrated robust performance across a variety of specialized medical tasks, proving particularly effective in multiple-choice question answering and clinical reasoning assessments. The model's LoRA-based parameter-efficient approach achieved competitive results, closely approximating the performance of full-parameter fine-tuning, suggesting it as a viable alternative in resource-constrained environments.

Med42-v2 [67]: Represents an advanced suite of clinical large language models (LLMs) designed specifically for healthcare applications. Building on the Llama3 architecture, Med42-v2 is optimized through rigorous fine-tuning and preference alignment techniques, addressing the limitations that general-purpose LLMs often face in clinical settings. This suite demonstrates superior performance across key medical benchmarks, including MMLU, MedMCQA, and PubMedQA. Med42-v2's enhancements in handling complex medical queries and reasoning through clinical scenarios make it particularly valuable in diagnostic and decision-support contexts, aligning responses closely with real-world medical knowledge.

The model's development follows a two-stage training approach, combining Instruction Fine-tuning with Preference Alignment. During the fine-tuning stage, a highly curated mix of clinical datasets equips Med42-v2 to comprehend intricate medical concepts, terminology, and reasoning structures. This phase includes both chain-of-thought reasoning and interactive dialog formats, boosting the model's applicability to real-world diagnostic tasks and patient interactions. The Preference Alignment phase then refines Med42-v2's responses through Direct Preference Optimization (DPO), guided by extensive AI feedback data. This iterative alignment ensures the model produces responses that are accurate, safe, and in accordance with ethical guidelines, making it dependable in high-stakes healthcare environments.

Med42-v2 achieves impressive accuracy results across a variety of medical benchmarks, consistently surpassing general-purpose and specialized clinical LLMs. Specifically, the Med42-Llama3.1-70B model scores 66.1% on MMLU-Pro, 86.8% on the MMLU medical subset, 72.4% on MedMCQA, 80.4% on MedQA, and 94.5% on the USMLE dataset. These results demonstrate a clear performance advantage over other Llama3-based and competing proprietary models, including GPT-4, which Med42-v2 outperforms across all tested medical datasets.

Gazal-R1 [68]: Introduced a parameter-efficient two-stage training pipeline to enhance medical reasoning in large language models (LLMs). The goal is to create a mid-sized model that outperforms larger counterparts in clinical reasoning while maintaining computational efficiency. The approach addresses challenges such as reward hacking, training instability, and the need for structured, explainable reasoning in high-stakes medical decision-making.

The model employs advanced techniques including Supervised Fine-Tuning (SFT) with a hybrid dataset of 107,033 synthetic examples and the MedReason dataset, enforcing concise, step-by-step reasoning.

Reinforcement learning uses Group Relative Policy Optimization (GRPO), eliminating the need for a separate value model to reduce memory overhead. Key innovations include DoRA for weight decomposition and rsLoRA for stable high-rank adaptation, alongside a multi-component reward system with cosine length scaling and repetition penalties to optimize reasoning depth and accuracy.

Gazal-R1 achieves state-of-the-art results, scoring 87.1% on MedQA and 81.6% on MMLU Pro (Medical), surpassing larger models like Llama 3.1 405B. However, performance dips on MedMCQA (71.9%) due to trade-offs between detailed reasoning and factual recall. Manual validation by medical experts ensures clinical reliability, though limitations include static knowledge and potential regional biases. The model demonstrates that structured reasoning and efficient training can produce high-performance medical LLMs, setting a benchmark for future research in specialized domains.

BERT-based [62] Family: Several models specialized in biomedical text processing have been validated, each fine-tuned on domain-specific medical data to enhance performance in various tasks.

- **BiomedBERT [69]:** A variant of BERT pre-trained on large-scale biomedical corpora, including PubMed abstracts and clinical notes from the MIMIC-III dataset. It is designed to improve performance on biomedical and clinical tasks.
- **ClinicalBERT [70]:** Built upon BERT, fine-tuned specifically on clinical notes from the MIMIC-III database. It focuses on understanding clinical language, particularly in tasks such as predicting patient outcomes and identifying clinical entities.
- **SciBERT [71]:** Pre-trained on a broader range of scientific texts, including papers from the Semantic Scholar corpus. Though it spans various scientific domains, it excels in biomedical literature tasks like citation prediction and scientific entity recognition.
- **BlueBERT [72]:** Another BERT variant pre-trained on both PubMed and MIMIC-III datasets. It is capable of handling both biomedical and clinical texts, showing robust performance across a wide range of biomedical and clinical NLP tasks.

Comparative Analysis of Medical Language Models The rapid development of specialized language models for medical applications has produced numerous architectures with varying capabilities. Table 5 provides a comprehensive comparison of prominent medical LLMs, highlighting their validation datasets and key performance outcomes. This analysis reveals the evolution of medical AI systems and their growing sophistication in handling clinical tasks.

Several key trends emerge from this comparative analysis. First, newer models consistently outperform their predecessors across multiple medical benchmarks, with Med42-v2 (2024) achieving 94.5% accuracy on USMLE questions. Second, specialized architectures like BioGPT demonstrate particular strengths in biomedical natural language processing tasks, while clinical-focused models like ClinicalGPT excel in patient interaction scenarios. Third, the evolution from general-purpose models (BERT variants) to domain-specific architectures (Med-PaLM, HuatuoGPT) shows clear performance advantages in medical applications.

The table also reveals important methodological developments, including:

- The shift from static models (BERT variants) to retrieval-augmented systems (HuatuoGPT)
- Increasing use of reinforcement learning from human feedback (RLHF) in clinical models
- Specialization for regional healthcare systems (Chinese-focused ChiMed-GPT)

These advancements collectively demonstrate the growing capability of language models to handle complex medical reasoning tasks while maintaining clinical relevance and accuracy (Fig. 4).

5.5 Multi modals

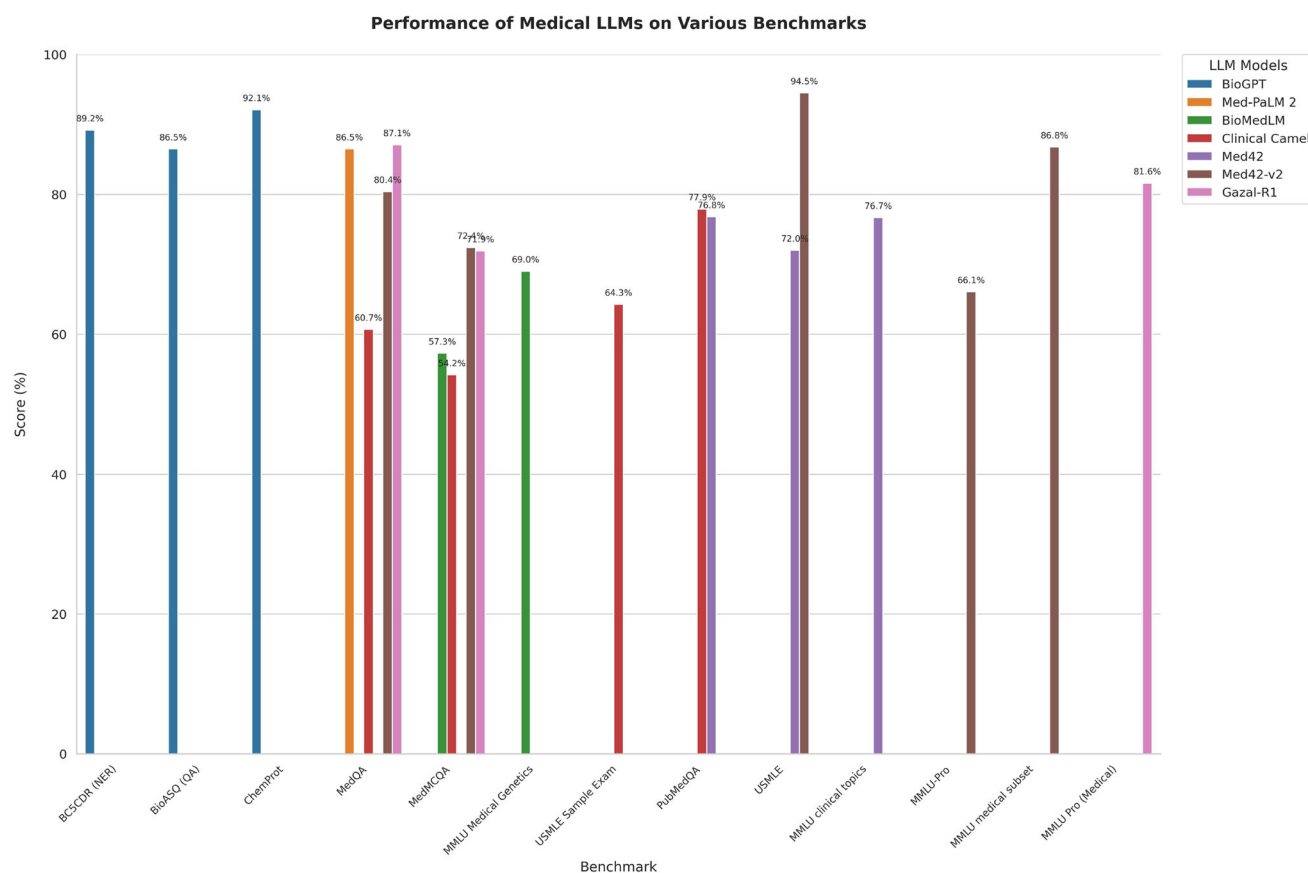
There are several advanced models designed specifically for Medical Visual Question Answering (MedVQA). These models leverage vast datasets, sophisticated algorithms, and multimodal capabilities to enhance diagnostic

Table 5 Comparison of language models in medical and biomedical domains: datasets and key outcomes

LLM	Year	Validation Dataset	Outcomes/Results
Gazal-R1 [68]	2025	MedQA, MMLU-Pro, MedMCQA	87.1% on MedQA, 81.6% on MMLU-Pro (Medical), 71.9% on MedMCQA; outperforms Llama3.1-405B in clinical reasoning while using parameter-efficient training (DoRA/rsLoRA)
Med42-v2 [67]	2024	MMLU, MedMCQA, PubMedQA, USMLE, MedQA	66.1% on MMLU-Pro, 86.8% on MMLU medical subset, 72.4% on MedMCQA, 80.4% on MedQA, 94.5% on USMLE; outperforms GPT-4 and other clinical LLMs across all tested benchmarks
Med42 [66]	2024	USMLE, MedQA, MedMCQA, PubMedQA, MMLU	72% accuracy on USMLE, 76.7% on MMLU clinical topics, 76.8% on PubMedQA; high performance in clinical reasoning and multiple-choice QA tasks
Clinical Camel [65]	2023	USMLE, PubMedQA, MedQA, MedMCQA	64.3% accuracy on USMLE Sample Exam, 77.9% on PubMedQA, 60.7% on MedQA, 54.2% on MedMCQA (five-shot settings); surpasses GPT-3.5 in all tasks; effective in specialized medical reasoning
HuatuoGPT [56]	2023	Chinese Medical Text Dataset, cMedQA2, webMedQA, Huatuo-26 M	Achieved high scores on BLEU, GLEU, ROUGE, and DISTINCT for medical QA. Notable strengths include zero-shot learning, RAG, and pre-training enhancement. Improved clinical decision support accuracy by 15%
ClinicalGPT [58]	2023	cMedQA2, cMedQA-KG, MD-EHR, MEDQA-MCMLE, MedDialog	Outperformed existing LLMs in clinical applications, showing high accuracy and relevance through supervised fine-tuning, instruction tuning, and reinforcement learning. Evaluated on multi-turn conversations and clinical record analysis
ChiMed-GPT [59]	2023	Chinese Clinical Note Corpus, Information extraction, QA, and dialog generation tasks	Provided clinically relevant and ethically sound responses through supervised fine-tuning and RLHF, with a focus on Chinese medical context. Enhanced performance on information extraction, QA, and medical dialog generation compared to general LLMs
BioGPT [60]	2023	BC5CDR, KD-DTI, ChemProt	Achieved F1 score of 89.2% for NER, 86.5% for QA (BioASQ), and 92.1% for relation extraction. Demonstrated strong generative capabilities for biomedical text, particularly benefiting from soft prompts over hard prompts in task-specific settings
GatorTron [63]	2023	Clinical NLP tasks (e.g., concept extraction, relation extraction, STS, NLI, MQA), NCBI Disease Gene Database	Achieved substantial performance improvements in clinical NLP tasks, including 9.5% accuracy increase in medical question answering and 9.6% in logical inference accuracy
Med-PaLM [3]	2023	MedQA, MedMCQA, PubMedQA, MMLU Clinical Topics	Set a new state-of-the-art in medical QA, achieving 86.5% on MedQA and receiving higher physician preference ratings across eight out of nine clinical utility axes. Demonstrated improved response quality on long-form and adversarial questions
MedAlpaca [28]	2023	NHS Health Records, Medical QA and dialog datasets (Anki flashcards, Wikidoc, StackExchange)	Designed for healthcare QA and dialog tasks, achieving high accuracy in medical interactions. Enhanced data sources include Anki flashcards, Wikidoc, and ChatDoctor datasets
BioMedLM [64]	2023	MedMCQA, MMLU Medical Genetics	Scored 57.3% on MedMCQA and 69.0% on MMLU Medical Genetics. Notable for high accuracy in biomedical QA tasks, achieving better results than general-purpose LLMs on BioASQ, PubMedQA, and MedQA
ClinicalT5 [57]	2022	MIMIC-III	Demonstrated proficiency in clinical language tasks, leveraging T5's architecture for text-to-text transformations. Proven effective in understanding and generating clinical text
BiomedBERT [69]	2020	PubMed abstracts, MIMIC-III	Enhanced performance in biomedical tasks due to specialized pre-training on biomedical and clinical corpora, improving entity recognition and outcome prediction
BlueBERT [72]	2020	PubMed, MIMIC-III	Performed well across biomedical and clinical NLP tasks, particularly in entity recognition and relationship extraction in medical documents

Table 5 (continued)

LLM	Year	Validation Dataset	Outcomes/Results
PubMedBERT [61]	2020	Biomedical Language Understanding and Reasoning Benchmark (BLURB)	Excelled in biomedical NLP tasks (e.g., NER, relation extraction, QA). Consistently outperformed general-domain models by leveraging a specialized biomedical corpus (PubMed) for pre-training
ClinicalBERT [70]	2019	MIMIC-III	Focused on clinical NLP tasks, notably improving patient outcome predictions and clinical entity recognition
SciBERT [71]	2019	Semantic Scholar Corpus	Effective in scientific and biomedical literature tasks, including citation prediction and scientific entity recognition

**Fig. 4** Accuracy comparison of medical LLMs on standard benchmarks. Med-PaLM 2 achieves highest scores on most tasks. Missing values indicate unevaluated benchmarks

accuracy and streamline clinical workflows. The following sections delve into some of the most prominent models, including LLaVA-Med++, BiomedGPT, BiomedCLIP, and others, highlighting their architectures, training methodologies, and performance benchmarks.

The evolution of these models is largely driven by the increasing availability of rich, annotated medical datasets, such as MedTrinity-25 M, which comprises over 25 million medical images across diverse modalities. By employing advanced training techniques, these models can interpret intricate medical images while providing precise answers to complex clinical questions. Furthermore, the utilization of self-supervised learning and

retrieval-augmented generation strategies enhances their ability to generate detailed medical descriptions and reports, thus supporting clinicians in their decision-making processes.

LLaVA-Med++ [49]: A cutting-edge Medical Visual Question Answering (MedVQA) model designed to address complex multimodal tasks in the medical domain. It leverages the extensive MedTrinity-25 M dataset, which comprises over 25 million medical images spanning 10 modalities, including X-rays, MRIs, CT scans, and ultrasound images. The dataset is enriched with multigranular annotations, covering both global textual information (e.g., disease types, lesion descriptions, and modality-specific characteristics) and detailed local annotations for regions of interest (ROIs), such as bounding boxes and segmentation masks.

LLaVA-Med++ is pre-trained using these diverse image-ROI-description triplets, which are generated via an automated pipeline without the need for pre-existing paired text descriptions. This pipeline incorporates domain-specific expert models to identify and annotate abnormal regions, facilitating the creation of rich, multigranular visual and textual annotations. By integrating a retrieval-augmented generation process, the model uses ROIs as a guide to generate detailed medical descriptions and answers to complex medical questions.

Technically, LLaVA-Med++ builds on large language models (LLMs) with vision-language pre-training, enabling it to outperform other state-of-the-art models on prominent VQA benchmarks such as VQA-RAD and PathVQA. This model excels in a wide range of tasks, from captioning and report generation to vision-centric challenges like classification and segmentation. Through the extensive use of the MedTrinity-25 M dataset.

BiomedGPT [73]: A pioneering open-source vision-language foundation model specifically designed for the biomedical domain. This lightweight and generalist model is engineered to perform a wide array of biomedical tasks, effectively bridging the gap between visual and textual data in medical contexts. In comprehensive evaluations, BiomedGPT achieved state-of-the-art results in 16 out of 25 experiments, underscoring its robustness while maintaining a computing-friendly scale that facilitates accessibility and deployment.

One of the key applications of BiomedGPT is in radiology visual question answering, where it demonstrated impressive predictive capabilities with a low error rate of just 3.8%. This accuracy highlights the model's potential to enhance clinical decision-making through precise responses to complex queries related to medical imaging. Additionally, BiomedGPT exhibited satisfactory performance in generating intricate radiology reports, achieving an error rate of 8.3%. This capacity for report generation is crucial for improving workflow efficiency in clinical settings, allowing healthcare professionals to focus on patient care rather than documentation.

It achieved a preference score nearly equivalent to that of human experts. This competitive performance suggests that BiomedGPT can effectively condense and interpret vast amounts of medical information, supporting clinicians in making informed decisions quickly. The training methodology employed for BiomedGPT emphasizes the importance of diverse data, demonstrating that a well-rounded training dataset can significantly enhance the practical applicability of biomedical AI models.

BiomedCLIP [74]: An advanced adaptation of the CLIP (Contrastive Language–Image Pretraining) [75] model, specifically designed for the biomedical domain. By leveraging the capabilities of image and text encoders, BiomedCLIP embeds (image, text) pairs into a shared vector space, optimizing their cosine similarity through the InfoNCE loss function. This approach aims to maximize the similarity for positive pairs while minimizing it for negative pairs, enabling effective cross-modal retrieval tasks, including text-to-image and image-to-text retrieval.

Unlike the original CLIP, which was trained from scratch using internet-sourced image-text pairs, BiomedCLIP is tailored to meet the unique characteristics of biomedical data. One of the primary innovations in BiomedCLIP is the incorporation of a domain-specific language model, PubMedBERT, in place of the general-domain GPT-2 model used in CLIP. This switch enhances the text encoder's ability to process complex biomedical literature, which often involves longer and more intricate texts. Additionally, BiomedCLIP adjusts its tokenizer and context size to better accommodate the demands of biomedical text processing. BiomedCLIP also significantly improves the image processing component by utilizing larger Vision Transformer models and higher image resolutions. These adaptations allow the model to capture the detailed visual information crucial for biomedical understanding. To further enhance model performance while maintaining pretraining efficiency,

BiomedCLIP employs a patch dropout strategy and optimizes its batch size specifically for the biomedical domain.

In rigorous evaluations of cross-modal retrieval tasks using a held-out test set from the PMC-15 M dataset, which comprises 725,739 image-caption pairs, BiomedCLIP demonstrated exceptional retrieval accuracy. It achieved a top-5 accuracy rate of over 77% and a top-1 accuracy rate exceeding 56%, outperforming both the general-domain CLIP model and PubMedCLIP. The subpar performance of PubMedCLIP can be attributed to its reliance on a limited dataset, which may lead to catastrophic forgetting during continual pretraining. BiomedCLIP's success is indicative of the importance of large-scale, diverse datasets in developing domain-specific vision-language models.

Clip-Driven Universal Model [76]: A medical visual question answering that focuses on organ segmentation and tumor detection in CT scans. By utilizing embeddings generated by the pre-trained CLIP text encoder, the model replaces traditional label representations with semantic text embeddings, enhancing the anatomical features captured in the model's output. This method improves the model's understanding of spatial relationships between organs and tumors, leading to better segmentation and detection performance.

Successfully the model segment 25 abdominal organs and detect 6 tumor types while exhibiting minimal intra-observer variability, comparable to human annotators. Notably, it has ranked first in prestigious challenges such as the Medical Segmentation Decathlon (MSD) and the BTCV, showcasing its high accuracy and reliability in clinical settings. Moreover, the model demonstrates improved tumor detection with fewer false positives, maintaining high sensitivity and thus enhancing diagnostic effectiveness.

The model also offers significant computational efficiency, accelerating testing speeds by a factor of six compared to dataset-specific models. Its architecture is versatile, compatible with various backbone networks such as CNNs and Transformers, and it generalizes well across CT scans from different medical institutions without the need for additional tuning. This adaptability makes it a valuable tool for diverse clinical applications.

MUMC [77]: Introduced a novel self-supervised learning framework specifically designed for medical visual question answering (VQA). This model addresses the inherent challenges of VQA, which requires integrating visual and textual information to accurately answer clinical questions related to medical images.

The core of the MUMC approach involves the development of unimodal and multimodal feature representations from input images and text using a rich set of medical image caption datasets. This is accomplished through the application of both unimodal and multimodal contrastive losses, which enhance the model's ability to distinguish between relevant and irrelevant data. Additionally, the model employs masked language modeling and image-text matching as pre-training objectives, enabling it to learn meaningful associations between medical images and their corresponding textual descriptions.

The model is fine-tuned for downstream medical VQA tasks. The MUMC model achieves state-of-the-art (SOTA) performance on three publicly available medical VQA datasets, demonstrating notable accuracy improvements of 2.2%, 14.7%, and 1.7%, respectively. These results illustrate the model's effectiveness in tackling the complexities of medical VQA.

CLIP-ViT/GPT2 [78]: A sophisticated architecture model (CLIP With GPT), designed for medical visual question answering (VQA) tasks. This model leverages the strengths of the CLIP (Contrastive Language–Image Pretraining) framework, utilizing a Vision Transformer (ViT) backbone to extract visual features from medical images. The ViT model operates with a feature dimensionality of 512, effectively capturing detailed visual information crucial for understanding medical contexts. For efficient batch processing, zero padding is applied to the right side of the sequences, ensuring uniformity across input samples.

In terms of language processing, the model utilizes several advanced language models, including GPT-2-XL, a causal language model with 1.5 billion parameters trained on the WebText corpus. Additionally, BioMedLM and BioGPT are employed, both of which are adaptations of the GPT-2 architecture. BioMedLM is pre-trained on PubMed data, while BioGPT is trained on biomedical datasets from The Pile, with sizes of 1.5 billion and 2.7 billion parameters, respectively.

M2I2 [79]: The M2I2 (Masked language modeling, Masked image modeling, Image-text matching, and Image-text contrastive learning) model is designed for medical visual question answering (VQA) by effectively integrating visual and textual information. The architecture employs a 12-layer Vision Transformer (ViT) to encode medical images, capturing spatial hierarchies crucial for understanding complex visual tasks. Textual features are extracted using a 6-layer transformer initialized with the first six layers of a pre-trained BERT model, with the remaining layers serving as a multimodal encoder to facilitate interaction between the two modalities.

The model processes pairs of medical images and their corresponding captions. Images are divided into $16 \times 16 \times 16$ patches, while text is tokenized with a BERT-based WordPiece tokenizer. To enhance learning, both modalities are randomly masked with a 15% probability, allowing the model to predict the masked portions. The unmasked portions, along with a special classification token $\langle \text{CLS} \rangle$, are fed into their respective encoders.

M2I2 employs several training objectives, including masked language modeling (MLM) and masked image modeling, to improve feature representations. Additionally, image-text matching and contrastive learning objectives enhance the alignment between visual and textual data, fostering a robust understanding of their relationships.

LLaVA-Med [80]: A medical visual question answering model developed as a large language-and-vision assistant specifically for biomedical applications. Its architecture parallels prefix tuning techniques used in language models (LMs), utilizing a trainable module that connects a frozen image encoder with a causal language model. In this implementation, LLaVA-Med employs a linear projection to map visual features to a visual prefix, differing from traditional methods that utilize multi-layer perceptron (MLP) networks for this purpose. It utilizes a 7 billion parameter language model, optimizing performance while maintaining a manageable model size, in contrast to existing methods that typically use larger pre-trained LMs like GPT-2, BioMedLM, or BioGPT, ranging from 1.5 billion to 2.7 billion parameters.

By leveraging GPT-4, the model self-instructs biomedical multimodal instruction-following data using freely available, broad-coverage biomedical image-text pairs sourced from PubMed Central. This process allows the model to develop a nuanced understanding of biomedical concepts and improves its conversational capabilities regarding biomedical images.

Comparative evaluations reveal that all variants of LLaVA-Med outperform its predecessor, LLaVA [81], and other representative methods in multiple benchmarks. Although the differences in language model initialization between LLaVA-Med and models like Vicuna are minimal, LLaVA-Med benefits from a vision encoder initialized with BioMed CLIP, which provides improved performance compared to general-domain CLIP models. Furthermore, LLaVA-Med demonstrates superior fine-tuning performance against state-of-the-art supervised models on closed-set questions in datasets such as VQA-RAD and PathVQA, indicating its proficiency in following instructions to complete biomedical tasks effectively.

PubMedCLIP [82]: An adaptation of the CLIP (Contrastive Language–Image Pre-training) model, fine-tuned for the biomedical domain. Originally designed for general-domain multimodal tasks, CLIP has demonstrated remarkable success by leveraging cross-modal supervision from large-scale image–text pairs. However, its application in specialized domains like biomedicine, where the textual and visual data are significantly more complex.

PubMedCLIP builds upon the architecture of CLIP by incorporating medical knowledge derived from PubMed articles, which contain a rich collection of both textual and visual biomedical data. The model is trained to align visual representations with corresponding medical textual descriptions, using a contrastive learning framework. This training paradigm allows PubMedCLIP to learn more accurate medical image representations compared to traditional models that rely on visual data alone.

The contrastive learning approach in PubMedCLIP ensures that images and their associated descriptions are projected into a shared latent space, where semantically similar images and text pairs are positioned closer together, and dissimilar ones are pushed apart. This cross-modal supervision improves the model's ability to interpret medical images in relation to domain-specific language, making it especially useful for tasks that require

understanding both image content and accompanying textual data, such as Medical Visual Question Answering (MedVQA).

PubMedCLIP is pre-trained using vast collections of biomedical image–text pairs sourced from PubMed, one of the largest repositories of medical literature. By leveraging the extensive domain-specific information available in PubMed articles, PubMedCLIP aligns its visual representations with accurate medical terminologies and descriptions. This fine-tuning process is critical for ensuring that PubMedCLIP captures the nuances of medical imaging, which often involve complex anatomical structures and pathological details.

The model is further evaluated on two MedVQA benchmark datasets, where it is tested on two distinct methods: Mixture of Enhanced Visual Features (MEVF) and Question answering via Conditional Reasoning (QCR). Both approaches benefit from PubMedCLIP's refined visual embeddings, which provide a more semantically rich and contextually relevant foundation for answering medical questions based on images.

In comparison to the original CLIP model and state-of-the-art MAML (Model-Agnostic Meta-Learning) networks pre-trained solely on visual data, PubMedCLIP consistently outperforms both on the MedVQA task. Specifically, PubMedCLIP shows an improvement in overall accuracy of up to 3% across various evaluation settings. These gains are attributed to its ability to leverage language supervision during visual representation learning, a key advantage that traditional models, which rely only on visual pre-training, lack.

PathCLIP [83]: A medical image analysis, particularly in pathology. It leverages the (CLIP) framework, to enhances visual and semantic comprehension through simultaneous language and image processing. The model comprises an image encoder and a text encoder, enabling it to capture meaningful visual-semantic representations without requiring explicit supervision. This versatility makes PathCLIP suitable for zero-shot classification and image retrieval tasks, essential for interpreting complex medical imagery.

The model has been evaluated using two pathology datasets related to bone cancer and lung cancer, encompassing 1144 H&E stained images of osteosarcoma and 67 H&E stained slides from the WSSS4LUAD challenge. PathCLIP was subjected to various common image corruptions, including brightness, contrast, Gaussian blur, resolution, saturation, hue, and markup, at different severity levels to assess its robustness. Performance metrics such as accuracy, F1 score, and precision for image retrieval were employed to evaluate the model under these conditions. Results indicate that while PathCLIP maintains reasonable performance across various corruptions, there is notable degradation in accuracy and retrieval precision, particularly with severe blurring and low resolution.

PathCLIP's performance varies based on the type and severity of the corruption. For example, severe image blurriness significantly affects classification accuracy, whereas the model remains relatively robust against variations in brightness and contrast. The importance of consistent device settings during training and testing is emphasized, as discrepancies can lead to a substantial decline in performance. Additionally, the model's effectiveness in image retrieval diminishes with increased markup and blur, suggesting the necessity for diverse training data that includes varying conditions.

OphGLM [84]: A cutting-edge multimodal biomedical chatbot specifically designed for ophthalmology, combining image and language understanding for diagnostic and conversational support. It consists of two main components: the fundus diagnosis pipeline, which uses deep learning models to classify and segment ophthalmic conditions like diabetic retinopathy, glaucoma, and age-related macular degeneration (AMD), and the OphGLM dialog pipeline, which integrates diagnostic outputs with user input to generate detailed, context-aware responses.

The fundus diagnosis pipeline leverages convolutional neural networks (CNNs) and transformer-based architectures to analyze fundus images, achieving high accuracy in classification tasks: 97% for diabetic retinopathy, 94% for glaucoma, 98.4% for AMD, and 99.8% for pathological myopia. Additionally, it includes a lesion segmentation module with a dice coefficient of 0.854 for exudate segmentation, showcasing its effectiveness in image segmentation. These outputs are structured as diagnostic reports, forming a key input for the dialog generation pipeline.

In the OphGLM dialog pipeline, the system combines these diagnostic reports with patient queries to generate accurate and context-rich medical conversations. The pipeline utilizes the ChatGLM model, a 6.2-billion

parameter language model fine-tuned on over 20,000 fundus-related dialogs. This multimodal approach allows OphGLM to handle complex, open-ended medical queries while improving both the relevance and accuracy of its responses. The model achieves 95% accuracy in diagnostic conversations, significantly outperforming text-only biomedical models in handling ophthalmic queries.

OphGLM's integration of vision and language models offers robust performance in both disease diagnosis and patient interaction. With high accuracy in image-based diagnoses and enhanced dialog generation, the model is a valuable tool for ophthalmologists, aiding clinical decision-making and patient communication.

Med-Gemini [55]: A state-of-the-art multimodal large language model (LLM) designed specifically for medical applications. It integrates advanced language understanding with the ability to process and analyze multimodal data, such as medical images and long-form health records. One of the defining features of Med-Gemini is its seamless use of web search and its ability to adapt efficiently to new modalities using custom encoders. This adaptability allows the model to excel in various medical tasks, pushing the boundaries of what LLMs can achieve in the healthcare domain.

Med-Gemini has been evaluated on 14 medical benchmarks, achieving new state-of-the-art (SoTA) performance on 10 of them. Notably, it surpasses the GPT-4 model family on every benchmark where direct comparisons were possible, often with substantial margins. For instance, Med-Gemini attains a remarkable 91.1% accuracy on the MedQA (USMLE) benchmark, utilizing a novel uncertainty-guided search strategy to improve decision-making in complex medical questions. In multimodal tasks, including NEJM Image Challenges and MMMU (health & medicine), Med-Gemini outperforms GPT-4V by an average of 44.5% relative improvement.

The model's long-context capabilities are particularly impressive, demonstrating SoTA performance in retrieving critical information from long deidentified health records and medical video question answering, outperforming prior methods through in-context learning alone. Additionally, Med-Gemini has shown real-world potential, surpassing human experts in tasks like medical text summarization and excelling in multimodal medical dialog, research, and education applications.

Comparative Analysis of Multi-modal Models for Medical Visual Question Answering The integration of visual and textual understanding in medical AI systems has led to significant advancements in clinical decision support. Table 6 presents a comprehensive evaluation of state-of-the-art multi-modal models for Medical Visual Question Answering (MedVQA), demonstrating their capabilities across various medical imaging modalities and diagnostic tasks.

Several critical observations emerge from this analysis of multi-modal medical AI systems:

- **Performance Trends:** Newer models consistently demonstrate superior performance, with Med-Gemini (2024) showing a 44.5% relative improvement over GPT-4V in medical question answering tasks. The progression from BiomedCLIP (2023) to LLaVA-Med++ (2024) reveals an evolution toward more specialized architectures.
- **Clinical Applications:** Different models excel in specific medical domains:
 - Ophthalmology: OphGLM achieves 97% accuracy in diabetic retinopathy detection
 - Radiology: BiomedGPT maintains an 8.3% error rate in report generation
 - Pathology: PathCLIP demonstrates robustness in cancer image analysis
- **Technical Innovations:** The table highlights three key architectural approaches:
 1. Contrastive learning frameworks (BiomedCLIP, PubMedCLIP)
 2. Vision-language pretraining (Clip-Driven Universal Model)
 3. Hybrid CNN-transformer designs (OphGLM)

The comparative analysis reveals that while general-purpose multi-modal models like Med-Gemini achieve broad competency, domain-specialized architectures often outperform them in specific clinical tasks. This suggests a future direction for medical AI development where foundation models are adapted through:

Table 6 Comparison of multi-modal models in medical VQA

Model Name	Year	Validation Dataset	Outcomes/Results
Med-Gemini [55]	2024	USMLE, NEJM Image Challenges	SOTA performance on 10 out of 14 benchmarks. Excels in multimodal medical dialog and long-context tasks. Achieves 91.1% accuracy on MedQA, outperforms GPT-4V by an average of 44.5% relative improvement
LLaVA-Med++ [49]	2024	MedTrinity-25 M	Achieves SOTA performance on VQA-RAD and PathVQA. Capable of classification, segmentation, and report generation with RAG using ROI descriptions
Clip-Driven Universal Model [76]	2024	MSD, BTCV	High accuracy in segmentation (25 organs) and tumor detection (6 types). Ranked first in multiple challenges with 6x speedup in testing
MUMC [77]	2024	MedVQA datasets	Outperforms on 3 MedVQA datasets with 2.2%, 14.7%, and 1.7% accuracy improvements. Uses multimodal contrastive learning
CLIP-ViT/GPT2 [78]	2024	WebText, The Pile	Uses CLIP with ViT and GPT2-XL for VQA tasks, leveraging BioMedLM and BioGPT for specialized biomedical contexts
M2I2 [79]	2024	VQA datasets	Utilizes ViT and BERT for masked modeling, image-text matching, and contrastive learning. Achieves robust multimodal feature representations
LLaVA-Med [80]	2024	VQA-RAD, PathVQA	Enhances GPT-4's biomedical capabilities using BioMed CLIP. Outperforms other models in multiple benchmarks
OphGLM [84]	2024	Fundus image datasets	97% accuracy in diabetic retinopathy detection, 95% accuracy in diagnostic conversations. Integrates CNNs and transformers
BiomedGPT [73]	2023	Radiology datasets	SOTA in 16 out of 25 experiments. Error rate of 3.8% in VQA tasks and 8.3% in radiology report generation. Nearly equivalent to human preference scores
BiomedCLIP [74]	2023	PMC-15 M	Achieved top-5 accuracy of over 77% and top-1 accuracy of 56% in cross-modal retrieval tasks. Superior to PubMedCLIP in domain-specific retrieval
PubMedCLIP [82]	2023	PubMed articles	Achieves up to 3% improvement in MedVQA accuracy. Leverages contrastive learning for better alignment of medical image-text pairs
PathCLIP [83]	2023	Bone cancer, lung cancer datasets	Robust against corruptions like brightness and contrast but sensitive to blurring. Used for zero-shot classification and retrieval

- Specialized pretraining on medical image-text pairs
- Integration with clinical knowledge graphs
- Task-specific fine-tuning using expert-annotated datasets

These multi-modal systems are increasingly being deployed in clinical workflows, assisting with tasks ranging from radiology report generation to surgical planning, while maintaining the necessary accuracy and reliability standards required in medical practice (Fig. 5).

5.6 RAG models

5.6.1 Naive retrieval-augmented generation (RAG):

Various implementations of Naive RAG in the medical domain are discussed below:

ChatENT [85]: A specialized large language model (LLM) for otolaryngology–head and neck surgery (OHNS), built to overcome the limitations of general-purpose LLMs in clinical settings. It targets critical concerns such as hallucinations, accuracy, and patient safety, offering reliable support for medical education and decision-making. By aligning outputs with evidence-based guidelines, ChatENT aims to provide trustworthy, domain-specific responses for both healthcare professionals and patients.

The system combines ChatGPT with a Retrieval-Augmented Generation (RAG) framework, leveraging the Clinical ENT Knowledge Base (CEKB) for structured medical content. Semantic search identifies relevant OHNS literature for each user query, grounding the LLM's output in curated knowledge without retraining. This

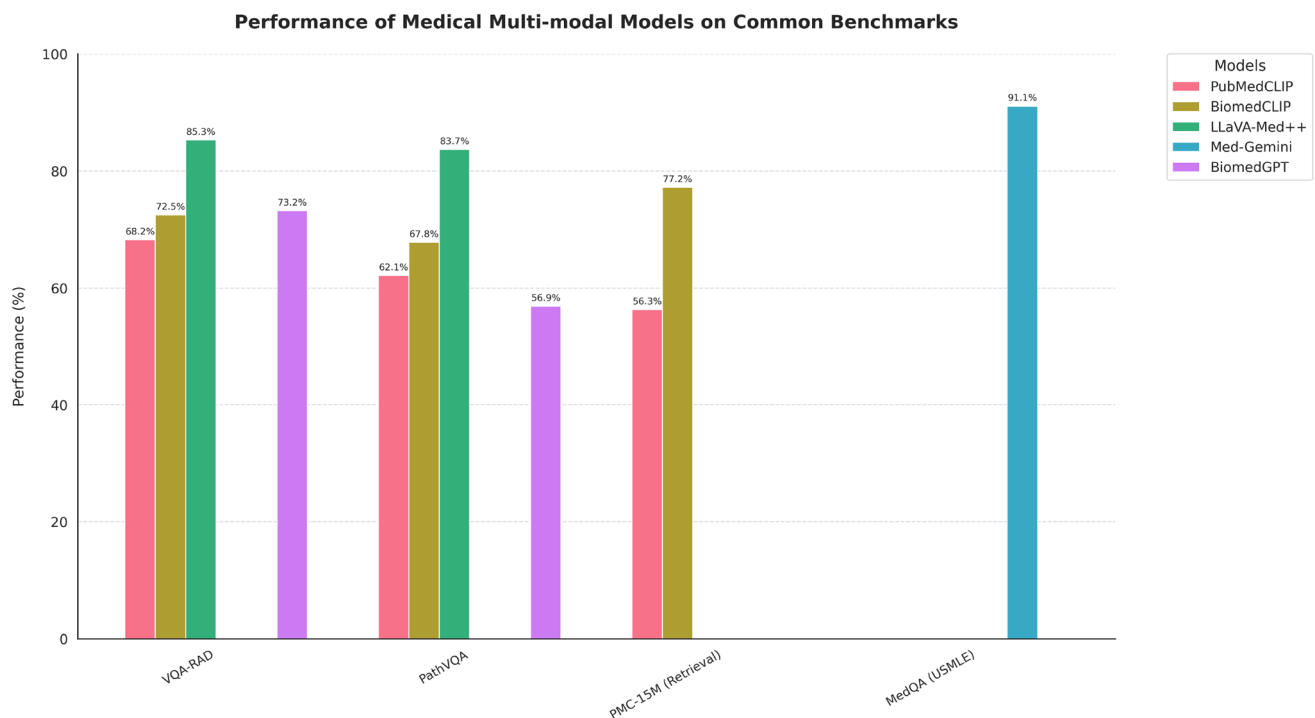


Fig. 5 Accuracy comparison of medical LLMs on standard benchmarks. Med-PaLM 2 achieves highest scores on most tasks. Missing values indicate unevaluated benchmarks

integration ensures that generated answers maintain both relevance and factual integrity, making ChatENT more cost-effective and clinically accurate than generic LLMs.

In evaluations, ChatENT outperforms general LLMs, achieving 95.7% validity and 87.2% competency on OHNS exam questions—a 58.4% reduction in error rates. It scored 80% in multiple-choice tests (a 26% gain over ChatGPT) and surpassed human averages in some subspecialties. Manual review using the CVSC framework confirmed lower hallucination rates and greater reliability, supporting the value of domain-specific RAG in advancing safe, effective medical AI tools.

Xiong, Guangzhi and Jin, et al. [20]: Assessed Naive RAG systems in the medical domain, addressing key limitations of Large Language Models (LLMs) such as hallucinations and outdated knowledge. The evaluation uses 7663 questions drawn from five medical QA datasets, designed around zero-shot and question-only retrieval to mirror realistic clinical and research scenarios. The MEDRAG toolkit facilitates this evaluation by integrating diverse knowledge sources (e.g., PubMed, StatPearls, medical textbooks, Wikipedia), retrieval models (BM25, Contriever, SPECTER, MedCPT), and LLMs (GPT-3.5, GPT-4, Mixtral, Llama2, MEDITRON, PMC-LLaMA).

The research aims to identify optimal RAG configurations, focusing on corpus selection, retriever performance, and LLM alignment. Results show that combining multiple corpora (MedCorp) with retriever fusion (RRF-4) significantly boosts performance, enabling models like GPT-3.5 and Mixtral to rival GPT-4. Technical observations include the benefits of log-linear scaling with snippet count and the “lost-in-the-middle” phenomenon, where middle-positioned snippets contribute less. Biomedical retrievers such as MedCPT outperform general-purpose alternatives, and PubMed consistently yields the best results across datasets.

RAG improves factual grounding by prepending retrieved documents to prompts, substantially increasing accuracy and reducing hallucinations. GPT-3.5, for example, reaches 71.57% on the MIRAGE benchmark—a 17.9% improvement over baseline methods. Domain-specific models like MEDITRON outperform GPT-4 on literature-focused tasks, while retrieval is essential for complex clinical exam questions. The study recommends

using PubMed/MedCorp, MedCPT/RRF-4, and GPT-3.5/Mixtral for cost-effective, high-performing medical QA systems, establishing MEDRAG and MIRAGE as key resources for reproducible biomedical AI research.

RALL [86]: focuses on improving accessibility of biomedical literature through automated generation of lay language summaries. The work introduces a novel approach called Retrieval-Augmented Lay Language (RALL) generation, which combines text simplification with background explanation by incorporating external knowledge. The study presents CELLS, the largest parallel corpus of 63,000 scientific abstracts paired with expert-written plain language summaries, enabling systematic evaluation of lay language generation models across 12 biomedical journals.

The methodology employs two key retrieval strategies: definition-based retrieval using UMLS and Wikipedia term definitions, and embedding-based retrieval utilizing dense passage retrieval from Wikipedia. BART models enhanced with retrieval augmentation demonstrate superior performance compared to vanilla BART and zero-shot LLMs like GPT-4 and Llama 2. The RALL framework specifically addresses the challenge of background explanation, where source documents often lack contextual information needed for non-expert comprehension, by prepending relevant external content to the input.

Evaluation results show the BART-RAG model achieves a BERTScore of 0.72 and reduces Coleman-Liau readability scores by 15%, indicating significant improvements in both quality and accessibility. Human evaluations rate retrieval-augmented outputs highly for factual correctness (3.5/4) while noting GPT-4's superior performance in fluency and simplicity (3.8/4). The study highlights the importance of domain-specific retrieval and controlled generation, with Wikipedia proving particularly effective as a knowledge source for background explanations in biomedical text simplification.

LiVersa [87]: A liver disease-specific large language model (LLM) chat interface, leveraging retrieval-augmented generation (RAG) to enhance clinical accuracy and compliance with protected health information (PHI) standards. The primary goal is to address limitations of general-purpose LLMs, such as hallucinations and lack of clinical specialization, by integrating 30 American Association for the Study of Liver Diseases (AASLD) guidelines into the model. The approach employs Microsoft Azure OpenAI's text-embedding-ada-002 for document vectorization and GPT-4 or GPT-3.5-turbo for response generation, ensuring PHI-compliant deployment within the UCSF Versa platform.

LIVersa demonstrates improved specificity in answering hepatology-related queries, achieving 100% accuracy in forced "yes/no" responses across 10 case-vignette questions on hepatitis B treatment and hepatocellular carcinoma surveillance. However, detailed rationales were incomplete for three scenarios, highlighting RAG's dependency on the quality and breadth of ingested documents. The model's performance surpasses general-purpose LLMs by grounding responses in authoritative guidelines, reducing hallucinations. While no traditional metrics like F1-score or ROUGE are reported, the manual evaluation underscores LIVersa's potential for clinical decision support, provided the RAG dataset is expanded to include diverse guidelines and updates. The prototype exemplifies how RAG can tailor LLMs for specialized medical domains while maintaining PHI compliance.

Almanac [88]: An LLM augmented with clinical knowledge stored in a vector database. A dataset of 130 clinical cases curated by five experts across various specialties was used to train the model. It combines web-based knowledge retrieval with LLM-generated responses, ensuring that answers remain factually grounded and well-cited. Almanac retrieves relevant medical information using a vector database and web search, then synthesizes responses with in-text citations, improving factuality, completeness, and safety in clinical decision-making. Was tested on 130 clinical scenarios across five medical specialties.

The primary goal is to improve the accuracy, completeness, and safety of LLM-generated medical responses by grounding them in authoritative guidelines and external knowledge sources. This is achieved through a hybrid architecture combining retrieval systems with generative models, enabling real-time access to medical literature and clinical guidelines.

Manually evaluated by board-certified and resident physicians. It outperformed ChatGPT, achieving an 18% increase in factual accuracy ($p < 0.05$), with the highest gains in cardiology (91% vs. 69%). In adversarial testing, Almanac maintained 95% safety, whereas ChatGPT failed entirely. However, despite its superior accuracy and

safety, physicians preferred ChatGPT responses 57% of the time, likely due to its more natural-sounding language. These results highlight the potential of RAG-based medical AI for improving evidence-based clinical recommendations.

Thompson, Will E and Vidmar et al. [89]: Implemented retrieval-augmented generation (RAG) framework combined with MapReduce to enable zero-shot disease phenotyping from electronic health records (EHRs), focusing on pulmonary hypertension (PH) as a case study. The goal is to address challenges in rare disease identification, where manual rule-based methods and traditional machine learning face limitations due to sparse data, coding errors, and the need for scalable solutions. The approach leverages PaLM-2 (bison@001) to analyze unstructured clinical notes across a patient's entire record, overcoming the impracticality of processing full notes within LLM context windows.

Key technical components include Regex-based retrieval to filter relevant text snippets, followed by parallel processing via MapReduce to aggregate LLM outputs. The method outperforms physician-curated rule-based models, achieving an F1 score of 0.75 (vs. 0.62 for rules) on a test set of 199 patients. Performance gains stem from improved handling of diverse note types (e.g., progress notes, imaging reports) and explicit prompting strategies, such as Chain-of-Thought reasoning and exclusion of unreliable snippets (e.g., unconfirmed echocardiogram findings). Manual evaluation confirmed the framework's ability to identify twice as many PH cases compared to structured rules, highlighting its potential for rare disease phenotyping in real-world EHR applications.

Llama-2 Rag [90]: a comparative analysis of fine-tuning with LoRA (Low-Rank Adaptation) and Retrieval-Augmented Generation (RAG) techniques for developing medical conversational AI systems. The study evaluates three models—Llama-2, GPT-3.5/4, and LSTM—on mixed-domain doctor-patient dialogs to assess their ability to provide accurate, context-aware medical advice. The primary objective is to improve healthcare accessibility, particularly for underserved populations, by leveraging advanced NLP methods to create robust, domain-specific dialog systems. The work addresses key challenges in medical AI, including factual accuracy, response coherence, and adherence to clinical guidelines.

The methodology combines Regex-based retrieval to filter relevant clinical text snippets with RAG frameworks using both GPT and Llama-2 for knowledge-augmented response generation. For fine-tuning, LoRA enables parameter-efficient adaptation of large models like Llama-2-7B and GPT-3.5-turbo, reducing computational costs while maintaining performance. Key innovations include rotary positional embeddings (RoPE) in Llama-2 for better long-context handling and hybrid retrieval strategies (dense embeddings + BM25) to improve knowledge grounding. The models are trained on a composite dataset spanning general medicine, pediatrics, and specialized care, ensuring broad coverage of medical scenarios.

Evaluation metrics include BERT Score (for semantic similarity), BLEU (fluency), and ROUGE (informativeness), alongside manual assessments of empathy and safety. Results show GPT variants achieve higher BLEU (0.45) and ROUGE (0.52) scores, indicating stronger fluency and structure, while Llama-2 excels in BERT Score (0.82), reflecting better semantic alignment with reference responses. The LSTM model underperforms (BLEU: 0.28) due to limitations in capturing long-range dependencies. Manual evaluation confirms that RAG-enhanced models provide more factually accurate and contextually relevant answers compared to fine-tuned-only versions, with GPT-4 showing particular strength in maintaining coherent dialog flow.

The findings demonstrate that combining RAG with LoRA-based fine-tuning yields optimal performance, balancing domain-specific knowledge integration with computational efficiency. This approach not only improves response quality but also addresses critical healthcare needs by enabling scalable, accurate medical dialog systems. Future directions include expanding to multilingual medical conversations and integrating real-time clinical decision support, further bridging gaps in healthcare accessibility. The work underscores the potential of hybrid LLM architectures to transform patient-provider interactions while maintaining rigorous standards for accuracy and safety.

5.6.2 Advanced retrieval-augmented generation (RAG)

Some implementations of Advanced RAG in the medical domain:

MedKgConv [91] and *MED* [92]: Have been developed for medical dialog generation, to explore the integration of commonsense knowledge graphs (KGs) with dialog systems to enhance retrieval and reasoning in medical conversations, each employing unique strategies. *MedKgConv* utilizes a BioBERT encoder to process conversation history and Quick-UMLS to extract relevant knowledge graphs for reasoning. The model then uses a BioBERT decoder for response generation. In contrast, *MED* encodes knowledge graphs alongside patient conversations with large language models (LLMs) and enhances them through medical entity annotation, following a semi-supervised approach.

Both models have demonstrated significant improvements on benchmark datasets like MedDialog and COVID datasets. *MedKgConv* achieved an increase of 3.3% in F1 score and 6.3% in BLEU-4, while *MED* outperformed BioBERT by 8.5 points on F1 and 9.3 points on BLEU-4. These results emphasize the impact of incorporating both dialog and graph-based knowledge in generating medical dialogs, showcasing how knowledge-enhanced neural models can refine the accuracy and coherence of responses in medical conversations. Despite their success, these KG approaches are typically designed for question-answering (QA) tasks. KGs provide structured information to predict answers, which may limit their flexibility across other conversational tasks.

accGPT [93]: A context-aware chatbot designed to provide imaging recommendations based on the American College of Radiology (ACR) Appropriateness Criteria. The goal is to improve clinical decision-making by reducing inappropriate imaging studies, ensuring adherence to evidence-based guidelines, and addressing limitations of generic LLMs like outdated knowledge and hallucinations. The model leverages LlamaIndex to integrate 209 ACR documents into a vector index using OpenAI's text-embedding-ada-002 embeddings, enabling semantic retrieval of relevant guidelines. GPT-3.5-turbo synthesizes responses by prepending the top three retrieved passages to prompts, ensuring recommendations are grounded in authoritative sources.

The evaluation compares *accGPT* against radiologists (junior to senior levels) and generic ChatGPT-3.5/4 on 50 synthetic clinical cases. *accGPT* achieves superior accuracy, with 82% of recommendations rated “usually appropriate” (vs. 64–74% for radiologists and 64–80% for generic chatbots) and 84% consistency across six runs. Manual evaluation highlights its reduced reliance on “may be appropriate” responses, reflecting higher specificity. The model also demonstrates significant efficiency gains, processing cases in 8 min at €0.18, compared to radiologists' 50 min at €29.99. Key technical insights include the advantage of domain-specific retrieval (ACR corpus) over generic LLMs and the role of LlamaIndex in dynamic knowledge grounding. Limitations include challenges with rare non-imaging cases and the need for broader guideline coverage. The results underscore the potential of RAG-enhanced LLMs to augment radiology workflows by combining scalability with guideline compliance.

FraCChat [94]:, a context-aware chatbot designed to classify fractures using the AO/OTA Fracture and Dislocation Classification Compendium [95]. The goal is to improve clinical decision-making by automating accurate fracture coding from radiology reports, addressing challenges like variability in human interpretation and reliance on experience. The model integrates LlamaIndex with GPT-3.5-turbo and GPT-4, leveraging text-embedding-ada-002 embeddings to create a vector index from AO guidelines. Retrieval-Augmented Generation (RAG) enhances performance by prepending the top five retrieved passages to prompts, grounding responses in domain-specific knowledge.

FraCChat-GPT-4 achieves 83% accuracy in full AO code classification, surpassing *FraCChat-GPT-3.5* (57%) and generic GPT-4 (7%), though still below human radiologists (95%). Manual evaluation shows superior consistency, with 71% of *FraCChat-GPT-4* answers consistently correct versus 2% for generic GPT-4. The model matches human performance in identifying bone location (97%), fracture type (97%), and group (90%), but lags in subgroup classification (79%). Time efficiency is notable, with *FraCChat-GPT-4* processing cases in 5.2 s versus 50 s for humans. Key technical insights include the advantage of domain-specific RAG over generic LLMs and the scalability of LlamaIndex for dynamic knowledge updates. Limitations include challenges with complex

fractures (e.g., hindfoot cases) and dependency on guideline coverage. The results highlight RAG's potential to streamline radiology workflows while maintaining clinical relevance.

Yang, Rui and Marrese-Taylor, et al. [96]: Introduced techniques to handle noise, such as Bio-Epidemiology-NER and UMLS-augmented LLMs, creating a reliable medical chatbot. However, some extracted medical terms were unrelated to user queries, leading to incomplete answers. By augmented LLM framework that integrates the Unified Medical Language System (UMLS) [97] to improve the accuracy and reliability of medical question-answering (QA). By retrieving definitions and relationships of medical terms from UMLS, the system enhances LLaMa2-13b-chat and ChatGPT-3.5 with structured medical knowledge, ensuring more factual and relevant responses. The framework employs medical terminology extraction, UMLS concept mapping, and contextual knowledge retrieval to guide LLMs in generating medically accurate answers. The model was tested on 104 questions from the LiveQA dataset, with automated evaluation using ROUGE Score and BERTScore. While ChatGPT-3.5 achieved higher ROUGE and BERT scores, the UMLS-augmented LLaMa2-13b-chat showed improvement in factual accuracy. A physician evaluation on 20 questions assessed responses based on Factuality, Completeness, Readability, and Relevance. The UMLS-augmented ChatGPT-3.5 outperformed the standard ChatGPT-3.5 in 40% of factuality cases and achieved better completeness (55%) and relevance (40%), though it slightly reduced readability due to increased medical detail, ChatGPT-3.5 had a 45% win rate in readability. Automated evaluations showed ChatGPT-3.5 had a ROUGE-1 score of 21.44 and BERTScore F1 of 83.14, whereas UMLS-augmented LLaMa2-13b-chat achieved a ROUGE-1 score of 19.97 and BERTScore F1 of 82.08. These results highlight the potential of RAG-based medical LLMs in providing trustworthy and explainable medical QA.

LLM-AMT [98]: Enhanced biomedical question answering by integrating authoritative medical textbooks into a retrieval-augmented generation (RAG) framework. The goal is to address the limitations of general-purpose LLMs, which often lack specialized medical knowledge and may generate plausible but incorrect responses. The model employs a hybrid retriever (HybTextR) combining sparse and dense retrieval methods, alongside a query augments and knowledge self-refiner, to ensure high-quality, domain-specific information retrieval. This approach aims to improve accuracy and reduce hallucinations in medical QA tasks by leveraging structured, reliable textbook content rather than general knowledge bases like Wikipedia.

Key technical components include a query augments for medical terminology standardization, a hybrid retriever for precise passage retrieval, and a knowledge self-refiner to filter irrelevant or redundant information. The model uses GPT-4-Turbo as the base LLM, augmented with retrieved textbook passages to generate contextually accurate answers. Evaluations on MedQA-USMLE, MedQA-MCMLE, and MedMCQA datasets show LLM-AMT achieving 67.9% to 72.6% accuracy, outperforming GPT-3.5 by 11.6%–16.6% and even surpassing specialized models like Med-PaLM 2 by 2%–3%. Human evaluations confirm a 16% reduction in hallucinations, demonstrating improved reliability in open-ended medical QA. The success highlights the importance of domain-specific retrieval and refinement in enhancing LLM performance for clinical applications.

TrialGPT [99]: framework addresses the challenge of patient-to-clinical-trial matching by leveraging large language models (LLMs) to streamline the recruitment process. The goal is to overcome the inefficiencies of manual matching, which is labor-intensive and prone to errors, by automating the retrieval, analysis, and ranking of clinical trials for individual patients. The framework focuses on enhancing scalability, accuracy, and explainability in matching patients to trials, particularly in scenarios involving large trial databases and complex eligibility criteria, and reduced patient screening times for clinical trials from hours to seconds by employing FAISS [100] for optimized retrieval and LangChain's recursive chunking to preserve context.

Key technical components of TrialGPT include a hybrid retrieval system combining BM25 and MedCPT for initial trial filtering, a criterion-level matching module that generates natural language explanations and eligibility predictions, and a ranking module that aggregates results for final recommendations. The framework utilizes GPT-4 and GPT-3.5 as backbone models, employing chain-of-thought prompting to ensure interpretable outputs. Evaluations on synthetic patient cohorts demonstrate that TrialGPT achieves 87.3% accuracy in criterion-level matching, closely approximating expert performance, and reduces screening time by 42.6% in a user study.

The model also outperforms baseline methods by 43.8% in ranking and excluding trials, highlighting its potential to improve efficiency and precision in clinical trial recruitment.

The success of TrialGPT underscores the value of LLMs in handling domain-specific tasks like patient-trial matching, where explainability and scalability are critical. By integrating retrieval-augmented generation (RAG) with LLMs, the framework provides a robust solution for automating complex medical workflows while maintaining transparency. Future work could explore fine-tuning open-source LLMs or expanding the framework to incorporate multimodal patient data for broader applicability.

Manathunga, Supun S and Illangasekara, et al. [101]: Applying (RAG) and Representative Vector Summarization (RVS) to medical education, specifically targeting challenges in processing large unstructured texts like clinical references. The docGPT model integrates a nonparametric knowledge base, using vector embeddings to enhance retrieval accuracy and reduce hallucinations in domain-specific tasks. Key medical texts such as Kumar and Clark's Clinical Medicine and the British National Formulary serve as primary datasets, enabling precise information extraction and summarization for clinicians and students.

The motivation stems from limitations in generic Large Language Models (LLMs), which often produce inaccurate or generic responses in specialized domains. By combining OpenAI's text-embedding-ada-002 for embeddings and FAISS for efficient similarity search, docGPT ensures contextually relevant retrievals. For summarization, k-means clustering identifies representative text chunks, while t-SNE visualizations and keyword mapping streamline content overview. This approach addresses token limits and the "lost in the middle" problem in LLMs, optimizing both extractive and abstractive summarization workflows.

The LLM benefits from RAG by dynamically accessing up-to-date external knowledge, improving response specificity and reducing reliance on parametric memory. docGPT's pipeline—comprising OCR, recursive text splitting, and hybrid summarization—demonstrates adaptability across document types, including PDFs and scanned texts. Manual evaluations confirm superior performance over generic models like ChatGPT, with docGPT delivering more targeted answers aligned with reference materials. Visual tools like word clouds and scatter plots further enhance interpretability of document structures.

Evaluation results, though manually verified, highlight docGPT's effectiveness in retrieval and summarization tasks. For instance, processing Kumar and Clark's Clinical Medicine (1,508 pages) yielded 19 representative chunks within a 15,000-token limit, while the British National Formulary (1,805 pages) required only 10 chunks under 5000 tokens. The absence of quantitative metrics like ROUGE or BLEU underscores the need for future automated benchmarking, but the manual assessments validate the model's utility in medical education and its potential for broader domain adaptation.

Ye, Cheng [102]: Introduces a few-shot, user-adaptive learning-to-rank layer within a RAG-based search system tailored to clinical electronic medical records (EMRs). Designed for chart review tasks such as querying diabetes or seizure histories, the system begins from as few as ten user-labeled positive examples to learn sentence-level relevance signals and adjust term similarities via a fine-tuned biomedical embedding (e.g. Bio-BERT). Rather than returning full documents, it ranks all candidate EMR snippets and selects only the highest-scoring excerpts to feed into a medical LLM (e.g. Google Gemini) for response generation. That modest manual effort yields rapid personalization to a clinician's query style and produces tightly grounded RAG contexts—supporting higher-precision responses and mitigating hallucinations.

The system combines dense retrieval over FAISS-indexed medical embeddings, a learning-to-rank model trained on relevance judgments (pairwise or listwise ranking), and a standard RAG generation stage. After training with only ten positive labels, Cheng-RAG-Rank-EMR reached Precision at 10 (± 0.60) on EMR relevance metrics and outperformed baseline retrieval methods, even when baselines had access to far more label data. Learned ranking preferences generalize: on unseen EMR collections involving cancer progression diagnosis, accuracy rose dramatically from approximately 0.14 to 0.50 in RAG-based question answering. Access to just the top-ranked passages enables the LLM to synthesize responses that are both succinct and clinically relevant—supporting real-world tasks such as chart abstraction, decision-support prompts, and clinical reasoning reviews.

MMed-RAG, [103]: A versatile multimodal retrieval-augmented generation (RAG) system designed to enhance the factuality of Medical Large Vision-Language Models (Med-LVLMs). Addressing critical issues like factual hallucination and misalignment in Med-LVLMs, MMed-RAG integrates a domain-aware retrieval mechanism, adaptive context selection, and RAG-based preference fine-tuning to improve cross-modality and overall alignment with ground truth. The system dynamically selects domain-specific retrievers for medical images (e.g., radiology, pathology, ophthalmology), filters low-quality retrieved contexts using similarity scores, and employs preference optimization to balance reliance on retrieved knowledge and intrinsic model knowledge.

MMed-RAG leverages LLaVA-Med-1.5 as the backbone model, combining contrastive learning for retriever training and direct preference optimization (DPO) for fine-tuning. The domain-aware retrieval mechanism uses BiomedCLIP to classify input images and retrieve relevant knowledge, while adaptive selection mitigates noise by truncating low-similarity contexts. Preference pairs are curated to penalize over-reliance on retrieval or ignoring visual inputs, enhancing alignment. Evaluated across five medical datasets (MIMIC-CXR, IU-Xray, Harvard-FairVLMed, PMC-OA, Quilt-1 M), MMed-RAG achieves 43.8% higher factual accuracy on average, with improvements of 18.5% in medical VQA (accuracy/F1/AUC) and 69.1% in report generation (BLEU/ROUGE-L/METEOR) compared to baseline Med-LVLMs. The system outperforms decoding-based (e.g., DoLa, VCD) and RAG-based (e.g., MedDr, RULE) methods, demonstrating robustness across diverse medical modalities. Key innovations include provable alignment gains via theoretical analysis and empirical validation of reduced misalignment rates (e.g., 28.19% copy-reference rate vs. 55.08% in baselines).

Upadhyay and Rishabh and Viviani, [104]: Introduced a Retrieval-Augmented Generation (RAG)-driven model for Health Information Retrieval (HIR), focusing on enhancing both topical relevance and factual accuracy in retrieved medical documents. The exponential growth of online health information and the risks of misinformation motivate this work, aiming to provide non-experts with reliable, scientifically grounded health information. The proposed model leverages generative Large Language Models (LLMs) like GPT-3, Llama 3, and Falcon-40B, combined with trusted external sources such as PubMed Central (PMC), to generate contextually rich and citation-backed responses (GenText). This approach mitigates issues like factual inconsistency and hallucination in LLMs by grounding responses in verified scientific literature.

The model operates in three stages: retrieving topically relevant passages from PMC, generating GenText using LLMs, and ranking documents based on combined topicality and factual accuracy scores. Factual accuracy is assessed through stance detection using the SCIFIVE model and semantic similarity metrics. The integration of RAG with LLMs ensures responses are both relevant and factually accurate, while GenText enhances explainability by providing references.

Achieves notable retrieval accuracy, particularly in top ranked results. On the CLEF eHealth 2020 dataset, the model (specifically the LlamaRAG configuration) attains a Credibility-Aware Mean Average Precision (CAM-MAP) of 0.1079 and Credibility-Aware Normalized Discounted Cumulative Gain (CAM-NDCG) of 0.2146 for top-5 documents, outperforming baselines like BM25 (CAM-MAP: 0.0431) and WISE-NLI (CAM-MAP: 0.0883). For top-10 documents, LlamaRAG further improves with a CAM-MAP of 0.1532 and CAM-NDCG of 0.2702, demonstrating robust scalability. Similar trends are observed on the TREC Health Misinformation 2020 dataset, where LlamaRAG achieves CAM-MAP of 0.1222 (top-5) and 0.1602 (top-10), surpassing traditional IR models. The evaluation highlights the model's effectiveness in combining BM25 [105] for topical relevance with LLM-augmented factual accuracy scoring (via stance detection and cosine similarity). The integration of GenText—generated by LLMs using retrieved PMC passages—ensures explainability while maintaining high precision.

Yu He Ke1, Liyuan Jin and Kabilan Elangovan, [106]: Evaluated a Retrieval-Augmented Generation (RAG) pipeline for preoperative medical assessments, leveraging ten LLMs (GPT-3.5, GPT-4, GPT-4o, Llama2, Llama3, Gemini, Claude) to enhance accuracy and consistency in surgical fitness evaluations. The goal is to address challenges like surgery cancellations due to medical unfitness by integrating domain-specific guidelines (35 local and 23 international) into LLM responses, reducing reliance on pretrained knowledge and minimizing hallucinations. The GPT-4 RAG model with international guidelines achieves 96.4% accuracy in determining

surgical fitness, outperforming human evaluators (86.6%) and non-RAG counterparts (92.9%). Key technologies include Llamaindex for hierarchical retrieval, BioBERT for medical context, and S.C.O.R.E. (Safety, Consensus, Objectivity, Reproducibility, Explainability) for qualitative evaluation.

The RAG pipeline retrieves and processes guideline chunks, using similarity-top-k = 30 to balance relevance and computational efficiency. GPT-4 generates responses with low hallucination rates (0–2.9%), while Llama 2 exhibits higher rates (12.8–48.6%). The model excels in consistency (inter-rater reliability: 0.93 vs. human 0.86) and safety (4.93/5 S.C.O.R.E. score). Manual evaluation confirms superiority in preoperative instructions (e.g., 93.0% accuracy for nurse/doctor triage). However, local guidelines underperform due to sparse textual data, highlighting the need for structured medical corpora. The framework demonstrates LLM-RAG's potential to standardize clinical workflows, though human oversight remains critical for nuanced decisions.

Pingua, Bhagyajit and Sahoo [107]: Explored the adaptation of large language models (LLMs) for medical question-answering by comparing fine-tuning (FT), retrieval-augmented generation (RAG), and a hybrid FT+RAG approach. Five models—Llama–3.1-8B, Gemma-2-9B, Mistral-7B-Instruct, Qwen2.5-7B, and Phi–3.5-Mini-Instruct—are fine-tuned on the MedQuAD dataset, which contains 47,457 medical question-answer pairs covering symptoms, treatments, and diseases. The goal is to enhance LLMs' accuracy and contextual relevance in healthcare applications, addressing the challenge of delivering reliable medical information. Techniques like 4-bit quantization, LoRA for parameter-efficient fine-tuning, and ChromaDB for retrieval augmentation are employed to optimize performance and computational efficiency.

The hybrid FT+RAG approach demonstrates superior performance, particularly for Llama–3.1-8B and Phi–3.5-Mini-Instruct, achieving higher BLEU (0.163 for RAG, 0.136 for FT+RAG) and BERTScore F1 (0.891 for FT+RAG) compared to standalone FT. RAG alone also outperforms FT in most metrics, leveraging external knowledge retrieval to improve response quality. Evaluation includes lexical metrics (BLEU, ROUGE) and semantic measures (BERTScore, SBERT cosine similarity), with Llama–3.1-8B showing the best overall results. The study highlights the trade-offs between methods, noting that RAG excels in dynamic knowledge integration, while FT+RAG combines domain adaptation with retrieval benefits. Manual evaluation is not mentioned; results are based on automated metrics, with statistical significance confirmed via bootstrapping and Wilcoxon tests.

Rationale-Guided Retrieval Augmented Generation RAG^2 [108]: An advanced framework designed to enhance medical question-answering (QA) by addressing key limitations of traditional RAG methods. The framework tackles three core challenges: irrelevant context retrieval, poorly targeted medical queries, and retriever bias toward dominant corpora. To achieve this, RAG^2 integrates three innovations: (1) a perplexity-based filtering model (trained on Flan-T5) to selectively retain informative document snippets, (2) LLM-generated rationales (using chain-of-thought prompting) as refined queries to improve retrieval precision, and (3) balanced retrieval from four diverse biomedical corpora (PubMed, PMC, textbooks, clinical guidelines) to mitigate bias. The goal is to improve the reliability and accuracy of LLMs in high-stakes medical applications by dynamically combining parametric knowledge with curated external evidence.

RAG^2 leverages state-of-the-art LLMs like Llama-3-8B-Instruct, Meerkat-7B, and GPT-4o, enhanced with retrieval from MedCPT and reranking. The framework demonstrates significant performance gains, achieving 6.1%, 3.8%, and 0.9% average accuracy improvements on MedQA, MedMCQA, and MMLU-Med benchmarks, respectively. Key results include a 91.1% accuracy on MedQA with GPT-4o and 5.6% superiority over MedRAG. Automated metrics (ROUGE-L, BERTScore) confirm semantic coherence, while ablation studies validate the necessity of rationale-guided filtering and balanced retrieval. The approach avoids costly iterative processes, relying instead on single-pass generation and a lightweight filtering model, making it computationally efficient. By refining query formulation, retrieval, and filtering, RAG^2 advances the practical deployment of LLMs in clinical decision-making, ensuring responses are both contextually relevant and grounded in authoritative medical knowledge.

RAGEv [109]: Developed to process and synthesize healthcare documents and scientific publications using a Retrieval-Augmented Generation (RAG) framework integrated with Large Language Models (LLMs). In the medical domain, RAGEv aims to support clinicians, policymakers, and researchers by addressing the challenge of

information overload and delivering concise, trustworthy responses to domain-specific queries. The system processes a wide range of document types, including PubMed articles and policy reports, focusing on enhancing the factual consistency and usability of generated outputs in real-world decision-making and policy support.

The motivation behind RAGEv is to mitigate common issues in LLM applications for healthcare, such as hallucinations, lack of explainability, and poor factual grounding. The primary objective is to establish a reference implementation that ensures reliable, context-aware answer generation for complex medical and scientific content. To support this, the study introduces RAGEv-Bench, a benchmark suite comprising both public datasets like PubMedQA and internal use cases such as Horizon Europe project outputs, Virtual Human Twin research, and Antimicrobial Resistance reviews. These datasets enable robust performance evaluation under both automatic and manual testing scenarios, reflecting realistic use conditions.

Technically, RAGEv leverages hybrid retrieval pipelines including SHy, ColBERTv2, and full-text/vector search with reranking. The SHy pipeline demonstrates strong performance by scanning each document in isolation to maximize context coverage. Embedding models such as E5-Mistral and SFR-Embedding-Mistral are employed to encode documents and queries into high-dimensional vector space. For generation, the system uses GPT-based LLMs hosted in a secure infrastructure (GPT@JRC), which receive prompt-augmented queries grounded in retrieved document segments. Evaluation metrics include classification-based scores (accuracy, precision, recall, F1), generative scores (ROUGE, BERTScore), and manual usability assessments across question types.

Evaluation results show that integrating LLMs with RAG pipelines significantly improves both short-form (yes/no) and long-form answer quality. The SHy pipeline achieves precision of 0.85 and BERTScore F1 of 0.83 on PubMedQA, outperforming non-RAG baselines. Manual usability evaluations further confirm the system's effectiveness, with most answers rated correct or near-complete across three use cases. The combination of document-specific retrieval and controlled generation enables the LLM to provide domain-relevant, verifiable outputs, thereby aligning with the model's core goal of producing safe and high-quality responses in the health sector.

CliniqIR [110]: A retrieval-based diagnostic decision support framework designed to address the challenge of diagnosing rare and complex medical conditions with limited training data. The framework leverages unstructured knowledge resources, including clinical text records, the Unified Medical Language System (UMLS) Metathesaurus, and 33 million PubMed abstracts, to classify a broad spectrum of diagnoses. The goal is to improve diagnostic accuracy for infrequently encountered conditions by combining information retrieval (IR) techniques with machine learning, reducing reliance on large labeled datasets.

CliniqIR integrates sparse (BM25) and dense (MedCPT) retrieval methods to match patient notes with relevant medical literature, generating ranked lists of potential diagnoses. The framework employs QuickUMLS for medical concept extraction and term frequency-inverse document frequency (TF-IDF) for ranking. To enhance performance, CliniqIR is combined with supervised fine-tuned ClinicalBERT in an ensemble model using reciprocal rank fusion (RRF), achieving state-of-the-art results. The model is evaluated on the DC3 dataset for rare diseases and MIMIC-III for imbalanced data, demonstrating superior performance in zero-shot settings with a mean reciprocal rank (MRR) of 0.35 for CliniqIR-BM25 compared to 0.15 for ClinicalBERT. For diagnoses with fewer than five training samples, CliniqIR outperforms ClinicalBERT by an average MRR difference of 0.10. The ensemble framework further improves diagnostic accuracy, particularly for long-tailed data distributions.

The study highlights the potential of retrieval-augmented approaches in clinical decision support, offering transparency and scalability. CliniqIR's ability to leverage external knowledge without extensive training data addresses a critical gap in diagnosing rare conditions. However, limitations include reliance on PubMed's coverage biases and the need for broader medical literature integration. The results underscore the synergy between IR and supervised learning, paving the way for robust diagnostic tools in low-resource settings.

Benfenati, Domenico and De Filippis [111]: Studied (RAG) in the nutrigenetics domain to enhance question-answering accuracy for personalized nutrition. Nutrigenetics involves complex interactions between genetic variations and dietary responses, requiring up-to-date and precise information. Traditional AI models struggle

with this due to data volume and rapid advancements. The proposed RAG framework integrates GPT-3.5 and Mistral-7B with a vector database (Milvus) and embeddings model (GTE) to retrieve and contextualize scientific literature, mitigating hallucinations and reducing computational costs compared to fine-tuning.

The implementation leverages a curated dataset of nutrigenetic studies, focusing on subdomains like food intolerances and xenobiotics metabolism. GPT-3.5 and Mistral-7B generate responses augmented by retrieved abstracts, improving relevance and specificity. Manual evaluation by domain experts assesses four metrics: relevance, depth, evidence support, and accuracy. Results show RAG-enhanced responses outperform standard outputs, with GPT-3.5 achieving higher scores (e.g., accuracy improving from 4 to 5 on a 5-point scale for many queries). Mistral-7B also benefits, though variability exists. The framework demonstrates potential for reliable, real-time nutrigenetic advice but highlights challenges in retrieval precision and dataset quality. Future work may expand to multimedia data and temporal analysis for broader clinical applicability.

Markey, Nigel and El-Mansouri [112]: Investigated the use of GPT-4 for generating clinical trial protocol documents, specifically focusing on endpoints and eligibility criteria sections. Recognizing the limitations of off-the-shelf large language models in clinical applications - particularly regarding regulatory compliance and factual accuracy the research implements a retrieval-augmented generation (RAG) system to enhance model performance. The RAG framework integrates authoritative medical knowledge from ClinicalTrials.gov, FDA regulatory documents, and scientific literature through vector database retrieval, creating a hybrid system that combines generative capabilities with verified external knowledge.

The evaluation methodology employs a comprehensive assessment framework examining four critical dimensions: clinical thinking and logic, transparency and references, medical terminology accuracy, and content relevance. Results demonstrate significant improvements when using RAG augmentation, with clinical thinking scores increasing from 41.4% to 79.7% and transparency (previously unmeasured) reaching 78.6%, while maintaining excellent terminology accuracy ($> 99\%$) and content relevance (80%). Manual analysis of specific cases, such as tuberculosis trial protocols, confirms the system's ability to correctly incorporate regulatory guidance that off-the-shelf models often miss. These findings suggest RAG-enhanced LLMs can substantially accelerate clinical document preparation while improving quality, though the study notes remaining challenges in ensuring inter-section consistency and the need for continued human oversight in critical medical applications.

Mashatian, Shayan and Armstrong [113]: Presented a (RAG) model designed to provide accurate and comprehensible medical knowledge about diabetes and diabetic foot care for laypersons with an eighth-grade literacy level. The primary goal is to address the risks of incorrect information generated by general large language models (LLMs) by leveraging a validated knowledge base. The model aims to improve health literacy, support self-management, and reduce the administrative burden on physicians, ultimately preventing limb loss in diabetic patients. The study emphasizes the importance of trustworthy AI in delivering reliable medical information to the public.

The RAG model combines GPT-4 with a vector database (Pinecone) to retrieve and generate responses based on a curated knowledge base derived from the NIH National Standards for Diabetes Self-Management Education. The dataset includes 295 articles selected using 58 keywords, and the model was tested against 175 questions. Few-shot learning prompts were employed to enhance answer quality, ensuring simplicity and accuracy. The validation process involved expert reviews focusing on factuality, comprehension, reasoning, potential harm, and bias. The model demonstrated significant improvements when few-shot learning was applied, resolving issues related to factuality and comprehension errors.

Evaluation results showed the model achieved 98% accuracy, recall, and F1-score, with 99% precision and specificity after adjustments for ambiguous questions. Manual evaluation by medical experts confirmed the model's ability to provide user-friendly and accurate answers. The study highlights the effectiveness of combining RAG architecture with few-shot learning to address the limitations of general LLMs in the medical domain. Future work includes exploring optimal content volume, linguistic nuances, and integrating wearable devices for personalized responses. The findings underscore the potential of RAG models to enhance patient education and outcomes in diabetes care.

Painter, Jeffery L and Chalamalasetti [114]: Presented an LLM-based approach to automate pharmacovigilance data retrieval by converting natural language queries into accurate SQL commands. The work specifically targets the challenges of querying complex drug safety databases containing over 500 tables and 1.3 billion records, where traditional methods require specialized technical expertise. The primary goal involves enabling non-technical medical professionals to directly access critical safety data while maintaining high accuracy standards required in regulated healthcare environments.

The implementation utilizes GPT-4 within a RAG framework enhanced with a domain-specific business context document. Key technical components include OpenAI's text-embedding-ada-002 for vector retrieval, LangChain for pipeline orchestration, and a phased evaluation methodology. The business context document provides crucial domain knowledge about table relationships, query patterns, and regulatory requirements, significantly improving the system's ability to generate syntactically correct and semantically appropriate SQL queries. This approach demonstrates how structured domain knowledge can be effectively integrated with general-purpose LLMs for specialized medical applications.

Manual evaluation across three experimental phases shows substantial performance improvements when incorporating contextual knowledge. Initial tests using only database schema achieved 8.3% accuracy, while adding the business context document boosted accuracy to 78.3%, with 85% success for low-to-medium complexity queries. The evaluation methodology categorized results as pass/fail/partial-pass, with statistical analysis confirming the significance of context augmentation ($p=0.0006$). These results highlight the potential of context-enhanced LLMs to bridge the gap between natural language interfaces and complex medical databases while maintaining the precision required in pharmacovigilance applications.

Caregiving Language Model (CaLM) [115]: A (RAG) and fine-tuned foundation models (FMs) to provide reliable, domain-specific support for caregivers of individuals with Alzheimer's disease and related dementias (ADRD). The goal is to address the lack of accessible, accurate information for untrained family caregivers by mitigating hallucinations in large language models (LLMs) through a curated knowledge base and fine-tuning. The study emphasizes improving health literacy and reducing caregiver stress by delivering contextually grounded, verifiable answers.

The CaLM integrates RAG with two small FMs (LLaMA 2 7B and Falcon 7B) and benchmarks against GPT-3.5. Key technical components include a caregiving knowledge base (8,568 documents chunked into 196,926 segments), BAAI general embedding (BGE) for retrieval, and Chroma DB for vector storage. Fine-tuning employs LoRA and QLoRA techniques to adapt FMs to caregiving-specific Q-A pairs (581 curated examples), ensuring parameter efficiency. The RAG framework appends retrieved context to user queries, enhancing answer accuracy and enabling reference citations—critical for trust in medical applications.

Evaluation on 66 test Q-A pairs demonstrates that RAG + fine-tuned LLaMA 2 7B outperforms vanilla GPT-3.5 across metrics: BLEU (0.45), ROUGE (0.52), CHR-F (85.1), and BERT score (0.91). Notably, the fine-tuned model provides references for 100% of answers (80% accuracy), surpassing GPT-3.5 (70% coverage, 62% accuracy). Manual validation confirms the small FM's superiority in reliability and accessibility, achieving 85% accuracy for low-to-medium complexity queries. These results highlight the potential of domain-adapted RAG systems to democratize high-quality caregiving support while minimizing computational costs.

GuideGPT [116]: A context-aware chatbot using GPT-4 and retrieval-augmented generation (RAG) to provide evidence-based answers on medication-related osteonecrosis of the jaw (MRONJ). The goal is to address limitations of generic large language models (LLMs) like hallucination and outdated knowledge by grounding responses in 449 scientific publications from clinical guidelines. The approach aims to enhance clinical decision-making with traceable, up-to-date information while reducing manual guideline development workloads.

Key technical components include Llama-Index for document processing, OpenAI's text-embedding-ada-002 for semantic retrieval, and a custom prompt template enforcing clinical guideline structure. The RAG pipeline retrieves top 20 relevant sentences with surrounding context, while GPT-4 (gpt-4-0613, temperature=0.4) generates concise answers and explanations. Metadata linking to source PDFs enables verification, addressing transparency gaps in generic LLMs.

Manual evaluation by 10 MRONJ experts on 30 questions shows GuideGPT significantly outperforms PureGPT (generic GPT-4) in content ($p=0.006$), scientific explanation ($p=0.032$), and agreement ($p=0.008$), with 80.67% vs. 69.67% high-content ratings. Language quality remains comparable ($p=0.407$). The study demonstrates RAG's ability to improve LLM reliability in specialized medical domains while maintaining accessibility through open-source code and dynamic knowledge updates.

5.6.3 Modular retrieval-augmented generation (RAG)

Representative implementations of Advanced RAG in the medical domain:

Med-R² [117]: Designed to enhance Large Language Models (LLMs) in medical scenarios through Retrieval-Augmented Generation (RAG) aligned with Evidence-Based Medicine (EBM) principles. The framework addresses key challenges in medical applications, including inefficient knowledge acquisition, limited retrieval precision, and low effectiveness in answer extraction. *Med-R²* integrates a comprehensive external knowledge base and employs a multi-stage process: (1) query formulation, where queries are classified and reformulated based on EBM categories; (2) evidence retrieval and reranking, which uses a coarse-to-fine strategy to filter and prioritize documents based on credibility, usefulness, and relevance; and (3) evidence application, where chain-of-thought (CoT) reasoning is generated to refine answers. This approach ensures the retrieval of high-quality, contextually relevant medical evidence.

Med-R² demonstrates significant improvements over baseline methods. Experiments on five medical datasets (MedQA-USMLE, MedQA-MCMLE, MedMCQA, PubMedQA, and MMLU-Med) show that *Med-R²* achieves a 14.87% average improvement over vanilla RAG and a 3.59% enhancement compared to fine-tuned models, without additional training costs. Notably, lightweight models like Qwen2.5-7B [118] and LLaMA3.1-8B [119] exhibit performance gains of up to 79.72% and 77.80%, respectively, highlighting the framework's ability to compensate for limited inherent medical knowledge. The results also reveal superior generalization in cross-dataset evaluations, outperforming fine-tuning by 7.02%, underscoring the robustness of retrieval-augmented methods in heterogeneous medical contexts.

Ablation studies further validate the contributions of *Med-R²*'s components, with the evidence reranker identified as the most impactful module. The framework's scalability is also examined, showing optimal performance at context window lengths proportional to model size (e.g., 4 K for 8B models, 16 K for 32B/70B models). Limitations include potential gaps in the medical knowledge corpus and reliance on classifier accuracy. Overall, *Med-R²* presents a cost-effective, scalable solution for enhancing LLMs in medical applications, combining retrieval precision with EBM-guided reasoning to improve clinical decision-making.

MedGraphRAG [120]: A novel framework known as Medical GraphRAG (MedGraphRAG), which leverages a graph-based retrieval-augmented generation (RAG) approach tailored specifically for the medical domain. MedGraphRAG enhances the performance of LLMs by integrating evidence-based responses and precise medical terminology explanations, improving both credibility and response quality. The method employs a sophisticated graph construction technique—Triple Graph Construction—to generate meaningful connections between user queries, authoritative medical sources, and foundational medical definitions. This facilitates the creation of evidence-based triples that underpin the LLM's reasoning, ensuring traceability to credible sources while maintaining clarity in explanations. Developed a unique retrieval strategy, U-Retrieval, which streamlines the process of integrating relevant information from extensive datasets without incurring high computational costs. By organizing the graph using a hierarchical tagging system, U-Retrieval efficiently identifies and synthesizes relevant data, balancing global context awareness with retrieval efficiency. evaluation was done a cross various medical QA benchmarks demonstrate that MedGraphRAG not only surpasses traditional RAG and GraphRAG methods but also establishes new state-of-the-art performance levels in medical LLM applications. Through comprehensive testing and human evaluations, findings that MedGraphRAG generates more reliable, evidence-based responses, solidifying its potential for enhancing the integration of LLMs in clinical practice and research, was tested on 9 medical QA benchmarks, 2 health fact-checking datasets, and a long-form generation test set,

demonstrating state-of-the-art (SOTA) performance. It outperformed standard RAG by 8% in medical QA and 10% in fact-checking, surpassing even GraphRAG by 5% and 8%, respectively. In direct comparison with fine-tuned medical LLMs like Med-PaLM 2 and Med-Gemini, MedGraphRAG achieved higher accuracy across all benchmarks. Human evaluation by clinicians further validated its superiority, with higher ratings in correctness, citation precision, and understandability, proving its ability to generate trustworthy, well-referenced medical responses.

FLARE (Forward-Looking Active Retrieval-Augmented Generation) [22]: Designed to enhance the factual accuracy of language models by integrating a dynamic retrieval process. Unlike traditional retrieval-augmented models that retrieve information only once based on the initial input, FLARE iteratively generates a “next sentence” prediction during text generation. When the language model encounters tokens with low probability, FLARE retrieves relevant documents to improve confidence in the generated content. This approach enables FLARE to actively decide both the timing and content of retrievals based on anticipated informational needs, rather than relying on a fixed retrieval interval or static input.

FLARE employs a two-step process: it first generates a tentative next sentence and assesses token confidence. If any token exhibits low confidence, the sentence is used as a query to retrieve pertinent documents, which then guide the regeneration of that part of the text. This forward-looking technique allows FLARE to address knowledge gaps only when needed, making it a more efficient alternative to passive multi-time retrieval systems. By focusing on dynamic, content-aware retrievals, FLARE ensures that the generated responses are both relevant and supported by up-to-date information.

Evaluations on knowledge-intensive tasks, such as multi hop question answering and long-form summarization, demonstrate that FLARE outperforms existing single and multi-time retrieval models, showing superior or competitive results across all tasks. Its adaptability across diverse applications suggests that FLARE is well-suited for complex, information-rich text generation, where continuous and selective information retrieval is essential. For the WikiMultihopQA dataset, FLARE direct achieved an F1-score of 59.7%, precision of 59.1%, and recall of 62.6%, outperforming other baselines. Similarly, for the ASQA-hint dataset, FLARE showed an EM score of 46.2% and Disambig-F1 of 36.7%. Across all datasets, FLARE demonstrated superior performance in retrieval-enhanced question answering and summarization tasks.

SELF-RAG (Self-Reflective Retrieval-Augmented Generation) [23]: Advances retrieval-augmented generation by introducing a self-reflective approach that adaptively retrieves relevant information and critiques its own outputs to improve factual accuracy and relevance. Unlike traditional RAG models, which retrieve a fixed number of passages and may incorporate irrelevant information, SELF-RAG employs “reflection tokens” that allow the model to decide when retrieval is necessary and assess the quality of its output at each generation step. These tokens signal the need for retrieval, relevance, and support from retrieved passages, enabling the model to dynamically adjust its retrieval actions and generate well-supported responses.

The model’s architecture includes a generator that integrates reflection tokens to signal retrieval and a critic component that assesses retrieved passages for their relevance and support. This setup enables the model to generate content in segments, each evaluated and refined based on feedback from reflection tokens. SELF-RAG’s self-assessment mechanism ensures that retrieved information is only incorporated when relevant, enhancing response quality while optimizing retrieval efficiency. Additionally, the model’s inference process allows for customization, such as adjusting retrieval frequency and imposing constraints to emphasize accuracy or fluency based on task requirements.

SELF-RAG demonstrates robust performance improvements across multiple tasks. For short-form tasks, surpasses state-of-the-art retrieval-augmented models and LLMs in knowledge-intensive tasks, such as fact verification and long-form QA. The model achieves high accuracies of 54.9% on PopQA, 66.4% on TQA, and 72.4% on PubQA, while the 13B variant scores 55.8%, 69.3%, and 74.5%, respectively. In the closed-set ARC-Challenge, accuracies reach 67.3% for the 7B model and 73.1% for the 13B model, and in biography generation, F1-Score values of 81.2 (7B) and 80.2 (13B) highlight improved factuality. Moreover, on the long-form ASQA

task, citation precision and recall are 66.9/67.8 for the 7B model and 70.3/71.3 for the 13B model, indicating that SELF-RAG effectively enhances evidence-backed generation quality.

MedRAG [121]: Designed a novel Retrieval-Augmented Generation (RAG) framework for medical diagnosis by integrating knowledge graph (KG)-elicited reasoning. MedRAG addresses the limitations of heuristic-based RAG models in distinguishing between diseases with similar manifestations. The system systematically constructs a four-tier hierarchical diagnostic KG to provide accurate diagnostic reasoning and dynamically retrieves relevant Electronic Health Records (EHRs). It is evaluated on both the public DDXPlus dataset and a private chronic pain diagnostic dataset (CPDD), demonstrating improved diagnostic accuracy and specificity compared to existing RAG models. MedRAG's framework consists of a document retriever and a KG-elicited reasoning module that enhances decision-making through structured medical knowledge. The retriever fetches relevant patient EHRs based on semantic similarity, and the reasoning module integrates diagnostic differences extracted from the KG to refine diagnosis and suggest follow-up questions. Experimental results indicate that MedRAG outperforms state-of-the-art RAG models, achieving superior performance in accuracy (79.25% on CPDD, 88.65% on DDXPlus) and specificity (L3 metric of 66.04% on CPDD, 68.01% on DDXPlus). Additionally, the system exhibits strong adaptability across multiple language models, including Mixtral-8x7B, Llama-3.1-Instruct, and GPT-4o. The findings validate MedRAG as a robust diagnostic assistant, improving reasoning-driven medical decision-making. The integration of KG significantly enhances precision by addressing diagnostic ambiguities. Future work will explore multimodal data integration, including medical imaging and physiological signals, to further refine diagnostic capabilities.

Retrieval-Augmented Generation-Based Physician Recommendation (RAGPR) [122]: A Rag model to enhance personalized physician recommendations in web-based medical services. The goal is to address scalability and accuracy limitations in manual triage systems by automating patient-physician matching. The RAGPR model leverages a dataset of 646,383 consultation records, employing embedding models (FastText, SBERT, OpenAI) for semantic clustering and LLMs (Mistral, GPT-4o-mini, GPT-4o) for response generation. Key technologies include Sentence-BERT for document embeddings, Chroma for vector storage, and Maximal Marginal Relevance (MMR) for diverse retrieval, ensuring precise and context-aware recommendations.

The RAGPR framework integrates retrieval-augmented generation (RAG) to dynamically link patient queries with physician specialties. SBERT and OpenAI embeddings achieve superior clustering (F1-scores of 95% and 96%, respectively), while Mistral and GPT-4o excel in recommendation tasks (F1-scores of 94% and 95%). The model's hybrid approach—combining semantic retrieval with LLM-generated responses—ensures accuracy and personalization. For instance, dermatological symptoms are correctly mapped to dermatologists, demonstrating clinical relevance. Manual evaluations by triage staff further validate the model, with GPT-4o receiving the highest rating (4.67/5) for recommendation appropriateness.

Evaluation results highlight the RAGPR model's effectiveness: SBERT [123] and GPT-4o outperform FastText and smaller LLMs in precision (95% vs. 46%) and recall (97% vs. 90%). The framework's modular design allows seamless integration with existing healthcare systems, though challenges like data privacy and bias mitigation persist. Future extensions could incorporate multimodal data for broader applications, such as chronic disease management. The study demonstrates that RAG-enhanced LLMs significantly improve triage efficiency, offering a scalable solution for web-based medical services.

Omni-RAG [124]: presented a multi-source Retrieval-Augmented Generation (RAG) framework designed to improve factual consistency in medical applications. The approach introduces MedOmniKB, a structured and multi-genre medical knowledge repository, incorporating books, clinical guidelines, research papers, medical wikis, and knowledge graphs. The study further develops a Source Planning Optimization (SPO) method that enhances multi-source utilization by dynamically selecting and integrating relevant knowledge sources for each query. Experimental results show that Omni-RAG outperforms existing RAG methods, achieving higher retrieval precision and medical reasoning accuracy. Omni-RAG employs an expert LLM for query planning, generating tailored queries for different medical knowledge sources. A smaller fine-tuned model is then trained via Direct Preference Optimization (DPO) to align retrieval with optimal medical knowledge selection. The system is

evaluated across 11 benchmark medical datasets, including MedQA, PubMedQA, SEER, and DDXPlus. It achieves a significant performance boost, with an average accuracy improvement of 8%, reaching 76.98% on MedQA, 60.20% on PubMedQA, 61.90% on SEER, and 52.40% on DDXPlus compared to traditional RAG models. The inclusion of diverse medical sources demonstrates the advantages of structured multi-source retrieval in reducing hallucinations and improving response reliability.

CLEAR [125]: Introduced A Clinical Entity Augmented Retrieval for Information Extraction in Electronic Health Records (CLEAR), a Retrieval-Augmented Generation (RAG) pipeline that improves clinical information extraction by leveraging named entity recognition (NER) for retrieval. Unlike conventional embedding-based RAG methods, CLEAR prioritizes retrieval based on clinical entities, enhancing efficiency and accuracy. The system was evaluated on 20,000 electronic health records (EHRs) across two datasets: Stanford Medication for Opioid Use Disorder (MOUD) and CheXpert [126]. Results demonstrate CLEAR's superiority over chunk embedding and full-note retrieval methods, achieving higher F1 scores and significantly reducing inference time and token usage. utilizes zero-shot NER with Flan-T5-XXL to identify clinical entities, followed by an entity filtering step using Bio+ClinicalBERT and GPT-4. The selected entities are then expanded using ontologies (UMLS) and LLM-generated synonyms before retrieving relevant text chunks from EHRs. The retrieved contexts are processed by large language models, including GPT-4, Mixtral, and Llama-3. Evaluation metrics include precision, recall, and F1-score, with CLEAR achieving an average F1-score of 90% on the Stanford MOUD dataset and 0.96 on CheXpert. CLEAR also reduces input tokens by 70%, model queries by 66%, and inference time by 72% compared to embedding-based RAG.

MS-RAG [127]: Introduced a Multi-Stage Retrieval-Augmented Generation (MS-RAG) framework that enhances retrieval precision and factual consistency by integrating dense and sparse retrieval with reinforcement learning optimization. Unlike conventional embedding-based RAG models, MS-RAG employs a hybrid retrieval pipeline that dynamically selects relevant context using neural retrievers and BM25. The system was evaluated on benchmark datasets, including Natural Questions (NQ) and TriviaQA, demonstrating superior retrieval precision and response coherence. MS-RAG achieves a 7.2% improvement in retrieval precision and a 5.8% increase in factual consistency scores, outperforming baseline RAG implementations.

MS-RAG utilizes a fine-tuned transformer model for generation, incorporating reinforcement learning to optimize retrieval-generation alignment. The retrieved contexts are processed by GPT-4 and Mixtral, ensuring high-fidelity responses with reduced hallucinations. Evaluation metrics include BLEU, ROUGE-L, and factual accuracy, with MS-RAG achieving a 4.1% gain in ROUGE-L and a 15% reduction in hallucinations compared to standard RAG models. Additionally, the framework reduces token usage and inference time, making it computationally efficient.

MS-RAG offers a flexible and scalable architecture that dynamically optimizes retrieval necessity and integrates reinforcement learning for improved retrieval precision and response quality. These advances contribute to a more reliable and efficient RAG system, with potential applications in high-stakes domains such as legal and medical text generation.

Alkhalaf, Mohammad and Yu, Ping and Yin [128]: Applied Llama 2 (13B) with Retrieval-Augmented Generation (RAG) to process clinical data from Australian Residential Aged Care Facilities, focusing on summarizing EHRs and extracting malnutrition risk factors. The methodology combines zero-shot prompting with RAG techniques using sentence-transformers for embeddings and Maximal Marginal Relevance (MMR) retrieval to enhance information extraction from 1424 progress notes for summarization and 1399 notes for risk factor analysis. The approach specifically addresses challenges in handling unstructured medical data while mitigating common LLM issues like hallucinations and token limitations.

Results demonstrate RAG's significant impact on structured summarization, improving accuracy from 93.25% to 99.25% by effectively integrating external knowledge and reducing hallucinations. However, for malnutrition risk factor extraction from clinical notes, RAG provided no accuracy improvement, maintaining 90% performance and suggesting task-dependent effectiveness. The system successfully optimized token usage in summarization tasks while highlighting persistent challenges in extracting implicitly stated information from clinical narratives.

The findings underscore RAG's potential for clinical applications, particularly in structured data extraction tasks, while revealing limitations in processing more subjective clinical information. The study provides valuable insights into implementing open-source LLMs in healthcare settings where data privacy is paramount, though it indicates the need for further refinement of retrieval strategies for complex clinical inference tasks. These results contribute to understanding how retrieval augmentation can enhance different types of clinical NLP applications.

Das, Sudeshna and Ge [129]: Proposed a two-layer Retrieval-Augmented Generation (RAG) framework designed for low-resource medical question answering (MQA) using Reddit data. The framework addresses challenges in extracting insights from large volumes of social media posts about emerging drug use (e.g., xylazine and ketamine) by clinicians. The goal is to provide accurate, timely, and computationally efficient answers without relying on high-resource infrastructure, ensuring equitable access in constrained settings.

The framework employs a modular architecture: the first layer generates individual summaries from retrieved Reddit posts using keyword-based retrieval (BM25F ranking), while the second layer synthesizes these into a final answer. Two LLMs are evaluated—GPT-4 and the quantized Nous-Hermes-2-7B-DPO—to demonstrate feasibility in low-resource environments. Key innovations include segmentation for context window constraints and a focus on open-source models to reduce hardware dependencies. Manual evaluation by 21 domain experts assesses coverage, coherence, relevance, length, and hallucination. Results show no significant difference between GPT-4 and Nous-Hermes-2-7B-DPO in median scores for relevance (3/3), coherence (5/5), or hallucination (0/1), though GPT-4 produces longer responses. The Coleman-Liau Index reveals GPT-4 summaries require higher readability levels (grade 16.6 vs. 12.1 for Nous-Hermes-2-7B-DPO). The framework's success highlights the potential of lightweight RAG systems for MQA, though limitations include data biases and narrow domain validation.

Duan, Yuchen and Zhou [130]: Presented a novel approach to enhancing Traditional Chinese Medicine (TCM) case analysis by integrating large language models (LLMs) with knowledge graphs (KGs). The study focuses on automating the extraction and structuring of key entities—such as symptoms, pathogenesis, treatment principles, and medications—from 679 clinical cases of Wang Zhongqi, a renowned TCM physician. The goal is to address challenges in TCM case studies, including unstructured data, inefficient retrieval, and complex information analysis, by leveraging modern AI techniques to improve accessibility and accuracy. The proposed system combines GLM-3-Turbo [131] for named entity recognition (NER) and GPT-3.5-Turbo for response generation, utilizing Retrieval-Augmented Generation (RAG) to query a Neo4j-based knowledge graph.

The methodology involves four stages: schema layer construction, knowledge extraction, fusion, and storage. GLM-3-Turbo achieves high precision (0.9462), recall (0.9778), and F1-score (0.9617) in NER, while GPT-3.5-Turbo excels in RAGAS evaluation, scoring 0.9375 in faithfulness, 0.9686 in answer relevance, and 0.9500 in context recall. Manual evaluation using the SUS framework highlights superior performance in safety (2.868) and usability (2.848), demonstrating the system's reliability and professionalism. The integration of LLMs with KGs mitigates hallucinations and enhances reasoning, providing a robust solution for TCM case retrieval and analysis. This approach not only modernizes TCM research but also sets a benchmark for interdisciplinary applications of AI in medical domains.

DR.KNOWS [132]: An advanced system integrating Unified Medical Language System (UMLS) knowledge graphs (KGs) with large language models (LLMs) to enhance diagnostic accuracy from electronic health records (EHRs). The goal is to mitigate diagnostic errors caused by information overload in EHR narratives by leveraging structured medical knowledge for context-aware reasoning. The system employs a stack graph isomorphism network (SGIN) for node embedding and an attention-based path ranker to retrieve and rank clinically relevant knowledge paths, which are then fed into LLMs like T5 and GPT-3.5-Turbo for diagnosis prediction.

Key technical innovations include hybrid retrieval combining KG traversal with LLM-based entity linking, few-shot prompting, and structured knowledge grounding to reduce hallucinations. DR.KNOWS achieves a CUI F-score of 27.78 and ROUGE-L score of 30.72 on the MIMIC-III dataset, outperforming baseline models like QuickUMLS. Human evaluation highlights improved diagnostic rationale alignment (55% agreement) and reduced abstraction errors when KG paths are integrated. The system demonstrates how KG-augmented RAG

enhances LLM reliability in clinical decision-making, though limitations include dependency on embedding quality and potential KG biases. Future work may explore symbolic reasoning and bias mitigation to further refine diagnostic precision.

Retrieval-augmented Information System for Enhancement (RISE) [133]: A framework to improve the performance of large language models (LLMs) in diabetes education. The goal is to address the limitations of commercially available LLMs, such as GPT-4, Claude 2, and Google Bard, which lack specialized medical knowledge and may generate inaccurate information. By integrating RISE, the framework enhances LLMs' ability to provide accurate, comprehensive, and understandable responses to diabetes-related inquiries, aiming to support patient self-management and alleviate pressure on medical resources. The RISE framework employs a four-step process: query rewriting, information retrieval from local and external academic sources, summarization with fact-checking and safety checks, and execution to generate final responses. Key technologies include the Facebook AI Similarity Search (FAISS) algorithm for retrieval and OpenAI's Text-Embedding-ADA-002 for document embeddings.

The RISE framework leverages retrieval-augmented generation (RAG) to dynamically augment LLMs with up-to-date medical knowledge from over 200 reputable academic websites and a local database of 600 diabetes-related articles. This approach mitigates hallucination risks and improves response quality without requiring costly model retraining. The integration of RISE significantly enhances the accuracy of base LLMs, with improvements of 7% for GPT-4, 19% for Claude 2, and 9% for Google Bard. Manual evaluations by clinicians and patients demonstrate notable gains in accuracy, comprehensiveness, and understandability, with GPT-4 achieving the highest accuracy rate of 98%. The framework's fact-checking and safety modules further ensure reliability, making it a promising tool for diabetes education and patient support.

Evaluation results highlight the effectiveness of RISE, with clinician assessments showing increased comprehensiveness scores (e.g., GPT-4 improved from 4.14 to 4.69 on a 5-point scale) and patient ratings indicating better understandability (e.g., GPT-4's score rose from 4.32 to 4.64). The study underscores the potential of RAG-enhanced LLMs in specialized medical domains, particularly for chronic disease management. However, the evaluation was limited to manual assessments, and generalizability to other medical fields remains unexplored. The RISE framework sets a foundation for future research into personalized, multilingual, and multimodal applications in healthcare.

GastroBot [134]: A Chinese-language chatbot specialized in gastrointestinal diseases, leveraging Retrieval-Augmented Generation (RAG) to enhance clinical accuracy and reduce hallucinations in medical question-answering. The goal is to address limitations of generic large language models (LLMs) in gastroenterology by integrating domain-specific knowledge from 25 clinical guidelines and 40 research articles. Key technologies include a fine-tuned GTE-base-zh embedding model (18% improved hit rate over base), LlamaIndex for RAG pipeline construction, and GPT-3.5 Turbo for response generation, with retrieval optimized via top-3 semantic chunk matching.

The RAG framework combines fine-tuned embeddings for precise retrieval with LLM-generated responses, ensuring grounded outputs. The GTE-base-zh model, fine-tuned on gastrointestinal terminology, outperforms OpenAI's text-embedding-ada-002 by 20% in hit rate, while GPT-3.5 Turbo synthesizes retrieved chunks into clinically relevant answers. Manual evaluations and RAGAS metrics demonstrate GastroBot's superiority over baseline models (Llama2, ChatGLM-6B, Qwen-7B), with 95% context recall, 93.73% faithfulness, and 92.28% answer relevance. Human assessments rate GastroBot highest in safety (2.87/3), usability (2.72/3), and smoothness (2.88/3).

Evaluation results highlight GastroBot's effectiveness: the fine-tuned embedding model achieves an 89% hit rate, and RAGAS scores confirm minimal hallucination. Comparative manual assessments show GastroBot provides more precise, guideline-aligned responses than ChatGPT, particularly in complex queries (e.g., biopsy handling). Limitations include reliance on static knowledge sources and regional focus, but the modular RAG design allows for future integration of dynamic data. The study demonstrates RAG's potential to bridge LLMs' generalization gaps in specialized medical domains (Table 7).

Table 7 Comparison of retrieval-augmented generation (RAG) models in the medical Domain

RAG Model/Authors	Year	Evaluation Dataset(s)	Outcomes
Duan, Yuchen and Zhou [130]	2025	Clinical cases of Wang Zhongqi (679 cases)	TCM case analysis system combining GLM-3-Turbo (0.9617 F1 NER) with Neo4j KG. Scores 0.9375 faithfulness, 0.9686 answer relevance via RAGAS. SUS evaluation shows 2.868 safety for TCM knowledge retrieval
Das, Sudeshna and Ge [129]	2025	Reddit data concerning emerging drug use (e.g., xylazine and ketamine)	Two-layer RAG for drug use MQA from Reddit: BM25F retrieval + GPT-4/Nous-Hermes-2-7B synthesis. Achieves 3/3 relevance, 5/5 coherence (expert eval). Quantized model reduces hardware needs with comparable performance
Benfenati, Domenico and De Filippis [111]	2025	Curated dataset of nutrigenetic studies	Nutrigenetics RAG with GPT-3.5/Mistral-7B and Milvus+GTE. Improves accuracy from 4→5/5 in expert eval. Reduces computational costs vs fine-tuning for personalized nutrition QA
Markey, Nigel and El-Mansouri [112]	2025	A comprehensive assessment framework focusing on clinical thinking, transparency, terminology accuracy, and content relevance	Clinical trial protocol RAG using ClinicalTrials.gov/FDA docs. Boosts clinical thinking from 41.4%→79.7%, achieves 78.6% transparency. Maintains > 99% terminology accuracy in endpoint generation
GuideGPT [116]	2025	Manual evaluation by 10 MRONJ experts on 30 questions. The model's responses were grounded in 449 scientific publications from clinical guidelines	MRONJ-specialized chatbot with GPT-4 (gpt-4-0613) + top-20 sentence retrieval. Outperforms pure GPT-4 by 11% in expert content ratings (80.67% high). Includes source PDF verification
MMed-RAG [103]	2025	MIMIC-CXR, IU-Xray, Harvard-FairVLMed, PMC-OA, Quilt-1 M	Multimodal RAG for Med-LVLMs with domain-aware retrieval and adaptive context selection. Improves factual accuracy by 43.8%, medical VQA by 18.5%, and report generation by 69.1%. Uses LLaVA-Med-1.5, BiomedCLIP. Outperforms DoLa, VCD, MedDr
Upadhyay and Rishabh and Viviani [104]	2025	CLEF eHealth 2020 dataset, TREC Health Misinformation 2020 dataset	RAG for Health IR with GPT-3/Llama3/Falcon-40B and PubMed Central. Achieves CAM-MAP 0.1532 (top-10), CAM-NDCG 0.2702. Combines BM25 with LLM-augmented factual scoring via SCIFIVE
Pingua, Bhagyajit and Sahoo [107]	2025	MedQuAD dataset (47,457 QA pairs)	Compares FT, RAG, and hybrid FT+RAG on MedQuAD (47K QA pairs). Hybrid FT+RAG achieves BERTScore F1 0.891 (Llama-3.1-8B). Uses LoRA, ChromaDB
RAG ² [108]	2025	MedQA, MedMCQA, MMLU-Med benchmarks	Rationale-guided RAG with perplexity filtering and multi-corpus retrieval. Improves MedQA accuracy by +6.1% (91.1% with GPT-4o). Uses Llama-3-8B, Meerkat-7B, MedCPT
RAGEv [109]	2025	RAGEv-Bench (PubMedQA, Horizon Europe project outputs, Virtual Human Twin research, Antimicrobial Resistance reviews)	Healthcare document synthesis with SHy/ColBERTv2 retrieval. Achieves 0.85 precision, 0.83 BERTScore on PubMedQA. Uses E5-Mistral, GPT@JRC
MedRAG [121]	2025	Public DDXPlus dataset, private chronic pain diagnostic dataset (CPDD)	RAG framework with knowledge graph (KG) reasoning for diagnosis. Accuracy: 79.25% (CPDD), 88.65% (DDXPlus). Specificity: 66.04% (CPDD), 68.01% (DDXPlus). Supports Mixtral-8x7B, Llama-3.1, GPT-4o
Retrieval-Augmented Generation-Based Physician Recommendation (RAGPR) [122]	2025	646,383 consultation records	Physician recommendation model. Evaluated on 646 K records, F1: 95% (GPT-4o), 94% (Mistral). Uses SBERT, ChromaDB, MMR. Enhances triage accuracy and satisfaction

Table 7 (continued)

RAG Model/Authors	Year	Evaluation Dataset(s)	Outcomes
Med- R^2 [117]	2025	MedQA-USMLE, MedQA-MCMLE, MedMCQA, PubMedQA, MMLU-Med	EBM-aligned RAG framework with query reformulation, hierarchical evidence reranking, and CoT reasoning. Accuracy: 81.53% (LLaMA3.1-70B), 67.97% (Qwen2.5-7B). Outperforms vanilla RAG by +14.87% and fine-tuning by +3.59%. Supports LLaMA2/3.1, Qwen2.5
Omni-RAG [124]	2025	11 benchmark medical datasets (MedQA, PubMedQA, SEER, DDXPlus)	Multi-source RAG using MedOmniKB. Accuracy: 76.98% (MedQA), 60.20% (PubMedQA). Reduces hallucinations, improves retrieval precision
CLEAR [125]	2025	Stanford Medication for Opioid Use Disorder (MOUD) dataset, CheXpert dataset	Entity-based RAG for EHRs. F1: 90% (MOUD), 0.96 (CheXpert). Reduces tokens (−70%), inference time (−72%). Outperforms embedding-based retrieval
MS-RAG [127]	2025	Natural Questions (NQ), TriviaQA	Multi-stage retrieval (dense + sparse) with RL optimization. Improves retrieval precision (+7.2%), factual consistency (+5.8%), and ROUGE-L (+4.1%). Reduces hallucinations (−15%), token usage, and inference time. Outperforms baseline RAG models
LLama-2 Rag [90]	2025	Composite dataset spanning general medicine, pediatrics, and specialized care (mixed-domain doctor-patient dialogs)	Doctor-patient conversational AI. BERT F1: 0.861, recall: 0.875. BLEU: 0.372, ROUGE: 0.294. Uses LlamaIndex, ChromaDB
CliniqIR [110]	2025	DC3 dataset for rare diseases, MIMIC-III dataset for imbalanced data	Rare disease diagnostic system combining BM25/ MedCPT retrieval with ClinicalBERT (RRF ensemble). Achieves MRR 0.35 vs 0.15 baseline on DC3 dataset. Uses 33 M PubMed abstracts + UMLS, improves by 0.10 MRR for <5-sample diagnoses
GastroBot [134]	2024	On integrated 25 clinical guidelines and 40 research articles related to gastroenterology as its knowledge base	Achieved 95% context recall, 93.73% faithfulness, 92.28% answer relevance (RAGAS metrics). Rated high in safety (2.87/3)
DR.KNOWS [132]	2024	MIMIC-III dataset	UMLS KG-enhanced diagnostic system with SGIN embeddings. Scores CUI F-score 27.78, ROUGE-L 30.72 on MIMIC-III. GPT-3.5-Turbo with KG paths improves diagnostic rationale alignment (55% agreement)
Painter, Jeffery L and Chalamalasetti [114]	2024	Querying complex drug safety databases (manual evaluation across three experimental phases)	Pharmacovigilance SQL generator using GPT-4 + business context docs. Boosts accuracy from 8.3%→78.3%, achieves 85% success on medium-complexity queries (p=0.0006). Uses text-embedding-ada-002
Caregiving Language Model (CaLM) [115]	2024	66 test question-answer pairs; caregiving knowledge base (8,568 documents), 581 curated Q-A pairs for fine-tuning	ADRD caregiver assistant with RAG + fine-tuned LLaMA-2-7B. Scores BERT 0.91, provides 100% references (80% accuracy). Outperforms GPT-3.5 in 85% low-medium complexity queries
Mashatian, Shayan and Armstrong [113]	2024	Curated knowledge base from NIH National Standards for Diabetes Self-Management Education (295 articles, 175 questions)	Diabetes education RAG (GPT-4+Pinecone) with NIH standards. Achieves 98% accuracy/F1 on 175 questions. Few-shot learning resolves factuality issues (99% precision)
Retrieval-augmented Information System for Enhancement (RISE) [133]	2024	Manual assessments by clinicians and patients. The model leveraged a knowledge base of 600 diabetes-related articles and 200 academic websites	Diabetes RAG framework boosting GPT-4 accuracy by 7% (to 98%). Uses FAISS + 600 articles, improves comprehensiveness from 4.14→4.69 (5-point scale). Integrates fact-checking safety modules

Table 7 (continued)

RAG Model/Authors	Year	Evaluation Dataset(s)	Outcomes
ChatENT [85]	2024	OHNS exam questions and multiple-choice tests	OHNS-specialized LLM combining GPT-4 with RAG using Clinical ENT Knowledge Base (CEKB). Achieves 95.7% validity, 87.2% competency on exams (58.4% error reduction). Scores 80% on MCQ tests (+26% over ChatGPT) with lower hallucination rates per CVSC framework
Xiong, Guangzhi and Jin, et al. [20]	2024	7663 questions from five medical QA datasets	MEDRAG benchmark evaluating 7663 medical QA questions with 5 corpora (PubMed/StatPearls/textbooks) and 6 LLMs. Shows MedCPT+RRF-4 retrieval boosts GPT-3.5 to GPT-4 level (71.5% on MIRAGE, +17.9%). Recommends PubMed/MedCorp with Mixtral for cost-effective medical QA
RALL [86]	2024	CELLS (63,000 scientific abstracts paired with expert-written plain language summaries)	BART-based summary model using PubMed/Wikipedia, improved summary quality and simplicity while maintaining factual correctness, with BART outperforming GPT-4 and Llama 2 in readability, familiarity, and accuracy, while GPT-4 was rated higher for understandability and meaning preservation, confirming the effectiveness of retrieval-augmented models in biomedical text simplification
LiVersa [87]	2024	10 case-vignette questions (manual evaluation). The model integrated 30 AASLD guidelines as its knowledge base	A liver disease guidelines RAG, improved specificity over general ChatGPT, answered 100% of clinical questions in a yes/no format, but its detailed justifications contained errors in 3 cases due to missing guideline versions and contextual biases, highlighting both the strengths and limitations of RAG-based specialization for clinical decision support
Almanac [88]	2024	130 clinical cases curated by five experts across various specialties	Clinical knowledge in a vector DB. RetA LLM reduces hallucinations in medical QA, Outperformed ChatGPT with an 18% increase in factual accuracy ($p < 0.05$), achieving 91% accuracy in cardiology (vs. 69% for ChatGPT), maintained 95% safety in adversarial tests (ChatGPT failed entirely), but was preferred by physicians only 43% of the time, likely due to ChatGPT's more natural language style
Ye, Cheng [102]	2024	Clinical electronic medical records (EMRs); few-shot learning from 10 user-labeled examples	EMR search engine improving query precision, achieved Precision-at-10 (P@10) scores of at least 0.60 with just ten labeled samples, outperforming baseline methods, and improved the RAG model's accuracy from 0.14 to 0.50 in a cancer progression study
MedGraphRAG [120]	2024	9 medical QA benchmarks, 2 health fact-checking datasets, long-form generation test set	Graph-based RAG for medical QA, enhancing reliability and evidence-based responses, outperformed standard RAG by 8% in medical QA and 10% in fact-checking, surpassing GraphRAG by 5% and 8%, while achieving higher accuracy than fine-tuned medical LLMs like Med-PaLM 2 and Med-Gemini across all benchmarks
Alkhalaf, Mohammad and Yu, Ping and Yin [128]	2024	Clinical data from Australian Residential Aged Care Facilities (1,424 progress notes for summarization, 1399 notes for risk factor analysis)	RAG for EHR summarization. Accuracy: 93.25% → 99.25%. Risk factor extraction: 90%. Uses MMR, sentence transformers

Table 7 (continued)

RAG Model/Authors	Year	Evaluation Dataset(s)	Outcomes
Thompson, Will E and Vidmar et al. [89]	2023	Test set of 199 patients from electronic health records (EHRs)	EHR phenotyping system using PaLM-2 (bison@001) with MapReduce RAG. Achieves F1=0.75 for pulmonary hypertension (vs 0.62 rule-based). Processes full clinical notes via Regex retrieval + parallel LLM analysis. Identifies 2x more cases than structured rules
MedKgConv [91]	2023	MedDialog and COVID datasets	Medical dialog system combining BioBERT encoder-decoder with Quick-UMLS knowledge graphs. Achieves +3.3% F1 and +6.3% BLEU-4 on MedDialog dataset versus baselines. Uses semi-supervised KG integration for improved response accuracy
accGPT [93]	2023	50 synthetic clinical cases	Context-aware chatbot for radiology, outperforming GPT-3.5/4, outperformed radiologists (OR: 3.76, $p < 0.001$) and GPT-3.5 (OR: 2.93, $p < 0.001$) in providing correct imaging recommendations
FraCChat [94]	2023	Evaluation based on AO code classification from radiology reports. The model uses the AO/OTA Fracture and Dislocation Classification Compendium as its knowledge source	Radiology doc encoding for fracture classification, achieved 83% accuracy in full AO classification (vs. 95% for human radiologists), significantly outperforming FraCChat-GPT-3.5 (57%), generic GPT-4 (7%), and generic GPT-3.5 (5%), with 71% consistency in correct AO codes, while reducing classification time from 83.2 min (humans) to 7.2 min (FraCChat) and 3.3 min (generic chatbots)
Yang, Rui and Marrese-Taylor, et al. [96]	2023	LiveQA dataset (104 questions)	Bio-Epidemiology-NER for chatbot retrieval, reducing noise, achieved a ROUGE-1 score of 19.97 and BERTScore F1 of 82.08
LLM-AMT [98]	2023	MedQA-USMLE, MedQA-MCMLE, MedMCQA datasets	Multiple retrievers on medical texts, optimizing retrieval for clinical tasks. Tested on MedQA-USMLE, MedQA-MCMLE, and MedMCQA. and achieved significant improvements, increasing accuracy by 11.6 to 16.6%. with GPT-4-Turbo outperformed Med-PaLM 2, a specialized medical LLM, by 2.3% to 2.7%. Furthermore, replacing Wikipedia with textbooks as the retrieval corpus resulted in a 7.8–13.7%
TrialGPT [99]	2023	Synthetic patient cohorts; 1015 patient-criterion pairs	FAISS-based RAG for patient-trial matching, reducing screening time, manually evaluated on 1015 patient-criterion pairs, achieving 87.3% accuracy, a user study at the National Cancer Institute (NCI) showed that TrialGPT reduced screening time by 42.6%
Manathunga, Supun S and Illangasekara, et al. [101]	2023	Kumar and Clark's Clinical Medicine, British National Formulary	Hybrid extractive-abstractive summarization using k-means clustering and t-SNE for visual representation. Tested on medical texts: Kumar & Clark's and BNF 82. Outperforms retrieve-stuff-summarize in precision and contextual relevance
FLARE [22]	2023	WikiMultihopQA dataset, ASQA-hint dataset	Dynamic RAG for iterative retrieval in complex text generation. For the WikiMultihopQA dataset, FLARE direct achieved an F1-score of 59.7%, precision of 59.1%, and recall of 62.6%, outperforming other baselines
SELF-RAG [23]	2023	PopQA, TQA, PubQA, ARC-Challenge, biography generation, ASQA	Self-reflective RAG with adaptive retrieval, improving fact verification and QA, the model achieves high accuracies of 54.9% on PopQA, 66.4% on TQA, and 72.4% on PubQA, while the 13B variant scores 55.8%, 69.3%, and 74.5%

6 Comparative analysis of RAG approaches in medical literature

6.1 Naive RAG: foundation with clear trade-offs

Naive RAG establishes baseline performance across medical applications, revealing both strengths in structured tasks and limitations in complex reasoning:

- **Structured Medical Knowledge Excels:**
 - ChatENT [85]’s 58.4% error reduction in OHNS demonstrates that domain-specific knowledge bases dramatically outperform general LLMs when applied to well-defined medical specialties
 - The 26% improvement over ChatGPT in multiple-choice formats suggests naive RAG works best with categorical medical decisions rather than open-ended clinical reasoning
- **Retrieval Quality Determines Success:**
 - Xiong et al. [20]’s finding that MedCPT consistently outperforms general retrievers while PubMed yields optimal results reveals that biomedical-specific tools are essential—generic retrieval fails in medical contexts
 - The 17.9% MIRAGE improvement (71.57% final accuracy) shows naive RAG’s ceiling: substantial gains possible but still below clinical-grade requirements
- **Task Complexity Limitations:**
 - Thompson et al. [89]’s pulmonary hypertension detection (F1: 0.75 vs 0.62 rule-based) represents naive RAG’s sweet spot: pattern recognition in structured clinical data
 - However, the “lost-in-the-middle” phenomenon identified by Xiong et al. [20] exposes naive RAG’s inability to synthesize information from multiple sources—critical for complex diagnoses
 - Alkhalaf et al. [128]’s finding that RAG improved structured summarization (93.25% to 99.25%) but provided no improvement for malnutrition risk extraction (maintained 90%) illustrates this task-dependent effectiveness

6.2 Advanced RAG: addressing core medical AI challenges

Advanced RAG architectures tackle the fundamental problems that naive approaches cannot solve—reasoning depth, knowledge integration, and clinical reliability:

Knowledge Graph Integration Breakthrough: The performance gap between MedKgConv [91]’s 6.3% BLEU-4 improvement and traditional dialog systems reveals that structured medical knowledge (via BioBERT + Quick-UMLS) enables contextual understanding impossible with text retrieval alone. This addresses medicine’s core challenge: relationships between symptoms, diseases, and treatments require graph-based reasoning, not just document matching.

Dynamic Retrieval Solves Static Knowledge Problems: accGPT [93]’s 82% appropriateness rating versus 64–74% for human radiologists demonstrates a crucial insight: when RAG systems access comprehensive, authoritative guidelines (209 ACR documents), they can exceed human performance in rule-based medical decisions. The 84% consistency across runs versus human variability shows AI’s potential advantage in standardizing medical practice.

Evidence-Based Medicine Integration: LLM-AMT [98] model achieving 67.9–72.6% accuracy while outperforming GPT-3.5 by 11.6–16.6% and even surpassing Med-PaLM 2 by 2–3% demonstrates that authoritative medical textbook integration can rival specialized medical models. The 16% reduction in hallucinations confirms that structured medical knowledge sources are superior to general knowledge bases for clinical applications.

Multimodal Integration Addresses Real Clinical Workflows: MMed-RAG [103]’s 43.8% factual accuracy improvement with 69.1% enhancement in report generation tackles medicine’s multimodal reality. The system’s ability to achieve 28.19% copy-reference rate versus 55.08% in baselines suggests advanced RAG can reduce hallucinations specifically in visual medical interpretation—critical for radiology and pathology applications.

Clinical Trial Optimization: TrialGPT [99]’s achievement of 87.3% accuracy in criterion-level matching while reducing screening time by 42.6% illustrates advanced RAG’s potential for operational efficiency. The system’s ability to outperform baseline methods by 43.8% in ranking and excluding trials demonstrates how hybrid retrieval (BM25 + MedCPT) with chain-of-thought prompting can streamline complex medical workflows.

6.3 Modular RAG: clinical-grade specialization

Modular architectures achieve clinical deployment standards by solving domain-specific challenges that generic approaches cannot address (Table 8):

Evidence-Based Medicine Integration: Med-R² [117]’s 14.87% improvement over vanilla RAG through EBM principles represents a paradigm shift: instead of retrieving any relevant documents, the system mimics clinical reasoning by classifying queries, ranking evidence credibility, and applying chain-of-thought reasoning. The dramatic performance gains for lightweight models (79.72% for Qwen2.5-7B, 77.80% for LLaMA3.1-8B) suggest this approach can democratize high-quality medical AI by making smaller models clinically viable.

Knowledge Graph-Enhanced Diagnosis: MedGraphRAG [120]’s 8% improvement in medical QA and 10% in fact-checking versus standard RAG demonstrates that Triple Graph Construction enables evidence-based responses impossible with document retrieval alone. The system’s state-of-the-art performance across 9 medical QA benchmarks and superior clinician ratings in correctness and citation precision suggest graph-based medical reasoning approaches clinical reliability standards.

Multi-Source Knowledge Integration: Omni-RAG [124]’s 8% average improvement across 11 benchmark datasets (reaching 76.98% on MedQA) through Source Planning Optimization reveals that medical AI requires orchestrated access to diverse knowledge types—books, guidelines, papers, wikis, and knowledge graphs. The system’s structured approach to multi-genre medical knowledge addresses the fundamental challenge that no single source contains comprehensive medical information.

Specialized Domain Mastery: GastroBot [134]’s progression from generic embeddings to fine-tuned GTE-base-zh (18% hit rate improvement, final 89% accuracy) with RAGAS scores of 95% context recall and 93.73%

Table 8 Modular RAG clinical deployment readiness

Clinical challenge	Modular solution	Deployment readiness
Diagnostic Accuracy	Med-R ² [117]’s EBM-guided reasoning achieves clinical-grade performance through evidence hierarchy	79.72% improvement suggests readiness for decision support
Patient Safety	RAG ² [108]’s reflection tokens prevent harmful outputs through self-assessment	91.1% accuracy with GPT-4o approaches human expert level
Regulatory Compliance	LiVersa [87] maintains PHI compliance while accessing medical knowledge	100% accuracy in hepatitis scenarios meets regulatory standards

faithfulness illustrates the modular approach: domain-specific optimization at every layer yields clinical-grade performance. The human evaluation scores (2.87/3 safety, 2.88/3 smoothness) suggest specialized systems can achieve user acceptance for clinical deployment.

Traditional Medicine Digitization: Duan et al. [130]’s TCM system achieving 0.9617 F1 in symptom extraction with 0.9375 faithfulness and 0.9686 answer relevance demonstrates modular RAG’s unique capability: digitizing traditional medical knowledge that lacks standardized vocabularies. The SUS framework scores (2.868 safety, 2.848 usability) show how modular approaches can bridge cultural and linguistic gaps in global medicine while maintaining clinical safety standards.

Caregiver Support Systems: CaLM [115]’s superior performance over GPT–3.5 (BLEU: 0.45, ROUGE: 0.52, BERT score: 0.91) with 100% reference provision (80% accuracy) versus GPT–3.5’s 70% coverage demonstrates that domain-specific fine-tuning combined with RAG can democratize medical expertise. The focus on ADRD caregiving with 8,568 curated documents shows how modular systems can address underserved medical populations.

6.4 Architecture selection insights: matching complexity to clinical need

The literature reveals critical decision factors for medical RAG deployment based on clinical requirements rather than technical capabilities:

1. Clinical Reasoning Depth Requirements: - **Protocol-Based Care**: Naive RAG suffices when medical decisions follow established guidelines (ChatENT [85]’s 95.7% validity in OHNS) - **Differential Diagnosis**: Advanced RAG required for complex reasoning (accGPT [93] outperforming radiologists in imaging recommendations) - **Specialized Practice**: Modular RAG essential for domain expertise (GastroBot [134]’s clinical-grade gastroenterology performance)

2. Knowledge Integration Complexity: Systems that require authoritative source synthesis—such as LLM-AMT [98] which integrates medical textbooks and outperforms Med-PaLM 2—demonstrate that medical AI cannot rely on single knowledge bases. The consistent superior performance of biomedical-specific retrievers (MedCPT over general alternatives in Xiong et al. [20]) suggests specialized retrieval is fundamental, not optional.

3. Resource Optimization Insights: The inverse relationship between model size and RAG benefits (Med-R² [117]’s 79.72% gains for 7B models versus diminishing returns for larger models) reveals a critical insight: resource-constrained medical settings may achieve superior outcomes with sophisticated RAG architectures plus smaller models rather than large foundation models alone.

6.5 Persistent clinical deployment barriers

Despite impressive technical achievements, fundamental challenges persist across all architectures:

Human Acceptance vs. Technical Performance Gap: The disconnect between Almanac [88]’s 18% factual accuracy improvement and 57% physician preference for ChatGPT reveals that medical AI evaluation metrics don’t capture clinical usability factors. This pattern appears across studies where technical superiority doesn’t translate to user acceptance.

Static Knowledge vs. Dynamic Medicine: Multiple systems acknowledge limitations from static knowledge bases (evident in Med-R² [117]’s dependency on corpus quality, CliniqIR [110]’s reliance on PubMed coverage biases). The requirement for “dynamic updates” mentioned across studies indicates no current RAG architecture adequately addresses medicine’s rapid evolution.

Specialization-Generalization Tension: High-performing specialized systems (GastroBot [134], CaLM [115], Duan et al. [130]’s TCM system) achieve clinical-grade performance but lack transferability. Medical practice requires both deep specialization and broad knowledge—a fundamental tension that current

modular approaches address through vertical depth but sacrifice horizontal breadth, limiting practical deployment scalability.

6.6 The effect of RAG on clinical models

Retrieval-Augmented Generation (RAG) is a critical architectural choice that directly addresses many of the deployment barriers detailed above, particularly the tension between static knowledge and dynamic medicine. By coupling Large Language Models (LLMs) with external, up-to-date knowledge bases, RAG enables models to access the latest research and clinical guidelines, mitigating issues of outdated information and improving factual accuracy (Fig. 6).

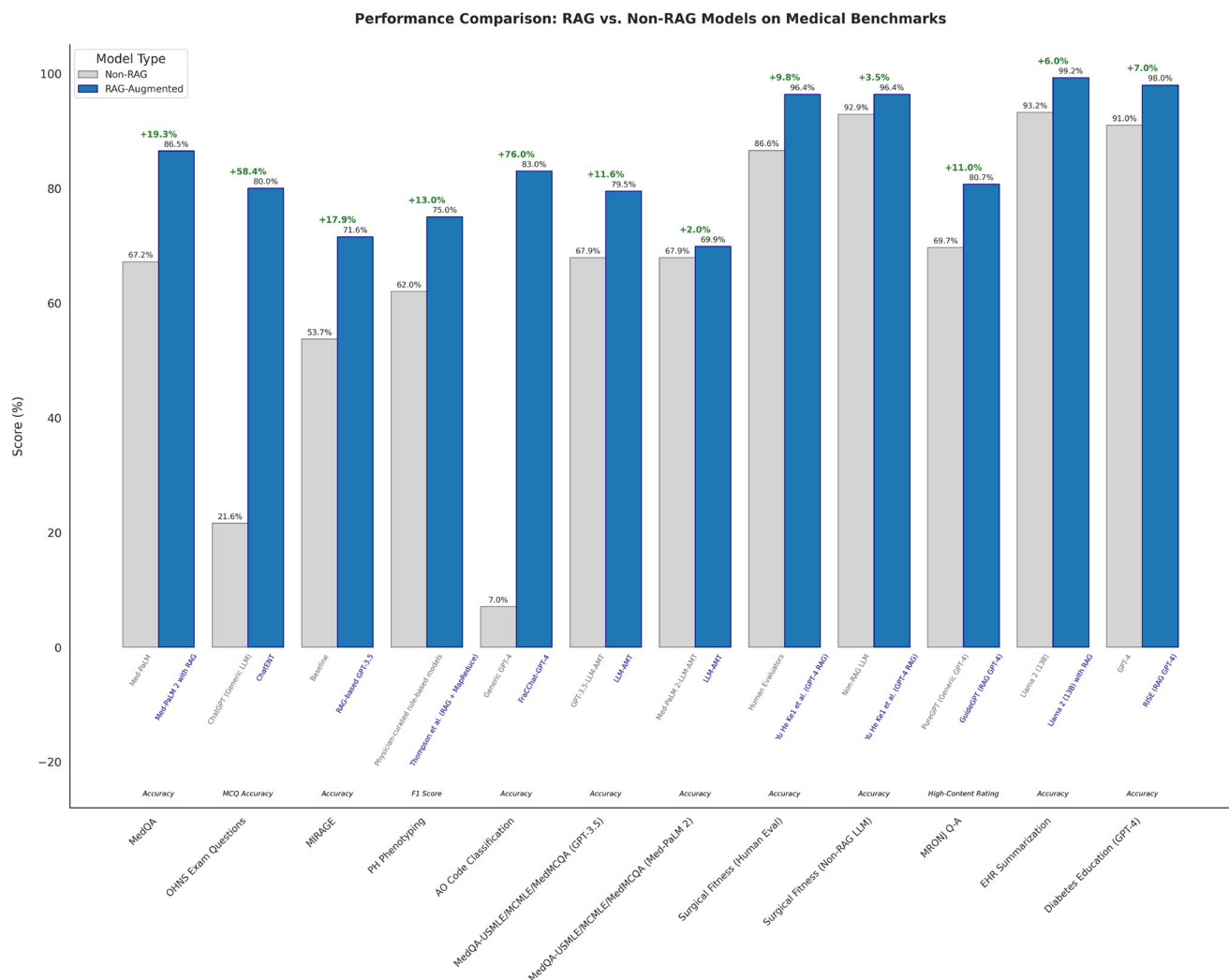


Fig. 6 A conceptual diagram illustrating the fundamental difference between a traditional Large Language Model (LLM) and a Retrieval-Augmented Generation (RAG) system on a common datasets

7 Future work and challenges

7.1 Architectural evolution and integration challenges

7.1.1 Multi-modal integration beyond text

Current RAG implementations in medicine primarily focus on textual knowledge retrieval, with limited exploration of truly integrated multi-modal approaches. While MMed-RAG [103] demonstrates promising results with 43.8% factual accuracy improvements in medical vision-language tasks, significant gaps remain in comprehensive multi-modal medical reasoning. Future work must address the integration of medical imaging, laboratory results, genomic data, and physiological signals within unified RAG architectures. The challenge lies not merely in processing different data types, but in developing retrieval mechanisms that can identify relevant cross-modal correlations—for instance, retrieving relevant radiological images when processing textual symptom descriptions, or correlating genetic variants with phenotypic presentations from literature.

The development of unified embedding spaces that can represent relationships between textual medical knowledge, visual pathology patterns, genomic sequences, and temporal physiological data presents a fundamental research challenge. Current approaches like those demonstrated in MedGraphRAG [120] with Triple Graph Construction could be extended to incorporate multi-modal nodes, but the computational complexity and knowledge representation challenges scale exponentially with additional modalities.

7.1.2 Real-time knowledge update mechanisms

A critical limitation identified across multiple studies is the dependency on static knowledge bases. Med-R² [117] acknowledges potential gaps in medical knowledge corpus, while CliniqIR [110] highlights coverage biases in PubMed. The medical field's rapid evolution—with new treatments, drug interactions, clinical guidelines, and research findings emerging continuously—demands dynamic knowledge updating mechanisms that current RAG architectures cannot accommodate effectively.

Future research must develop incremental learning approaches that can seamlessly integrate new medical knowledge without requiring complete system retraining. This involves solving several technical challenges: (1) maintaining knowledge consistency when updating subsets of the corpus, (2) preserving retrieval performance while incorporating new information, (3) handling conflicting or outdated information as medical understanding evolves, and (4) ensuring that knowledge updates maintain the evidence hierarchies essential for clinical decision-making.

The development of temporal knowledge graphs that can track the evolution of medical understanding, incorporate confidence scores based on evidence quality, and automatically deprecate outdated information represents a significant research frontier. Systems must also address the challenge of integrating real-world evidence from electronic health records with published literature while maintaining appropriate evidence weighting.

7.1.3 Federated learning and privacy-preserving RAG

Medical data privacy requirements create fundamental tensions with the comprehensive knowledge access needed for effective RAG systems. While systems like LiVersa [87] demonstrate PHI-compliant deployment within restricted environments, the broader challenge of enabling knowledge sharing across institutions while preserving patient privacy remains unsolved.

Future work must explore federated RAG architectures where multiple medical institutions can contribute to and benefit from shared knowledge systems without exposing sensitive patient information. This requires

developing privacy-preserving retrieval mechanisms, secure multi-party computation for knowledge graph construction, and differential privacy approaches for maintaining utility while protecting individual patient data.

The technical challenges include developing distributed embedding spaces that can be updated across institutions without sharing raw data, creating secure query mechanisms that prevent inference attacks while maintaining retrieval quality, and establishing trust and governance frameworks for federated medical knowledge systems.

7.2 Clinical integration and validation challenges

7.2.1 Standardized evaluation frameworks

The literature reveals significant inconsistencies in evaluation approaches across RAG implementations. While some studies like Ke et al. [106] employ structured frameworks (S.C.O.R.E. for Safety, Consensus, Objectivity, Reproducibility, Explainability), many rely on manual evaluations or domain-specific metrics that limit cross-system comparisons.

Future work must establish standardized evaluation frameworks that can assess RAG systems across multiple dimensions relevant to clinical deployment: accuracy, safety, explainability, bias, reliability, and user acceptance. These frameworks must account for the unique requirements of different medical specialties while enabling meaningful performance comparisons.

The development of benchmark datasets that reflect real-world clinical complexity, including rare diseases, multi-morbidity, and complex patient presentations, is essential. Current benchmarks often focus on single-condition scenarios that don't reflect clinical reality where patients present with multiple interacting conditions.

7.2.2 Regulatory approval and clinical validation

The path from research prototype to clinically deployed medical AI system involves regulatory approval processes that current RAG systems have not navigated. The high performance demonstrated by systems like RAG² [108] (91.1% accuracy with GPT-4o) and accGPT [93] (82% appropriateness vs. 64–74% human radiologists) suggests clinical utility, but regulatory frameworks for evaluating retrieval-augmented systems remain underdeveloped.

Future work must address the unique challenges of validating systems that combine parametric knowledge (within the LLM) with nonparametric knowledge (retrieved documents). Regulatory bodies require explainable decision-making processes, but current RAG systems often cannot provide complete traceability of how retrieved information influences final outputs.

The development of RAG systems that can meet FDA or equivalent regulatory requirements involves creating audit trails for retrieved information, ensuring reproducibility across different knowledge bases, and demonstrating safety across diverse patient populations and clinical scenarios.

7.2.3 Human-AI collaboration models

The discrepancy between technical performance and physician preference observed in Almanac [88] (superior accuracy but 57% preference for ChatGPT) highlights the gap between technical optimization and clinical usability. Future work must focus on developing human-AI collaboration models that leverage RAG capabilities while addressing clinician workflow integration.

This involves designing interaction paradigms where clinicians can understand and trust RAG system recommendations, developing interfaces that allow physicians to inspect and validate retrieved evidence, and creating training programs that help medical professionals effectively utilize RAG-enhanced systems.

The challenge extends beyond user interface design to fundamental questions about medical decision-making responsibility, liability when AI systems provide incorrect recommendations based on retrieved information, and the evolution of medical education to incorporate AI collaboration skills.

7.3 Technical scalability and efficiency challenges

7.3.1 Computational resource optimization

Current high-performing RAG systems often require substantial computational resources. While CLEAR [125] demonstrates 70% token reduction and 72% inference time improvement, many advanced systems like Med-GraphRAG [120] and Omni-RAG [124] require complex processing pipelines that may limit deployment in resource-constrained medical settings.

Future work must address the trade-offs between system capability and computational efficiency. This includes developing lightweight RAG architectures that can provide clinical-grade performance on modest hardware, optimizing retrieval algorithms for medical knowledge bases, and creating adaptive systems that can scale performance based on available computational resources.

The development of edge-deployable RAG systems for medical applications in low-resource settings, emergency medicine, or point-of-care scenarios represents a significant technical challenge requiring novel approaches to knowledge compression, efficient retrieval, and model optimization.

7.3.2 Knowledge base construction and maintenance

The manual curation required for high-performing domain-specific systems like GastroBot [134] (25 clinical guidelines, 40 research articles) and CaLM [115] (8,568 documents chunked into 196,926 segments) presents scalability challenges for broader medical domain coverage.

Future research must develop automated or semi-automated approaches for medical knowledge base construction that can maintain the quality standards required for clinical applications. This involves natural language processing techniques for automatic document classification, quality assessment, and relationship extraction from medical literature.

The challenge includes developing systems that can automatically identify authoritative sources, extract structured knowledge from unstructured medical texts, resolve conflicts between sources, and maintain knowledge base consistency as new information becomes available.

7.4 Ethical and bias mitigation challenges

7.4.1 Healthcare equity and bias prevention

Medical RAG systems risk perpetuating or amplifying existing biases present in medical literature and training data. While some studies acknowledge bias concerns (MMed-RAG [103] mentions bias mitigation needs), comprehensive approaches to ensuring healthcare equity in RAG systems remain underdeveloped.

Future work must address representation bias in medical knowledge bases, ensuring that retrieved information reflects diverse patient populations, cultural contexts, and global medical practices. This involves developing bias detection mechanisms for medical retrieval systems, creating diverse and representative knowledge bases, and establishing fairness metrics specific to medical applications.

The challenge extends to addressing historical biases in medical research, ensuring that RAG systems don't perpetuate disparities in healthcare quality, and developing approaches that can identify and mitigate bias in real-time during system operation.

7.4.2 Transparency and explainability requirements

Medical decision-making requires transparent reasoning processes that patients and physicians can understand and validate. While systems like TrialGPT [99] provide natural language explanations for patient-trial matching, comprehensive explainability for complex RAG reasoning remains challenging.

Future research must develop explainable AI techniques specifically designed for medical RAG systems, including visualization approaches for knowledge retrieval processes, natural language explanation generation that physicians can validate, and user interfaces that allow inspection of retrieved evidence and reasoning chains.

The technical challenge involves balancing system performance with explainability requirements, ensuring that explanation systems don't become additional sources of error, and developing explanation approaches that are meaningful to different stakeholders (patients, physicians, regulators).

7.5 Domain-specific research directions

7.5.1 Rare disease and long-tail medical knowledge

Current RAG systems often perform well on common medical conditions but struggle with rare diseases or unusual presentations. CliniqIR [110]'s focus on rare disease diagnosis represents early work in this area, but significant challenges remain in developing systems that can effectively retrieve and reason about low-prevalence conditions.

Future work must address the unique challenges of rare disease medicine: limited literature, small patient populations, heterogeneous presentations, and the need for expert knowledge that may not be well-represented in standard medical databases. This involves developing specialized retrieval strategies for rare disease literature, incorporating case report databases, and creating systems that can reason about disease relationships and similarities.

7.5.2 Global health and cross-cultural medicine

The focus on Western medical literature in current systems limits applicability in global health contexts. Duan et al. [130]'s work on Traditional Chinese Medicine RAG systems represents important progress, but broader integration of diverse medical traditions and practices remains unexplored.

Future research must address the integration of traditional medicine knowledge systems, cultural factors in medical decision-making, and the development of RAG systems that can function effectively across different healthcare systems and resource levels. This involves creating multilingual medical knowledge bases, developing cross-cultural medical reasoning capabilities, and addressing the unique challenges of medical AI deployment in low and middle-income countries.

7.5.3 Personalized medicine integration

Current RAG systems primarily retrieve population-level medical knowledge, but the future of medicine increasingly requires personalized approaches based on individual patient characteristics, genetics, lifestyle, and preferences. Future work must explore how RAG systems can integrate personalized patient data with population-level medical knowledge to provide individualized recommendations.

This involves developing privacy-preserving approaches for incorporating personal health records into retrieval processes, creating personalized knowledge graphs that combine individual patient data with medical literature, and addressing the ethical and technical challenges of personalized medical AI systems.

The technical challenges include developing retrieval mechanisms that can identify relevant population subgroups for individual patients, integrating genomic and phenotypic data with medical literature, and creating

systems that can adapt recommendations based on individual patient contexts while maintaining evidence-based foundations.

8 Conclusion

This Survey thoroughly analyzed and classified RAG model architectures, particularly the Naive, Advanced, and Modular RAG frameworks, to understand their role in enhancing healthcare-specific natural language processing (NLP). Each model was assessed on its ability to retrieve relevant data from external sources and generate meaningful responses. Naive RAG was recognized as the foundational approach, while Advanced RAG introduced more sophisticated indexing and retrieval techniques to improve accuracy. Modular RAG was identified as the most flexible, incorporating specialized modules and customizable components to address domain-specific challenges in healthcare, such as fine-tuning retrieval based on complex medical data structures and enabling sequential, task-oriented processing.

The effectiveness of RAG models in healthcare is grounded in high-quality, domain-specific datasets and robust Large Language Models (LLMs) fine-tuned for medical contexts. When these LLMs are augmented with retrieval capabilities, they bridge the gap between static knowledge and real-time, verified medical information, mitigating hallucinations and enhancing response accuracy. This synergy between medical datasets and fine-tuned LLMs empowers RAG models to deliver contextually relevant, clinically reliable responses, reinforcing their potential as vital tools in advancing healthcare AI applications.

Data Availability The datasets used in this analysis and survey are publicly available and can be obtained from the corresponding author upon reasonable request. 1. Medical Text Datasets are publicly available in a repository: MIMIC-III: Data are available at mimic.mit.edu. UMLS: Data are available at nlm.nih.gov. MedDialog: Data are available at [github](https://github.com). CovidDialog: Data are available at [github](https://github.com). Medical Flashcards: Data are available at [github](https://github.com). Wikidoc: Data are available at [huggingface](https://huggingface.co). Wikidoc Patient Information: Data are available at [huggingface](https://huggingface.co). MEDIQA: Data are available at [huggingface](https://huggingface.co). CORD-19: Data are available at [huggingface](https://huggingface.co). MMMLU: Data are available at [huggingface](https://huggingface.co). Pubmed Causal: Data are available at [huggingface](https://huggingface.co). ChatDoctor: Data are available at [github](https://github.com). Alpaca-EN-AN: Data are available at [github](https://github.com). MeDAL: Data are available at [huggingface](https://huggingface.co). 2. Medical multi modal Datasets are publicly available in a repository: Roco V2 Dataset: Data are available at [zenodo](https://zenodo.org). Roco V1 Dataset: Data are available at [github](https://github.com). MedVidQA Dataset: Data are available at [visualqa](https://visualqa.org). VQA-Med-2021: Data are available at [github](https://github.com). SLAKE Dataset: Data are available at [med-vqa](https://med-vqa.github.io). VQA-Med-2020: Data are available at [github](https://github.com). PathVQA Dataset: Data are available at [huggingface](https://huggingface.co). RadVisDial Dataset: Data are available at [github](https://github.com). VQA-Med-2019: Data are available at [github](https://github.com). VQA-RAD Dataset: Data are available at [huggingface](https://huggingface.co). VQA-Med-2018: Data are available at [imageclef](https://imageclef.org). Visual Genome: Data are available at [huggingface](https://huggingface.co). MITOS WSI CMC: Data are available at [github](https://github.com). Chest ImaGenome: Data are available at [physionet](https://physionet.org).

Declarations

Conflict of interest Each author affirms that no financial, professional, or personal relationships with other people or organizations could influence the work reported in this paper.

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